

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	1	of	98

NSP/004/042/001 - Conductor Sag Charts For Wood Pole Lines up to and including 33kV

1. Purpose

The purpose of this document is to provide a series of sag charts for use with NSP/004/042

This document supersedes the following documents, all copies of which should be destroyed.

Ref	Version	Title
NSP/004/042/001	1.0	Conductor Sag Charts For Wood Pole Lines up to and including 33kV
DSS/004/042	1.0	Specification for Yorkshire Electricity Overhead Lines to ESI 43-40
O.309	March 1996	High Voltage Single Circuit Lines on Wood Poles for Voltages up to and Including 33kV
TS 71-01 to TS 7113	Initial Version	Erection sag and tension tables for lines designed formally to DSS/004/042 version 1.0

2. Scope

This document contains sag charts for the erection of 3-phase and single-phase overhead lines on wood poles up to and including 33kV. It has been designed to be in accordance with ENA TS 43-40: Issue 2

Additionally this document contains sag charts to be used for the renovation of existing HV lines using various historical conductor types

Document reference		NSP/004/042/001			
Version:-	2.0	Date of Issue:-	Sept 10	Page	2 of 98

2.1 Contents

1. Purpose	1
2. Scope.....	1
2.1 Contents	2
3. Technical specification	4
3.1 Sag Charts.....	4
3.2 Pre-Stress Tensions for conductor erection	4
4. References	5
4.1 External Documentation	5
4.2 Internal documentation	5
4.3 Summary Of Changes	5
5 Definitions	5
6. Authority for issue	6
Appendix 1 - Schedule of sag charts.....	7
Section 1 - Sag charts for new construction.....	7
Appendix 2 - 50mm AAAC (Hazel) - (90m Basic) Normal altitude	8
Appendix 3 - 50mm AAAC (Hazel) - (120m Basic) Normal altitude	10
Appendix 3a - 50mm AAAC (Hazel) - (120m Basic) Normal altitude	11
Appendix 4 - 100mm AAAC (Oak) - (110m Basic) Normal altitude.....	12
Appendix 4a - 100mm AAAC (Oak) - (110m Basic) Normal altitude.....	13
Appendix 5 - 175mm AAAC (Elm) - (110m Basic) Normal altitude	14
Appendix 5a - 175mm AAAC (Elm) - (110m Basic) Normal altitude	15
Appendix 6 - 200mm AAAC (Poplar) - (120m Basic) Normal altitude	16
Appendix 6a - 200mm AAAC (Poplar) - (120m Basic) Normal altitude.....	17
Appendix 7 - 50mm AAAC (Hazel) - (75m Basic) High altitude	18
Appendix 7a - 50mm AAAC (Hazel) - (75m Basic) - High altitude	19
Appendix 8 - 50mm AAAC (Hazel) - (80m Basic) High altitude	20
Appendix 8a - 50mm AAAC (Hazel) - (80m Basic) High altitude	21
Appendix 9 - 50mm AAAC (Hazel) - (100m Basic) High altitude	22
Appendix 9a - 50mm AAAC (Hazel) - (100m Basic) High altitude	23
Appendix 10 - 100mm AAAC (Oak) - (80m Basic) High altitude.....	24
Appendix 10a - 100mm AAAC (Oak) - (80m Basic) High altitude.....	25
Appendix 11 - 100mm AAAC (Oak) - (90m Basic) High altitude.....	26
Appendix 11a - 100mm AAAC - (Oak) - (90m Basic) High altitude.....	27
Appendix 12 - 175mm AAAC (Elm) - (90m Basic)- High altitude	28
Appendix 12a - 175mm AAAC (Elm) - (90m Basic) High altitude	29
Appendix 13 - 175mm AAAC (Elm)- (100m Basic) - High altitude	30
Appendix 13a - 175mm AAAC (Elm)- (100m Basic) Severe altitude	31
Appendix 14 - 200mm AAAC (Poplar) - (90m Basic)- High altitude.....	32
Appendix 14a - 200mm AAAC (Poplar) - (90m Basic)- High altitude.....	33
Appendix 15 - 200mm AAAC (Poplar) - (100m Basic)- High altitude.....	34
Appendix 15a - 200mm AAAC (Poplar) - (100m Basic)- High altitude.....	35
Appendix 16 - Section 2 - Sag charts for the renovation of existing overhead lines.....	36
Appendix 17 - .017" or 13mm ² CACU - (All altitudes) - 80m Basic	37
Appendix 17a - .017" or 13mm ² CACU - (All altitudes) - 80m Basic	38
Appendix 18 - .017" or 13mm ² CACU - (All altitudes) - 90m Basic	39
Appendix 18a - .017" or 13mm ² CACU - (All altitudes) - 90m Basic	40
Appendix 19 - .017" or 13mm ² CACU - (All altitudes) - 100m Basic	41
Appendix 19a - .017" or 13mm ² CACU - (All altitudes) - 100m Basic	42
Appendix 20 - .017" or 13mm ² CACU - (All altitudes) - 120m Basic	43
Appendix 20a - .017" or 13mm ² CACU - (All altitudes) - 120m Basic	44
Appendix 21 - .05" or 32mm ² HDBC - (All altitudes) - 80m Basic	45
Appendix 21a - .05" or 32mm ² HDBC - (All altitudes) - 80m Basic	46
Appendix 22 - .05" or 32mm ² HDBC - (All altitudes) - 90m Basic	47
Appendix 22a - .05" or 32mm ² HDBC - (All altitudes) - 90m Basic	48

Document reference		NSP/004/042/001			
Version:-	2.0	Date of Issue:-	Sept 10	Page	3 of 98

Appendix 23 - .05" or 32mm ² HDBC - (All altitudes) - 100m Basic	49
Appendix 23a - .05" or 32mm ² HDBC - (All altitudes) - 100m Basic	50
Appendix 24 - .05" or 32mm ² HDBC - (All altitudes) - 120m Basic	51
Appendix 24a - .05" or 32mm ² HDBC - (All altitudes) - 120m Basic	52
Appendix 25 - .05" or 50mm ² AAAC - (All altitudes) - 80m Basic	53
Appendix 25a - .05" or 50mm ² AAAC - (All altitudes) - 80m Basic	54
Appendix 26 - .05" or 50mm ² AAAC - (All altitudes) - 100m Basic	55
Appendix 26a - .05" or 50mm ² AAAC - (All altitudes) - 100m Basic	56
Appendix 27 - .05" or 50mm ² AAAC - (All altitudes) - 120m Basic	57
Appendix 27a - .05" or 50mm ² AAAC - (All altitudes) - 120m Basic	58
Appendix 28 - .025" or 16mm ² HDBC - (All altitudes) - 90m Basic	59
Appendix 28a - .025" or 16mm ² HDBC - (All altitudes) - 90m Basic	60
Appendix 29 - .025" or 16mm ² HDBC - (All altitudes) - 100m Basic	61
Appendix 29a - .025" or 16mm ² HDBC - (All altitudes) - 100m Basic	62
Appendix 30 - 3 SWG HDBC - (All altitudes) - 90m Basic	63
Appendix 30a - 3 SWG HDBC - (All altitudes) - 90m Basic	64
Appendix 31 - 3 SWG HDBC - (All altitude) - 100m Basic	65
Appendix 31 - a3 SWG HDBC - (All altitudes) - 100m Basic	66
Appendix 32 - 5SWG HDBC - (All altitudes) - 90m Basic	67
Appendix 32a - 5SWG HDBC - (All altitudes) - 90m Basic	68
Appendix 33 - 5SWG HDBC - (All altitude) - 100m Basic	69
Appendix 33a - 5SWG HDBC - (All altitudes) - 100m Basic	70
Appendix 34 - 6SWG HDBC - (All altitudes) - 90m Basic	71
Appendix 34a - 6SWG HDBC - (All altitudes) - 90m Basic	72
Appendix 35 - 6SWG HDBC - (All altitudes) - 100m Basic	73
Appendix 35a - 6SWG HDBC - (All altitudes) - 100m Basic	74
Appendix 36 - 7SWG HDBC - (All altitudes) - 90m Basic	75
Appendix 36a - 7SWG HDBC - (All altitudes) - 90m Basic	76
Appendix 37 - 7SWG HDBC - (All altitude) - 100m Basic	77
Appendix 37a - 7SWG HDBC - (All altitude) - 100m Basic	78
Appendix 38 - .175" ACSR / 175mm ACSR (Lynx) - (All altitudes) - 76.2mm Basic.....	79
Appendix 38a - .175" ACSR / 175mm ACSR (Lynx) - (All altitudes) - 76.2mm Basic.....	80
Appendix 39 - .1" / 100mm ² ACSR (Normal altitude) - 90m Basic	81
Appendix 39a - .1" / 100mm ² ACSR (Normal altitude) - 90m Basic	82
Appendix 40 - .1" / 100mm ² ACSR (Normal altitude) - 100m Basic	83
Appendix 40a - .1" / 100mm ² ACSR (Normal altitude) - 100m Basic	84
Appendix 41 - .1" / 70mm ² HDBC (Normal altitude) - 90m Basic.....	85
Appendix 41a - .1" / 70mm ² HDBC (Normal altitude) - 90m Basic.....	86
Appendix 42 - .15" / 100mm ² HDBC (Normal altitude) - 90m Basic.....	87
Appendix 42a - .15" / 100mm ² HDBC (Normal altitude) - 90m Basic.....	88
Appendix 43 - .2" / 125mm ² HDBC (Normal altitude) - 100m Basic.....	89
Appendix 43a - .2" / 125mm ² HDBC (Normal altitude) - 100m Basic.....	90
Appendix 44 - .1" / 100mm ² ACSR (High altitude) - 90m Basic	91
Appendix 44a - .1" / 100mm ² ACSR (High altitude) - 90m Basic	92
Appendix 45 - .1" / 70mm ² HDBC (High altitude) - 90m Basic.....	93
Appendix 45a - .1" / 70mm ² HDBC (High altitude) - 90m Basic.....	94
Appendix 46 - .15" / 100mm ² HDBC (High altitude) - 90m Basic.....	95
Appendix 46a - .15" / 100mm ² HDBC (High altitude) - 90m Basic.....	96
Appendix 47 - .2" / 125mm ² HDBC (High altitude) - 100m Basic.....	97
Appendix 47a - .2" / 125mm ² HDBC (High altitude) - 100m Basic.....	98

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	4	of	98

3. Technical specification

3.1 Sag Charts

Each conductor type has two pages of information specific to a particular conductor and the altitude in which it is constructed.

Sheet 1

Describes the conductor technical data and applied weather loads applicable in the selected environment. This information is used to derive the applicable conductor MCT that is then used to create the sag charts.

Additionally this sheet contains the MCW, MCP and FPT loading values, which together with the MCT are used in the main specification to determine the structure, stay and crossarm requirements.

Sheet 2

The erection chart shall be used to determine the appropriate sags and tension for new conductors only, as it contains additional factors that increase the conductor tensions. These factors are designed to compensate for long term creep.

Design charts contain no additional factors, they shall be used for the re-sagging of existing conductors or to review the likely sags or tensions in existing overhead lines. This chart shall be used for any clearance enquiries.

Appendix 2 to 15 - New construction

Appendix 16 to 47 - Renovation or modification to existing lines

3.2 Pre-Stress Tensions for conductor erection

Conductor Type	Pre-stress Erection Tension
50mm ² AAAC	5884N (600kgf)
100mm ² AAAC	12258N (1250kgf)
175mm ² AAAC	14,710N (1500kgf)
200mm ² AAAC	14,710N (1500kgf)

Document reference		NSP/004/042/001				
Version:-	2.0	Date of Issue:-	Sept 10	Page	5	of 98

4. References

4.1 External Documentation

Reference	Title
ENA TS 43-40	High voltage single circuit overhead lines on wood poles

4.2 Internal documentation

Reference	Title
NSP/004/042	Specification for High Voltage Single Circuit Lines on Wood Poles for Voltages upto and including 33kV

4.3 Summary Of Changes

Clause	Changes or amendments
2.1	Table of contents inserted
4.3	Summary of changes table inserted
Appendix 1	Simplified list of allowable sag charts
Appendix 2 – 15	Revised sag charts to incorporate higher tension limits for the 175mm AAAC and 200mm AAAC conductors and revisions to the % reduction used in all ordinate tables.
Appendix 16 – 47	Revised Over tension and ordinate values applied to charts

5 Definitions

Term	Definition
Basic Span	The span length adopted for sag/tension calculations
Everyday Tension (EDT)	The design stress in an unloaded conductor at 5°C. Used in sag / tension calculations to limit harmful conductor vibrations
Freezing Point Tension (FPT)	The design tension of an unloaded conductor at 0°C in still air.
Maximum Conductor Pressure (MCP)	The maximum transverse component of applied conductor load when subjected to a wind load.
Maximum Conductor Weight (MCW)	The maximum vertical component of applied conductor load, including the weight of accreted ice, if present.
MCT	The maximum conductor tension under design loading conditions
Maximum Span	The maximum permitted length of any span
Maximum Working Tension (MWT)	The absolute maximum conductor tension assessed at -5.6°C with wind and ice loading.
Rated Breaking Strength (RBS)	The rating assigned to a component (usually conductors) that defines the maximum mechanical load that the component will withstand without damage.
Recommended Span	The average span length in any section to which the line shall be planned. Individual spans will normally be within +/- 20% of the chosen basic span.
Sag	The vertical distance, under any system of conductor loading, between the conductor and a straight line joining adjacent supporting points, measured at mid-span
Span	The horizontal distance between adjacent supports. Individual spans will normally be within 20% of the chosen basic span

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	6	of	98

6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
G Hammel	Overhead Line Standards Manager		

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager		

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
C Holdsworth	Standards and Assessment Manager		

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator		

6.5 Approval

Approval is given for the content of this document

		Sign	Date
D Sillito	Major Project Manager South		
S Gardner	OHL Construction Manager		

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
G Earl	Primary Engineering Projects Manager		

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	7	of	98

Appendix 1 - Schedule of sag charts

Section 1 - Sag charts for new construction

Conductor Type	Basic Span	Altitude or Loading	Appendix No.
50mm ² AAAC	90m	Normal	2
	120m		3
100mm ² AAAC	110m		4
175mm ² AAAC	110m		5
200mm ² AAAC	120m		6
50mm ² AAAC	75m	High	7
	80m		8
	100m		9
100mm ² AAAC	80m		10
	90m		11
175mm ² AAAC	90m		12
	100m		13
200mm ² AAAC	90m		14
	100m		15

Note:

See Appendix 16 for details of historical renovation conductors

Document reference	NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	8	of	98

Appendix 2 - 50mm AAAC (Hazel) - (90m Basic) Normal altitude

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	HAZEL
Greased Conductor Weight (kg/m)	0.1648
Cross Sectional Area of Conductor (mm²)	59.9
Conductor Overall Diameter (mm)	9.9
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm²)	6016.33
Rated Breaking Strength of Conductor (kgf)	1800.8
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m²)	380
Radial Ice Thickness (mm)	5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	720.3
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	360.2
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	641.0
Maximum Conductor Weight (MCW) (kg/m)			0.378
Maximum Conductor Pressure (MCP) (kg/m)			0.771
Freezing Point Tension (FPT) (kgf)	at	0°C	397.1

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	HAZEL	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.07	0.25	0.27	0.29	0.31	0.33	0.34
80	0.30	1.00	1.08	1.16	1.23	1.31	1.38
120	0.67	2.24	2.43	2.60	2.77	2.94	3.10
160	1.20	3.99	4.31	4.62	4.93	5.22	5.51
200	1.87	6.24	6.74	7.23	7.70	8.16	8.60
240	2.70	8.98	9.70	10.40	11.09	11.75	12.39
280	3.67	12.22	13.20	14.16	15.09	15.99	16.86
320	4.80	15.96	17.25	18.50	19.71	20.89	22.02
360	6.07	20.20	21.83	23.41	24.95	26.43	27.87
400	7.50	24.94	26.95	28.90	30.80	32.63	34.41
440	9.07	30.18	32.61	34.97	37.27	39.49	41.64
480	10.80	35.92	38.80	41.62	44.35	46.99	49.55
520	12.67	42.16	45.54	48.84	52.05	55.15	58.15

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	9	of	98	

Appendix 2a - 50mm AAAC (Hazel) - (90m Basic) Normal altitude

Common Design & Erection table

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	HAZEL
Basic / Recommended Span (m)	90
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	439.6	0.17	0.23	0.30	0.38	0.47	0.57	0.67	0.79	1.52
0	397.1	0.19	0.25	0.33	0.42	0.52	0.63	0.75	0.88	1.68
5	360.2	0.21	0.28	0.37	0.46	0.57	0.69	0.82	0.97	1.85
10	324.6	0.23	0.31	0.41	0.51	0.63	0.77	0.91	1.07	2.06
15	290.9	0.25	0.35	0.45	0.57	0.71	0.86	1.02	1.20	2.29
20	259.4	0.29	0.39	0.51	0.64	0.79	0.96	1.14	1.34	2.57
25	230.9	0.32	0.44	0.57	0.72	0.89	1.08	1.28	1.51	2.89
30	205.6	0.36	0.49	0.64	0.81	1.00	1.21	1.44	1.69	3.25
35	183.8	0.40	0.55	0.72	0.91	1.12	1.36	1.61	1.89	3.63
40	165.4	0.45	0.61	0.80	1.01	1.25	1.51	1.79	2.11	4.04
45	150.0	0.49	0.67	0.88	1.11	1.37	1.66	1.98	2.32	4.45
50	137.2	0.54	0.74	0.96	1.22	1.50	1.82	2.16	2.54	4.86
55	126.5	0.59	0.80	1.04	1.32	1.63	1.97	2.34	2.75	5.27
60	117.6	0.63	0.86	1.12	1.42	1.75	2.12	2.52	2.96	5.67
65	110.1	0.67	0.92	1.20	1.52	1.87	2.26	2.70	3.16	6.06
70	103.6	0.72	0.97	1.27	1.61	1.99	2.41	2.86	3.36	6.44
75	98.1	0.76	1.03	1.34	1.70	2.10	2.54	3.03	3.55	6.81
80	93.2	0.80	1.08	1.41	1.79	2.21	2.67	3.18	3.73	7.16

Document reference	NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	10	of	98

Appendix 3 - 50mm AAAC (Hazel) - (120m Basic) Normal altitude

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	HAZEL
Greased Conductor Weight (kg/m)	0.1648
Cross Sectional Area of Conductor (mm²)	59.9
Conductor Overall Diameter (mm)	9.9
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm²)	6016.33
Rated Breaking Strength of Conductor (kgf)	1800.8
Basic / Recommended Span (m)	120
Wind Pressure on Conductor (N/m²)	380
Radial Ice Thickness (mm)	5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	720.3
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	360.2
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1288.7
Maximum Conductor Weight (MCW) (kg/m)			0.968
Maximum Conductor Pressure (MCP) (kg/m)			1.279
Freezing Point Tension (FPT) (kgf)	at	0°C	786.7

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	HAZEL	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.08	0.21	0.22	0.24	0.25	0.26	0.27
80	0.30	0.84	0.89	0.95	1.00	1.05	1.10
120	0.68	1.89	2.01	2.13	2.25	2.36	2.47
160	1.22	3.35	3.57	3.78	3.99	4.20	4.39
200	1.90	5.24	5.58	5.91	6.24	6.55	6.86
240	2.73	7.54	8.03	8.51	8.98	9.44	9.88
280	3.72	10.27	10.94	11.59	12.23	12.85	13.45
320	4.86	13.41	14.28	15.14	15.97	16.78	17.57
360	6.15	16.98	18.08	19.16	20.21	21.24	22.24
400	7.60	20.96	22.32	23.65	24.95	26.22	27.45
440	9.19	25.36	27.01	28.62	30.19	31.73	33.22
480	10.94	30.18	32.14	34.06	35.93	37.76	39.53
520	12.84	35.42	37.72	39.97	42.17	44.31	46.40

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	11	of	98	

Appendix 3a - 50mm AAAC (Hazel) - (120m Basic) Normal altitude

Common Design and erection Tables DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	HAZEL	
Basic / Recommended Span (m)	120	
Temperature Shift for Creep (Deg. C)	0	Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0	at 15°C
Required Percentage Increase in Tension (%)	0	at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	433.9	0.17	0.23	0.30	0.38	0.47	0.57	0.68	0.80	0.93
0	394.1	0.19	0.26	0.33	0.42	0.52	0.63	0.75	0.88	1.02
5	360.2	0.21	0.28	0.37	0.46	0.57	0.69	0.82	0.97	1.12
10	328.0	0.23	0.31	0.40	0.51	0.63	0.76	0.90	1.06	1.23
15	298.1	0.25	0.34	0.44	0.56	0.69	0.84	1.00	1.17	1.35
20	270.7	0.27	0.37	0.49	0.62	0.76	0.92	1.10	1.29	1.49
25	246.1	0.30	0.41	0.54	0.68	0.84	1.01	1.21	1.41	1.64
30	224.3	0.33	0.45	0.59	0.74	0.92	1.11	1.32	1.55	1.80
35	205.4	0.36	0.49	0.64	0.81	1.00	1.21	1.44	1.69	1.97
40	189.1	0.39	0.53	0.70	0.88	1.09	1.32	1.57	1.84	2.14
45	175.0	0.42	0.58	0.75	0.95	1.18	1.42	1.69	1.99	2.31
50	163.0	0.46	0.62	0.81	1.02	1.26	1.53	1.82	2.14	2.48
55	152.6	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	2.65
60	143.6	0.52	0.70	0.92	1.16	1.43	1.74	2.07	2.42	2.81
65	135.8	0.55	0.74	0.97	1.23	1.52	1.83	2.18	2.56	2.97
70	129.0	0.57	0.78	1.02	1.29	1.60	1.93	2.30	2.70	3.13
75	123.0	0.60	0.82	1.07	1.36	1.68	2.03	2.41	2.83	3.28
80	117.6	0.63	0.86	1.12	1.42	1.75	2.12	2.52	2.96	3.43

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	12	of	98

Appendix 4 - 100mm AAAC (Oak) - (110m Basic) Normal altitude

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	OAK
Greased Conductor Weight (kg/m)	0.3273
Cross Sectional Area of Conductor (mm²)	118.9
Conductor Overall Diameter (mm)	14
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm²)	6016.33
Rated Breaking Strength of Conductor (kgf)	3576.1
Basic / Recommended Span (m)	110
Wind Pressure on Conductor (N/m²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1788.1
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	715.2
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	714.9
Maximum Conductor Weight (MCW) (kg/m)			0.378
Maximum Conductor Pressure (MCP) (kg/m)			0.771
Freezing Point Tension (FPT) (kgf)	at	0°C	394.1

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	OAK	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	10

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.08	0.23	0.24	0.26	0.27	0.29	0.30
80	0.30	0.92	0.98	1.03	1.09	1.15	1.20
120	0.68	2.06	2.20	2.33	2.46	2.58	2.71
160	1.21	3.66	3.90	4.14	4.37	4.59	4.81
200	1.89	5.72	6.10	6.47	6.83	7.18	7.52
240	2.72	8.24	8.78	9.31	9.83	10.33	10.82
280	3.71	11.21	11.95	12.67	13.38	14.06	14.73
320	4.84	14.64	15.61	16.55	17.48	18.37	19.24
360	6.13	18.53	19.76	20.95	22.12	23.25	24.35
400	7.56	22.88	24.39	25.87	27.30	28.70	30.06
440	9.15	27.68	29.51	31.30	33.04	34.73	36.37
480	10.89	32.94	35.12	37.25	39.32	41.33	43.29
520	12.78	38.66	41.22	43.71	46.15	48.51	50.80

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	13	of	98	

Appendix 4a - 100mm AAAC (Oak) - (110m Basic) Normal altitude

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	OAK
Basic / Recommended Span (m)	110
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	865.7	0.17	0.23	0.30	0.38	0.47	0.57	0.68	0.80	0.93
0	784.7	0.19	0.26	0.33	0.42	0.52	0.63	0.75	0.88	1.02
5	715.2	0.21	0.28	0.37	0.46	0.57	0.69	0.82	0.97	1.12
10	649.1	0.23	0.31	0.40	0.51	0.63	0.76	0.91	1.07	1.24
15	587.2	0.25	0.34	0.45	0.56	0.70	0.84	1.00	1.18	1.37
20	530.3	0.28	0.38	0.49	0.62	0.77	0.93	1.11	1.30	1.51
25	479.0	0.31	0.42	0.55	0.69	0.85	1.03	1.23	1.44	1.67
30	433.7	0.34	0.46	0.60	0.76	0.94	1.14	1.36	1.59	1.85
35	394.4	0.37	0.51	0.66	0.84	1.04	1.26	1.49	1.75	2.03
40	360.7	0.41	0.56	0.73	0.92	1.13	1.37	1.63	1.92	2.22
45	332.0	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	2.42
50	307.6	0.48	0.65	0.85	1.08	1.33	1.61	1.92	2.25	2.61
55	286.8	0.51	0.70	0.91	1.16	1.43	1.73	2.05	2.41	2.80
60	268.9	0.55	0.75	0.97	1.23	1.52	1.84	2.19	2.57	2.98
65	253.5	0.58	0.79	1.03	1.31	1.61	1.95	2.32	2.73	3.16
70	240.2	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.88	3.34
75	228.4	0.64	0.88	1.15	1.45	1.79	2.17	2.58	3.03	3.51
80	218.1	0.68	0.92	1.20	1.52	1.88	2.27	2.70	3.17	3.68

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	OAK
Basic / Recommended Span (m)	110
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	10.0 at 15°C
Required Percentage Increase in Tension (%)	10 at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	936.3	0.16	0.21	0.28	0.35	0.44	0.53	0.63	0.74	0.86
0	853.2	0.17	0.23	0.31	0.39	0.48	0.58	0.69	0.81	0.94
5	781.2	0.19	0.26	0.34	0.42	0.52	0.63	0.75	0.89	1.03
10	711.9	0.21	0.28	0.37	0.47	0.57	0.70	0.83	0.97	1.13
15	645.9	0.23	0.31	0.41	0.51	0.63	0.77	0.91	1.07	1.24
20	584.3	0.25	0.34	0.45	0.57	0.70	0.85	1.01	1.18	1.37
25	527.6	0.28	0.38	0.50	0.63	0.78	0.94	1.12	1.31	1.52
30	476.6	0.31	0.42	0.55	0.70	0.86	1.04	1.24	1.45	1.68
35	431.6	0.34	0.46	0.61	0.77	0.95	1.15	1.36	1.60	1.86
40	392.6	0.38	0.51	0.67	0.84	1.04	1.26	1.50	1.76	2.04
45	359.2	0.41	0.56	0.73	0.92	1.14	1.38	1.64	1.92	2.23
50	330.7	0.45	0.61	0.79	1.00	1.24	1.50	1.78	2.09	2.42
55	306.5	0.48	0.65	0.85	1.08	1.33	1.62	1.92	2.26	2.62
60	285.8	0.52	0.70	0.92	1.16	1.43	1.73	2.06	2.42	2.81
65	268.1	0.55	0.75	0.98	1.24	1.53	1.85	2.20	2.58	2.99
70	252.8	0.58	0.79	1.04	1.31	1.62	1.96	2.33	2.73	3.17
75	239.5	0.61	0.84	1.09	1.38	1.71	2.07	2.46	2.89	3.35
80	227.9	0.65	0.88	1.15	1.45	1.80	2.17	2.59	3.03	3.52

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	14	of	98

Appendix 5 - 175mm AAAC (Elm) - (110m Basic) Normal altitude

Conductor Code Name (if any)	ELM
Greased Conductor Weight (kg/m)	0.5922
Cross Sectional Area of Conductor (mm²)	211
Conductor Overall Diameter (mm)	18.8
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm²)	5710.41
Rated Breaking Strength of Conductor (kgf)	6346.7
Basic / Recommended Span (m)	110
Wind Pressure on Conductor (N/m²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	2379.3
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1269.3
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	2024.4
Maximum Conductor Weight (MCW) (kg/m)			1.363
Maximum Conductor Pressure (MCP) (kg/m)			1.465
Freezing Point Tension (FPT) (kgf)	at	0°C	1386.5

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	ELM	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	10

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.08	0.23	0.24	0.26	0.27	0.28	0.30
80	0.31	0.91	0.97	1.03	1.08	1.14	1.19
120	0.70	2.04	2.18	2.31	2.44	2.56	2.68
160	1.24	3.63	3.87	4.10	4.33	4.55	4.77
200	1.94	5.67	6.05	6.41	6.77	7.12	7.45
240	2.80	8.17	8.71	9.23	9.75	10.25	10.73
280	3.81	11.12	11.85	12.57	13.27	13.95	14.61
320	4.98	14.52	15.48	16.42	17.33	18.22	19.08
360	6.30	18.38	19.59	20.78	21.93	23.06	24.15
400	7.78	22.69	24.19	25.65	27.08	28.47	29.82
440	9.41	27.46	29.27	31.04	32.76	34.44	36.08
480	11.20	32.68	34.83	36.94	38.99	40.99	42.94
520	13.14	38.35	40.88	43.35	45.76	48.11	50.39

Document reference	NSP/004/042/001							
Version:-	2.0	Date of Issue:-	Sept 10	Page	15	of	98	

Appendix 5a - 175mm AAAC (Elm) - (110m Basic) Normal altitude

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	ELM
Basic / Recommended Span (m)	110
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	1522.8	0.18	0.24	0.31	0.39	0.49	0.59	0.70	0.82	0.95
0	1386.5	0.19	0.26	0.34	0.43	0.53	0.65	0.77	0.90	1.05
5	1269.3	0.21	0.29	0.37	0.47	0.58	0.71	0.84	0.99	1.14
10	1157.6	0.23	0.31	0.41	0.52	0.64	0.77	0.92	1.08	1.25
15	1052.4	0.25	0.34	0.45	0.57	0.70	0.85	1.01	1.19	1.38
20	955.0	0.28	0.38	0.50	0.63	0.78	0.94	1.12	1.31	1.52
25	866.6	0.31	0.42	0.55	0.69	0.85	1.03	1.23	1.44	1.67
30	787.7	0.34	0.46	0.60	0.76	0.94	1.14	1.35	1.59	1.84
35	718.5	0.37	0.50	0.66	0.83	1.03	1.25	1.48	1.74	2.02
40	658.5	0.40	0.55	0.72	0.91	1.12	1.36	1.62	1.90	2.20
45	606.9	0.44	0.60	0.78	0.99	1.22	1.48	1.76	2.06	2.39
50	562.8	0.47	0.64	0.84	1.07	1.32	1.59	1.89	2.22	2.58
55	524.9	0.51	0.69	0.90	1.14	1.41	1.71	2.03	2.38	2.76
60	492.2	0.54	0.74	0.96	1.22	1.50	1.82	2.17	2.54	2.95
65	464.0	0.57	0.78	1.02	1.29	1.60	1.93	2.30	2.70	3.13
70	439.4	0.61	0.83	1.08	1.36	1.68	2.04	2.43	2.85	3.30
75	417.8	0.64	0.87	1.13	1.44	1.77	2.14	2.55	2.99	3.47
80	398.7	0.67	0.91	1.19	1.50	1.86	2.25	2.67	3.14	3.64

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	ELM
Basic / Recommended Span (m)	110
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	10.0 at 15°C
Required Percentage Increase in Tension (%)	10 at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	1647.9	0.16	0.22	0.29	0.36	0.45	0.54	0.65	0.76	0.88
0	1508.0	0.18	0.24	0.31	0.40	0.49	0.59	0.71	0.83	0.96
5	1386.6	0.19	0.26	0.34	0.43	0.53	0.65	0.77	0.90	1.05
10	1269.4	0.21	0.29	0.37	0.47	0.58	0.71	0.84	0.99	1.14
15	1157.6	0.23	0.31	0.41	0.52	0.64	0.77	0.92	1.08	1.25
20	1052.4	0.25	0.34	0.45	0.57	0.70	0.85	1.01	1.19	1.38
25	955.1	0.28	0.38	0.50	0.63	0.78	0.94	1.12	1.31	1.52
30	866.6	0.31	0.42	0.55	0.69	0.85	1.03	1.23	1.44	1.67
35	787.7	0.34	0.46	0.60	0.76	0.94	1.14	1.35	1.59	1.84
40	718.5	0.37	0.50	0.66	0.83	1.03	1.25	1.48	1.74	2.02
45	658.5	0.40	0.55	0.72	0.91	1.12	1.36	1.62	1.90	2.20
50	607.0	0.44	0.60	0.78	0.99	1.22	1.48	1.76	2.06	2.39
55	562.8	0.47	0.64	0.84	1.07	1.32	1.59	1.89	2.22	2.58
60	524.9	0.51	0.69	0.90	1.14	1.41	1.71	2.03	2.38	2.76
65	492.2	0.54	0.74	0.96	1.22	1.50	1.82	2.17	2.54	2.95
70	464.0	0.57	0.78	1.02	1.29	1.60	1.93	2.30	2.70	3.13
75	439.4	0.61	0.83	1.08	1.36	1.68	2.04	2.43	2.85	3.30
80	417.8	0.64	0.87	1.13	1.44	1.77	2.14	2.55	2.99	3.47

Document reference	NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	16	of	98

Appendix 6 - 200mm AAAC (Poplar) - (120m Basic) Normal altitude

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	POPLAR
Greased Conductor Weight (kg/m)	0.68992
Cross Sectional Area of Conductor (mm²)	239.4
Conductor Overall Diameter (mm)	20.1
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm²)	5700
Rated Breaking Strength of Conductor (kgf)	7200.2
Basic / Recommended Span (m)	120
Wind Pressure on Conductor (N/m²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	2379.3
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1440.04
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	2291.7
Maximum Conductor Weight (MCW) (kg/m)			1.496
Maximum Conductor Pressure (MCP) (kg/m)			1.515
Freezing Point Tension (FPT) (kgf)	at	0°C	1567.6

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	POPLAR	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	10

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.08	0.22	0.23	0.24	0.26	0.27	0.28
80	0.32	0.87	0.93	0.98	1.03	1.08	1.12
120	0.72	1.96	2.08	2.20	2.31	2.42	2.53
160	1.29	3.49	3.70	3.91	4.11	4.31	4.50
200	2.01	5.46	5.79	6.11	6.42	6.73	7.03
240	2.89	7.86	8.33	8.80	9.25	9.69	10.12
280	3.94	10.70	11.34	11.97	12.59	13.19	13.78
320	5.14	13.97	14.81	15.64	16.45	17.23	18.00
360	6.51	17.68	18.75	19.79	20.81	21.81	22.78
400	8.04	21.83	23.15	24.44	25.70	26.92	28.12
440	9.72	26.42	28.01	29.57	31.09	32.58	34.03
480	11.57	31.44	33.33	35.19	37.00	38.77	40.50
520	13.58	36.90	39.12	41.30	43.43	45.50	47.53

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	17	of	98	

Appendix 6a - 200mm AAAC (Poplar) - (120m Basic) Normal altitude

Conductor Code Name **POPLAR**
 Basic / Recommended Span (m) **120**
 Temperature Shift for Creep (Deg. C) **0** Insert minus sign as necessary
 Equivalent Percentage Increase in Tension (%) **0.0** at 15°C
 Required Percentage Increase in Tension (%) **0** at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	1717.0	0.18	0.25	0.32	0.41	0.50	0.61	0.72	0.85	0.98
0	1567.6	0.20	0.27	0.35	0.45	0.55	0.67	0.79	0.93	1.08
5	1440.0	0.22	0.29	0.38	0.49	0.60	0.72	0.86	1.01	1.17
10	1319.1	0.24	0.32	0.42	0.53	0.65	0.79	0.94	1.10	1.28
15	1206.1	0.26	0.35	0.46	0.58	0.72	0.87	1.03	1.21	1.40
20	1102.2	0.28	0.38	0.50	0.63	0.78	0.95	1.13	1.32	1.53
25	1008.0	0.31	0.42	0.55	0.69	0.86	1.04	1.23	1.45	1.68
30	924.0	0.34	0.46	0.60	0.76	0.93	1.13	1.34	1.58	1.83
35	850.0	0.37	0.50	0.65	0.82	1.01	1.23	1.46	1.71	1.99
40	785.4	0.40	0.54	0.70	0.89	1.10	1.33	1.58	1.86	2.15
45	729.3	0.43	0.58	0.76	0.96	1.18	1.43	1.70	2.00	2.32
50	680.7	0.46	0.62	0.81	1.03	1.27	1.53	1.82	2.14	2.48
55	638.6	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	2.65
60	601.8	0.52	0.70	0.92	1.16	1.43	1.73	2.06	2.42	2.81
65	569.7	0.54	0.74	0.97	1.23	1.51	1.83	2.18	2.56	2.97
70	541.4	0.57	0.78	1.02	1.29	1.59	1.93	2.29	2.69	3.12
75	516.4	0.60	0.82	1.07	1.35	1.67	2.02	2.40	2.82	3.27
80	494.2	0.63	0.86	1.12	1.41	1.75	2.11	2.51	2.95	3.42

Conductor Code Name **POPLAR**
 Basic / Recommended Span (m) **120**
 Temperature Shift for Creep (Deg. C) **0** Insert minus sign as necessary
 Equivalent Percentage Increase in Tension (%) **10.0** at 15°C
 Required Percentage Increase in Tension (%) **10** at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	1864.1	0.17	0.23	0.30	0.37	0.46	0.56	0.67	0.78	0.91
0	1709.5	0.18	0.25	0.32	0.41	0.50	0.61	0.73	0.85	0.99
5	1576.1	0.20	0.27	0.35	0.44	0.55	0.66	0.79	0.92	1.07
10	1448.1	0.21	0.29	0.38	0.48	0.60	0.72	0.86	1.01	1.17
15	1326.8	0.23	0.32	0.42	0.53	0.65	0.79	0.94	1.10	1.27
20	1213.2	0.26	0.35	0.45	0.58	0.71	0.86	1.02	1.20	1.39
25	1108.6	0.28	0.38	0.50	0.63	0.78	0.94	1.12	1.31	1.52
30	1013.8	0.31	0.42	0.54	0.69	0.85	1.03	1.22	1.44	1.67
35	929.1	0.33	0.45	0.59	0.75	0.93	1.12	1.34	1.57	1.82
40	854.5	0.36	0.49	0.65	0.82	1.01	1.22	1.45	1.71	1.98
45	789.3	0.39	0.54	0.70	0.88	1.09	1.32	1.57	1.85	2.14
50	732.7	0.42	0.58	0.75	0.95	1.18	1.42	1.69	1.99	2.31
55	683.7	0.45	0.62	0.81	1.02	1.26	1.53	1.82	2.13	2.47
60	641.1	0.48	0.66	0.86	1.09	1.35	1.63	1.94	2.27	2.64
65	604.1	0.51	0.70	0.91	1.16	1.43	1.73	2.06	2.41	2.80
70	571.7	0.54	0.74	0.97	1.22	1.51	1.83	2.17	2.55	2.96
75	543.2	0.57	0.78	1.02	1.29	1.59	1.92	2.29	2.68	3.11
80	518.0	0.60	0.82	1.07	1.35	1.67	2.01	2.40	2.81	3.26

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	18	of	98

Appendix 7 - 50mm AAAC (Hazel) - (75m Basic) High altitude

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	HAZEL
Greased Conductor Weight (kg/m)	0.1648
Cross Sectional Area of Conductor (mm ²)	59.9
Conductor Overall Diameter (mm)	9.9
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm ²)	6016.33
Rated Breaking Strength of Conductor (kgf)	1800.8
Basic / Recommended Span (m)	75
Wind Pressure on Conductor (N/m ²)	380
Radial Ice Thickness (mm)	5
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	720.3
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	360.2
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	602.2
Maximum Conductor Weight (MCW) (kg/m)			0.378
Maximum Conductor Pressure (MCP) (kg/m)			0.771
Freezing Point Tension (FPT) (kgf)	at	0°C	398.4

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	HAZEL	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	0

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.07	0.27	0.29	0.32	0.34	0.37	0.39
80	0.30	1.08	1.18	1.28	1.38	1.47	1.56
120	0.67	2.42	2.65	2.88	3.10	3.31	3.52
160	1.19	4.31	4.72	5.12	5.51	5.89	6.25
200	1.86	6.73	7.37	8.00	8.61	9.20	9.76
240	2.68	9.69	10.62	11.52	12.40	13.25	14.06
280	3.65	13.19	14.45	15.69	16.88	18.03	19.14
320	4.77	17.23	18.88	20.49	22.04	23.55	25.00
360	6.04	21.80	23.89	25.93	27.90	29.80	31.64
400	7.46	26.92	29.50	32.01	34.44	36.79	39.06
440	9.02	32.57	35.69	38.73	41.68	44.52	47.26
480	10.74	38.76	42.47	46.09	49.60	52.98	56.24
520	12.60	45.49	49.85	54.10	58.21	62.18	66.01

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	19	of	98	

Appendix 7a - 50mm AAAC (Hazel) - (75m Basic) - High altitude

Common Design & Erection Table

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	HAZEL	
Basic / Recommended Span (m)	75	
Temperature Shift for Creep (Deg. C)	0	Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	5.0	at 15°C
Required Percentage Increase in Tension (%)	5	at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	458.2	0.16	0.22	0.29	0.36	0.45	0.54	0.65	0.76	0.88
0	414.2	0.18	0.24	0.32	0.40	0.50	0.60	0.72	0.84	0.97
5	375.7	0.20	0.27	0.35	0.44	0.55	0.66	0.79	0.93	1.07
10	338.1	0.22	0.30	0.39	0.49	0.61	0.74	0.88	1.03	1.19
15	301.7	0.25	0.33	0.44	0.55	0.68	0.83	0.98	1.15	1.34
20	267.2	0.28	0.38	0.49	0.62	0.77	0.93	1.11	1.30	1.51
25	235.1	0.32	0.43	0.56	0.71	0.88	1.06	1.26	1.48	1.72
30	206.2	0.36	0.49	0.64	0.81	1.00	1.21	1.44	1.69	1.96
35	180.9	0.41	0.56	0.73	0.92	1.14	1.38	1.64	1.92	2.23
40	159.5	0.46	0.63	0.83	1.05	1.29	1.56	1.86	2.18	2.53
45	141.9	0.52	0.71	0.93	1.18	1.45	1.76	2.09	2.45	2.85
50	127.5	0.58	0.79	1.03	1.31	1.62	1.95	2.33	2.73	3.17
55	115.9	0.64	0.87	1.14	1.44	1.78	2.15	2.56	3.00	3.48
60	106.4	0.70	0.95	1.24	1.57	1.94	2.34	2.79	3.27	3.80
65	98.5	0.75	1.02	1.34	1.69	2.09	2.53	3.01	3.53	4.10
70	92.0	0.81	1.10	1.43	1.81	2.24	2.71	3.23	3.79	4.39
75	86.4	0.86	1.17	1.53	1.93	2.38	2.88	3.43	4.03	4.67
80	81.7	0.91	1.24	1.61	2.04	2.52	3.05	3.63	4.26	4.94

Document reference	NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	20	of	98

Appendix 8 - 50mm AAAC (Hazel) - (80m Basic) High altitude

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	HAZEL
Greased Conductor Weight (kg/m)	0.1648
Cross Sectional Area of Conductor (mm²)	59.9
Conductor Overall Diameter (mm)	9.9
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm²)	6016.33
Rated Breaking Strength of Conductor (kgf)	1800.8
Basic / Recommended Span (m)	80
Wind Pressure on Conductor (N/m²)	380
Radial Ice Thickness (mm)	5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	720.3
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	360.2
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	615.2
Maximum Conductor Weight (MCW) (kg/m)			0.378
Maximum Conductor Pressure (MCP) (kg/m)			0.771
Freezing Point Tension (FPT) (kgf)	at	0°C	398.0

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	HAZEL	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	0

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.07	0.26	0.28	0.31	0.33	0.35	0.37
80	0.30	1.03	1.13	1.22	1.31	1.40	1.48
120	0.67	2.33	2.54	2.75	2.95	3.14	3.33
160	1.20	4.14	4.51	4.88	5.24	5.59	5.92
200	1.87	6.46	7.05	7.63	8.19	8.73	9.25
240	2.69	9.30	10.16	10.99	11.79	12.57	13.33
280	3.66	12.66	13.82	14.96	16.05	17.11	18.14
320	4.78	16.54	18.06	19.53	20.97	22.35	23.69
360	6.05	20.93	22.85	24.72	26.54	28.29	29.98
400	7.47	25.84	28.21	30.52	32.76	34.93	37.02
440	9.04	31.27	34.14	36.93	39.64	42.26	44.79
480	10.76	37.22	40.63	43.95	47.18	50.30	53.30
520	12.62	43.68	47.68	51.58	55.37	59.03	62.56

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	21	of	98	

Appendix 8a - 50mm AAAC (Hazel) - (80m Basic) High altitude

Common Design & Erection Table

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	HAZEL	
Basic / Recommended Span (m)	80	
Temperature Shift for Creep (Deg. C)	0	Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	5.0	at 15°C
Required Percentage Increase in Tension (%)	5	at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	457.7	0.16	0.22	0.29	0.36	0.45	0.54	0.65	0.76	0.88
0	414.1	0.18	0.24	0.32	0.40	0.50	0.60	0.72	0.84	0.98
5	375.9	0.20	0.27	0.35	0.44	0.55	0.66	0.79	0.93	1.07
10	338.7	0.22	0.30	0.39	0.49	0.61	0.74	0.88	1.03	1.19
15	302.9	0.24	0.33	0.44	0.55	0.68	0.82	0.98	1.15	1.33
20	269.1	0.28	0.38	0.49	0.62	0.77	0.93	1.10	1.29	1.50
25	237.8	0.31	0.42	0.55	0.70	0.87	1.05	1.25	1.46	1.70
30	209.6	0.35	0.48	0.63	0.80	0.98	1.19	1.42	1.66	1.93
35	185.0	0.40	0.55	0.71	0.90	1.11	1.35	1.60	1.88	2.18
40	164.1	0.45	0.61	0.80	1.02	1.25	1.52	1.81	2.12	2.46
45	146.9	0.50	0.69	0.90	1.14	1.40	1.70	2.02	2.37	2.75
50	132.7	0.56	0.76	0.99	1.26	1.55	1.88	2.24	2.62	3.04
55	121.1	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.88	3.34
60	111.5	0.67	0.91	1.18	1.50	1.85	2.24	2.66	3.12	3.62
65	103.5	0.72	0.97	1.27	1.61	1.99	2.41	2.87	3.36	3.90
70	96.9	0.77	1.04	1.36	1.72	2.13	2.57	3.06	3.59	4.17
75	91.2	0.81	1.11	1.45	1.83	2.26	2.73	3.25	3.82	4.43
80	86.3	0.86	1.17	1.53	1.93	2.39	2.89	3.44	4.03	4.68

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	22	of	98

Appendix 9 - 50mm AAAC (Hazel) - (100m Basic) High altitude

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	HAZEL
Greased Conductor Weight (kg/m)	0.1648
Cross Sectional Area of Conductor (mm²)	59.9
Conductor Overall Diameter (mm)	9.9
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm²)	6016.33
Rated Breaking Strength of Conductor (kgf)	1800.8
Basic / Recommended Span (m)	100
Wind Pressure on Conductor (N/m²)	380
Radial Ice Thickness (mm)	5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	720.3
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	360.2
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	666.2
Maximum Conductor Weight (MCW) (kg/m)			0.378
Maximum Conductor Pressure (MCP) (kg/m)			0.771
Freezing Point Tension (FPT) (kgf)	at	0°C	396.2

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	HAZEL	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	0

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.08	0.23	0.24	0.26	0.28	0.29	0.31
80	0.30	0.90	0.97	1.04	1.11	1.17	1.24
120	0.68	2.03	2.19	2.34	2.49	2.64	2.78
160	1.20	3.60	3.89	4.16	4.43	4.69	4.94
200	1.88	5.63	6.07	6.50	6.92	7.33	7.72
240	2.71	8.11	8.74	9.36	9.97	10.55	11.12
280	3.69	11.04	11.90	12.74	13.56	14.36	15.14
320	4.82	14.42	15.54	16.64	17.72	18.76	19.77
360	6.10	18.25	19.67	21.07	22.42	23.74	25.02
400	7.53	22.53	24.29	26.01	27.68	29.31	30.89
440	9.11	27.26	29.39	31.47	33.50	35.47	37.38
480	10.84	32.44	34.97	37.45	39.86	42.21	44.48
520	12.72	38.08	41.04	43.95	46.78	49.53	52.20

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	23	of	98	

Appendix 9a - 50mm AAAC (Hazel) - (100m Basic) High altitude

Common Design & Erection Table

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	HAZEL	
Basic / Recommended Span (m)	100	
Temperature Shift for Creep (Deg. C)	0	Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	5.0	at 15°C
Required Percentage Increase in Tension (%)	5	at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	455.3	0.16	0.22	0.29	0.37	0.45	0.55	0.65	0.76	0.89
0	413.1	0.18	0.24	0.32	0.40	0.50	0.60	0.72	0.84	0.98
5	376.6	0.20	0.27	0.35	0.44	0.55	0.66	0.79	0.92	1.07
10	341.4	0.22	0.30	0.39	0.49	0.60	0.73	0.87	1.02	1.18
15	307.9	0.24	0.33	0.43	0.54	0.67	0.81	0.96	1.13	1.31
20	276.7	0.27	0.36	0.48	0.60	0.74	0.90	1.07	1.26	1.46
25	248.2	0.30	0.41	0.53	0.67	0.83	1.00	1.20	1.40	1.63
30	222.8	0.33	0.45	0.59	0.75	0.92	1.12	1.33	1.56	1.81
35	200.5	0.37	0.50	0.66	0.83	1.03	1.24	1.48	1.74	2.01
40	181.5	0.41	0.56	0.73	0.92	1.14	1.37	1.63	1.92	2.22
45	165.4	0.45	0.61	0.80	1.01	1.25	1.51	1.79	2.11	2.44
50	151.8	0.49	0.66	0.87	1.10	1.36	1.64	1.95	2.29	2.66
55	140.4	0.53	0.72	0.94	1.19	1.47	1.78	2.11	2.48	2.88
60	130.7	0.57	0.77	1.01	1.28	1.58	1.91	2.27	2.66	3.09
65	122.5	0.61	0.82	1.08	1.36	1.68	2.04	2.42	2.84	3.30
70	115.4	0.64	0.87	1.14	1.45	1.79	2.16	2.57	3.02	3.50
75	109.3	0.68	0.92	1.21	1.53	1.89	2.28	2.72	3.19	3.70
80	103.9	0.71	0.97	1.27	1.61	1.98	2.40	2.85	3.35	3.89

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	24	of	98

Appendix 10 - 100mm AAAC (Oak) -(80m Basic) High altitude

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	OAK
Greased Conductor Weight (kg/m)	0.3273
Cross Sectional Area of Conductor (mm²)	118.9
Conductor Overall Diameter (mm)	14
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm²)	6016.33
Rated Breaking Strength of Conductor (kgf)	3576.1
Basic / Recommended Span (m)	80
Wind Pressure on Conductor (N/m²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1788.1
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	715.2
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1193.8
Maximum Conductor Weight (MCW) (kg/m)			0.968
Maximum Conductor Pressure (MCP) (kg/m)			1.279
Freezing Point Tension (FPT) (kgf)	at	0°C	790.3

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	OAK	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	10

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.07	0.28	0.30	0.32	0.35	0.37	0.39
80	0.30	1.12	1.21	1.30	1.39	1.47	1.55
120	0.67	2.51	2.72	2.92	3.12	3.31	3.49
160	1.20	4.47	4.84	5.20	5.54	5.88	6.20
200	1.87	6.98	7.56	8.12	8.66	9.19	9.69
240	2.69	10.05	10.88	11.69	12.48	13.23	13.96
280	3.66	13.68	14.81	15.91	16.98	18.01	19.00
320	4.78	17.86	19.35	20.79	22.18	23.52	24.82
360	6.05	22.61	24.49	26.31	28.07	29.77	31.41
400	7.47	27.91	30.23	32.48	34.65	36.75	38.78
440	9.04	33.77	36.58	39.30	41.93	44.47	46.92
480	10.76	40.19	43.53	46.77	49.90	52.92	55.84
520	12.62	47.17	51.09	54.89	58.56	62.11	65.53

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	25	of	98	

Appendix 10a - 100mm AAAC (Oak) - (80m Basic) High altitude

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	OAK
Basic / Recommended Span (m)	80
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	876.3	0.17	0.23	0.30	0.38	0.47	0.56	0.67	0.79	0.92
0	790.3	0.19	0.25	0.33	0.42	0.52	0.63	0.75	0.87	1.01
5	715.2	0.21	0.28	0.37	0.46	0.57	0.69	0.82	0.97	1.12
10	642.5	0.23	0.31	0.41	0.52	0.64	0.77	0.92	1.08	1.25
15	573.0	0.26	0.35	0.46	0.58	0.71	0.86	1.03	1.21	1.40
20	507.8	0.29	0.39	0.52	0.65	0.81	0.97	1.16	1.36	1.58
25	448.0	0.33	0.45	0.58	0.74	0.91	1.10	1.31	1.54	1.79
30	395.0	0.37	0.51	0.66	0.84	1.04	1.25	1.49	1.75	2.03
35	349.2	0.42	0.57	0.75	0.95	1.17	1.42	1.69	1.98	2.30
40	310.9	0.47	0.64	0.84	1.07	1.32	1.59	1.90	2.22	2.58
45	279.2	0.53	0.72	0.94	1.19	1.47	1.77	2.11	2.48	2.87
50	253.3	0.58	0.79	1.03	1.31	1.61	1.95	2.33	2.73	3.17
55	232.1	0.63	0.86	1.13	1.43	1.76	2.13	2.54	2.98	3.46
60	214.5	0.69	0.93	1.22	1.54	1.91	2.31	2.75	3.22	3.74
65	199.8	0.74	1.00	1.31	1.66	2.05	2.48	2.95	3.46	4.01
70	187.4	0.79	1.07	1.40	1.77	2.18	2.64	3.14	3.69	4.28
75	176.9	0.83	1.13	1.48	1.87	2.31	2.80	3.33	3.91	4.53
80	167.7	0.88	1.20	1.56	1.98	2.44	2.95	3.51	4.12	4.78

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	OAK
Basic / Recommended Span (m)	80
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	10.0 at 15°C
Required Percentage Increase in Tension (%)	10 at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	940.9	0.16	0.21	0.28	0.35	0.43	0.53	0.63	0.73	0.85
0	853.7	0.17	0.23	0.31	0.39	0.48	0.58	0.69	0.81	0.94
5	777.2	0.19	0.26	0.34	0.43	0.53	0.64	0.76	0.89	1.03
10	702.5	0.21	0.29	0.37	0.47	0.58	0.70	0.84	0.98	1.14
15	630.3	0.23	0.32	0.42	0.53	0.65	0.79	0.93	1.10	1.27
20	561.4	0.26	0.36	0.47	0.59	0.73	0.88	1.05	1.23	1.43
25	497.0	0.30	0.40	0.53	0.67	0.82	1.00	1.19	1.39	1.61
30	438.4	0.34	0.46	0.60	0.76	0.93	1.13	1.34	1.58	1.83
35	386.6	0.38	0.52	0.68	0.86	1.06	1.28	1.52	1.79	2.07
40	342.1	0.43	0.59	0.77	0.97	1.20	1.45	1.72	2.02	2.34
45	305.0	0.48	0.66	0.86	1.09	1.34	1.62	1.93	2.27	2.63
50	274.4	0.54	0.73	0.95	1.21	1.49	1.80	2.15	2.52	2.92
55	249.4	0.59	0.80	1.05	1.33	1.64	1.99	2.36	2.77	3.22
60	228.8	0.64	0.88	1.14	1.45	1.79	2.16	2.57	3.02	3.50
65	211.8	0.70	0.95	1.24	1.56	1.93	2.34	2.78	3.26	3.79
70	197.6	0.75	1.01	1.33	1.68	2.07	2.51	2.98	3.50	4.06
75	185.5	0.79	1.08	1.41	1.79	2.21	2.67	3.18	3.73	4.32
80	175.2	0.84	1.14	1.49	1.89	2.34	2.83	3.36	3.95	4.58

Document reference	NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	26	of	98

Appendix 11 - 100mm AAAC (Oak) -(90m Basic) High altitude

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	OAK
Greased Conductor Weight (kg/m)	0.3273
Cross Sectional Area of Conductor (mm²)	118.9
Conductor Overall Diameter (mm)	14
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm²)	6016.33
Rated Breaking Strength of Conductor (kgf)	3576.1
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m²)	570
Radial Ice Thickness (mm)	12.5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1788.1
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	715.2
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1533.8
Maximum Conductor Weight (MCW) (kg/m)			1.277
Maximum Conductor Pressure (MCP) (kg/m)			2.267
Freezing Point Tension (FPT) (kgf)	at	0°C	788.5

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	OAK	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	10

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.07	0.26	0.28	0.30	0.32	0.33	0.35
80	0.30	1.04	1.11	1.19	1.27	1.34	1.41
120	0.67	2.33	2.51	2.68	2.85	3.01	3.17
160	1.20	4.14	4.46	4.77	5.07	5.36	5.64
200	1.87	6.47	6.97	7.45	7.92	8.37	8.81
240	2.70	9.32	10.03	10.73	11.40	12.05	12.68
280	3.67	12.69	13.66	14.60	15.52	16.40	17.26
320	4.80	16.57	17.84	19.07	20.27	21.42	22.54
360	6.07	20.97	22.57	24.14	25.65	27.11	28.53
400	7.50	25.89	27.87	29.80	31.66	33.47	35.22
440	9.07	31.32	33.72	36.05	38.31	40.50	42.62
480	10.80	37.28	40.13	42.91	45.60	48.20	50.72
520	12.67	43.75	47.10	50.36	53.51	56.57	59.53

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	27	of	98	

Appendix 11a - 100mm AAAC - (Oak) - (90m Basic) High altitude

Conductor Code Name	OAK
Basic / Recommended Span (m)	90
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	873.0	0.17	0.23	0.30	0.38	0.47	0.57	0.67	0.79	1.52
0	788.5	0.19	0.25	0.33	0.42	0.52	0.63	0.75	0.88	1.68
5	715.2	0.21	0.28	0.37	0.46	0.57	0.69	0.82	0.97	1.85
10	644.6	0.23	0.31	0.41	0.51	0.63	0.77	0.91	1.07	2.06
15	577.7	0.25	0.35	0.45	0.57	0.71	0.86	1.02	1.20	2.29
20	515.3	0.29	0.39	0.51	0.64	0.79	0.96	1.14	1.34	2.57
25	458.6	0.32	0.44	0.57	0.72	0.89	1.08	1.28	1.51	2.89
30	408.4	0.36	0.49	0.64	0.81	1.00	1.21	1.44	1.69	3.25
35	365.1	0.40	0.55	0.72	0.91	1.12	1.36	1.61	1.89	3.63
40	328.5	0.45	0.61	0.80	1.01	1.25	1.51	1.79	2.10	4.04
45	297.9	0.49	0.67	0.88	1.11	1.37	1.66	1.98	2.32	4.45
50	272.5	0.54	0.74	0.96	1.22	1.50	1.82	2.16	2.54	4.86
55	251.4	0.59	0.80	1.04	1.32	1.63	1.97	2.34	2.75	5.27
60	233.6	0.63	0.86	1.12	1.42	1.75	2.12	2.52	2.96	5.67
65	218.6	0.67	0.92	1.20	1.52	1.87	2.26	2.69	3.16	6.06
70	205.8	0.72	0.97	1.27	1.61	1.99	2.41	2.86	3.36	6.44
75	194.8	0.76	1.03	1.34	1.70	2.10	2.54	3.02	3.55	6.81
80	185.2	0.80	1.08	1.41	1.79	2.21	2.67	3.18	3.73	7.16

Conductor Code Name	OAK
Basic / Recommended Span (m)	90
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	10.0 at 15°C
Required Percentage Increase in Tension (%)	10 at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	939.6	0.16	0.21	0.28	0.35	0.44	0.53	0.63	0.74	1.41
0	853.6	0.17	0.23	0.31	0.39	0.48	0.58	0.69	0.81	1.55
5	778.6	0.19	0.26	0.34	0.43	0.53	0.64	0.76	0.89	1.70
10	705.6	0.21	0.28	0.37	0.47	0.58	0.70	0.83	0.98	1.88
15	635.4	0.23	0.32	0.41	0.52	0.64	0.78	0.93	1.09	2.09
20	569.0	0.26	0.35	0.46	0.58	0.72	0.87	1.04	1.22	2.33
25	507.3	0.29	0.40	0.52	0.65	0.81	0.98	1.16	1.36	2.61
30	451.5	0.33	0.44	0.58	0.73	0.91	1.10	1.30	1.53	2.94
35	402.2	0.37	0.50	0.65	0.82	1.02	1.23	1.46	1.72	3.30
40	359.8	0.41	0.56	0.73	0.92	1.14	1.38	1.64	1.92	3.68
45	324.1	0.45	0.62	0.81	1.02	1.26	1.53	1.82	2.13	4.09
50	294.2	0.50	0.68	0.89	1.13	1.39	1.68	2.00	2.35	4.51
55	269.5	0.55	0.74	0.97	1.23	1.52	1.84	2.19	2.57	4.92
60	248.8	0.59	0.81	1.05	1.33	1.64	1.99	2.37	2.78	5.33
65	231.5	0.64	0.87	1.13	1.43	1.77	2.14	2.55	2.99	5.73
70	216.8	0.68	0.92	1.21	1.53	1.89	2.28	2.72	3.19	6.11
75	204.2	0.72	0.98	1.28	1.62	2.00	2.42	2.88	3.39	6.49
80	193.4	0.76	1.04	1.35	1.71	2.12	2.56	3.05	3.58	6.85

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	28	of	98

Appendix 12 - 175mm AAAC (Elm) - (90m Basic)- High altitude

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	ELM
Greased Conductor Weight (kg/m)	0.5922
Cross Sectional Area of Conductor (mm²)	211
Conductor Overall Diameter (mm)	18.8
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm²)	5710.41
Rated Breaking Strength of Conductor (kgf)	6346.7
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m²)	570
Radial Ice Thickness (mm)	12.5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1814.4
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1269.3
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1814.4
Maximum Conductor Weight (MCW) (kg/m)			1.714
Maximum Conductor Pressure (MCP) (kg/m)			2.546
Freezing Point Tension (FPT) (kgf)	at	0°C	749.5

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	ELM	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	10

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.14	0.37	0.38	0.40	0.42	0.43	0.45
80	0.56	1.47	1.54	1.60	1.66	1.72	1.78
120	1.26	3.31	3.46	3.60	3.74	3.88	4.01
160	2.23	5.89	6.15	6.40	6.65	6.89	7.12
200	3.49	9.20	9.61	10.01	10.39	10.76	11.13
240	5.02	13.25	13.84	14.41	14.96	15.50	16.02
280	6.84	18.04	18.84	19.61	20.37	21.10	21.81
320	8.93	23.56	24.60	25.62	26.60	27.56	28.49
360	11.30	29.81	31.14	32.42	33.67	34.88	36.06
400	13.96	36.81	38.44	40.02	41.56	43.06	44.51
440	16.89	44.54	46.51	48.43	50.29	52.10	53.86
480	20.10	53.00	55.35	57.63	59.85	62.00	64.10
520	23.58	62.20	64.96	67.64	70.24	72.77	75.23

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	29	of	98	

Appendix 12a - 175mm AAAC (Elm) - (90m Basic) High altitude

Conductor Code Name	ELM
Basic / Recommended Span (m)	90
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	848.7	0.31	0.43	0.56	0.71	0.87	1.06	1.26	1.47	2.83
0	749.5	0.36	0.48	0.63	0.80	0.99	1.20	1.42	1.67	3.20
5	672.4	0.40	0.54	0.70	0.89	1.10	1.33	1.59	1.86	3.57
10	606.3	0.44	0.60	0.78	0.99	1.22	1.48	1.76	2.06	3.96
15	550.5	0.48	0.66	0.86	1.09	1.34	1.63	1.94	2.27	4.36
20	503.6	0.53	0.72	0.94	1.19	1.47	1.78	2.12	2.48	4.76
25	464.3	0.57	0.78	1.02	1.29	1.59	1.93	2.30	2.69	5.17
30	431.2	0.62	0.84	1.10	1.39	1.72	2.08	2.47	2.90	5.56
35	403.1	0.66	0.90	1.18	1.49	1.84	2.22	2.64	3.10	5.95
40	379.1	0.70	0.96	1.25	1.58	1.95	2.36	2.81	3.30	6.33
45	358.4	0.74	1.01	1.32	1.67	2.07	2.50	2.97	3.49	6.69
50	340.4	0.78	1.07	1.39	1.76	2.17	2.63	3.13	3.67	7.05
55	324.6	0.82	1.12	1.46	1.85	2.28	2.76	3.28	3.85	7.39
60	310.6	0.86	1.17	1.53	1.93	2.38	2.88	3.43	4.03	7.72
65	298.2	0.89	1.22	1.59	2.01	2.48	3.00	3.57	4.20	8.04
70	287.0	0.93	1.26	1.65	2.09	2.58	3.12	3.71	4.36	8.36
75	276.9	0.96	1.31	1.71	2.17	2.67	3.23	3.85	4.52	8.66
80	267.8	1.00	1.35	1.77	2.24	2.76	3.35	3.98	4.67	8.96

Conductor Code Name	ELM
Basic / Recommended Span (m)	90
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	10.0 at 15°C
Required Percentage Increase in Tension (%)	10 at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	946.6	0.28	0.38	0.50	0.63	0.78	0.95	1.13	1.32	2.53
0	836.2	0.32	0.43	0.57	0.72	0.89	1.07	1.27	1.50	2.87
5	748.4	0.36	0.48	0.63	0.80	0.99	1.20	1.42	1.67	3.20
10	671.5	0.40	0.54	0.71	0.89	1.10	1.33	1.59	1.86	3.57
15	605.5	0.44	0.60	0.78	0.99	1.22	1.48	1.76	2.07	3.96
20	549.8	0.48	0.66	0.86	1.09	1.35	1.63	1.94	2.28	4.36
25	503.0	0.53	0.72	0.94	1.19	1.47	1.78	2.12	2.49	4.77
30	463.8	0.57	0.78	1.02	1.29	1.60	1.93	2.30	2.70	5.17
35	430.8	0.62	0.84	1.10	1.39	1.72	2.08	2.47	2.90	5.57
40	402.8	0.66	0.90	1.18	1.49	1.84	2.22	2.65	3.11	5.95
45	378.8	0.70	0.96	1.25	1.58	1.95	2.36	2.81	3.30	6.33
50	358.2	0.74	1.01	1.32	1.67	2.07	2.50	2.98	3.49	6.70
55	340.2	0.78	1.07	1.39	1.76	2.18	2.63	3.13	3.68	7.05
60	324.4	0.82	1.12	1.46	1.85	2.28	2.76	3.29	3.86	7.39
65	310.5	0.86	1.17	1.53	1.93	2.38	2.88	3.43	4.03	7.73
70	298.0	0.89	1.22	1.59	2.01	2.48	3.01	3.58	4.20	8.05
75	286.9	0.93	1.26	1.65	2.09	2.58	3.12	3.72	4.36	8.36
80	276.8	0.96	1.31	1.71	2.17	2.67	3.24	3.85	4.52	8.67

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	30	of	98

Appendix 13 - 175mm AAAC (Elm)- (100m Basic) - High altitude

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	ELM
Greased Conductor Weight (kg/m)	0.5922
Cross Sectional Area of Conductor (mm ²)	211
Conductor Overall Diameter (mm)	18.8
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm ²)	5710.41
Rated Breaking Strength of Conductor (kgf)	6346.7
Basic / Recommended Span (m)	100
Wind Pressure on Conductor (N/m ²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	2379.3
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1269.3
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1971.2
Maximum Conductor Weight (MCW) (kg/m)			1.363
Maximum Conductor Pressure (MCP) (kg/m)			1.465
Freezing Point Tension (FPT) (kgf)	at	0°C	1389.8

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	ELM	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	10

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.08	0.24	0.26	0.27	0.29	0.31	0.32
80	0.31	0.96	1.03	1.09	1.16	1.22	1.28
120	0.70	2.16	2.31	2.46	2.61	2.75	2.89
160	1.24	3.84	4.11	4.38	4.64	4.89	5.13
200	1.94	6.00	6.42	6.84	7.24	7.64	8.02
240	2.79	8.64	9.25	9.85	10.43	11.00	11.55
280	3.80	11.76	12.59	13.41	14.20	14.97	15.72
320	4.96	15.36	16.45	17.51	18.54	19.55	20.53
360	6.27	19.44	20.81	22.16	23.47	24.74	25.98
400	7.75	24.00	25.70	27.36	28.98	30.55	32.08
440	9.37	29.04	31.09	33.10	35.06	36.96	38.81
480	11.15	34.56	37.00	39.39	41.72	43.99	46.19
520	13.09	40.56	43.43	46.23	48.97	51.63	54.21

Document reference	NSP/004/042/001							
Version:-	2.0	Date of Issue:-	Sept 10	Page	31	of	98	

Appendix 13a - 175mm AAAC (Elm)- (100m Basic) Severe altitude

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	ELM
Basic / Recommended Span (m)	100
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	1529.1	0.17	0.24	0.31	0.39	0.48	0.59	0.70	0.82	0.95
0	1389.8	0.19	0.26	0.34	0.43	0.53	0.64	0.77	0.90	1.04
5	1269.3	0.21	0.29	0.37	0.47	0.58	0.71	0.84	0.99	1.14
10	1153.8	0.23	0.31	0.41	0.52	0.64	0.78	0.92	1.08	1.26
15	1044.4	0.26	0.35	0.45	0.57	0.71	0.86	1.02	1.20	1.39
20	942.6	0.28	0.38	0.50	0.64	0.79	0.95	1.13	1.33	1.54
25	849.7	0.31	0.43	0.56	0.71	0.87	1.05	1.25	1.47	1.71
30	766.7	0.35	0.47	0.62	0.78	0.97	1.17	1.39	1.63	1.89
35	694.1	0.38	0.52	0.68	0.86	1.07	1.29	1.54	1.80	2.09
40	631.6	0.42	0.57	0.75	0.95	1.17	1.42	1.69	1.98	2.30
45	578.2	0.46	0.63	0.82	1.04	1.28	1.55	1.84	2.16	2.51
50	532.9	0.50	0.68	0.89	1.13	1.39	1.68	2.00	2.35	2.72
55	494.5	0.54	0.73	0.96	1.21	1.50	1.81	2.16	2.53	2.93
60	461.7	0.58	0.79	1.03	1.30	1.60	1.94	2.31	2.71	3.14
65	433.6	0.61	0.84	1.09	1.38	1.71	2.07	2.46	2.88	3.35
70	409.4	0.65	0.89	1.16	1.46	1.81	2.19	2.60	3.06	3.54
75	388.2	0.69	0.93	1.22	1.54	1.91	2.31	2.75	3.22	3.74
80	369.7	0.72	0.98	1.28	1.62	2.00	2.42	2.88	3.38	3.92

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	ELM
Basic / Recommended Span (m)	100
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	10.0 at 15°C
Required Percentage Increase in Tension (%)	10 at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	1650.9	0.16	0.22	0.29	0.36	0.45	0.54	0.65	0.76	0.88
0	1508.5	0.18	0.24	0.31	0.40	0.49	0.59	0.71	0.83	0.96
5	1384.4	0.19	0.26	0.34	0.43	0.53	0.65	0.77	0.90	1.05
10	1264.1	0.21	0.29	0.37	0.47	0.59	0.71	0.84	0.99	1.15
15	1148.8	0.23	0.32	0.41	0.52	0.64	0.78	0.93	1.09	1.26
20	1039.7	0.26	0.35	0.46	0.58	0.71	0.86	1.03	1.20	1.40
25	938.3	0.28	0.39	0.50	0.64	0.79	0.95	1.14	1.33	1.55
30	845.8	0.32	0.43	0.56	0.71	0.88	1.06	1.26	1.48	1.72
35	763.3	0.35	0.48	0.62	0.79	0.97	1.17	1.40	1.64	1.90
40	691.2	0.39	0.52	0.69	0.87	1.07	1.30	1.54	1.81	2.10
45	629.0	0.42	0.58	0.75	0.95	1.18	1.42	1.69	1.99	2.31
50	576.0	0.46	0.63	0.82	1.04	1.29	1.55	1.85	2.17	2.52
55	531.1	0.50	0.68	0.89	1.13	1.39	1.69	2.01	2.36	2.73
60	492.9	0.54	0.74	0.96	1.22	1.50	1.82	2.16	2.54	2.94
65	460.4	0.58	0.79	1.03	1.30	1.61	1.95	2.32	2.72	3.15
70	432.5	0.62	0.84	1.10	1.39	1.71	2.07	2.46	2.89	3.35
75	408.4	0.65	0.89	1.16	1.47	1.81	2.19	2.61	3.06	3.55
80	387.4	0.69	0.94	1.22	1.55	1.91	2.31	2.75	3.23	3.75

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	32	of	98

Appendix 14 - 200mm AAAC (Poplar) - (90m Basic)- High altitude

Conductor Code Name (if any)	POPLAR
Greased Conductor Weight (kg/m)	0.68992
Cross Sectional Area of Conductor (mm²)	239.4
Conductor Overall Diameter (mm)	20.1
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm²)	5700
Rated Breaking Strength of Conductor (kgf)	7200.2
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m²)	570
Radial Ice Thickness (mm)	12.5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1814.4
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1440.04
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1814.4
Maximum Conductor Weight (MCW) (kg/m)			1.859
Maximum Conductor Pressure (MCP) (kg/m)			2.621
Freezing Point Tension (FPT) (kgf)	at	0°C	675.0

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	POPLAR	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	10

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.18	0.41	0.43	0.44	0.46	0.47	0.48
80	0.73	1.65	1.71	1.77	1.82	1.88	1.93
120	1.65	3.70	3.84	3.97	4.10	4.22	4.35
160	2.93	6.59	6.83	7.06	7.29	7.51	7.73
200	4.57	10.29	10.67	11.03	11.39	11.74	12.07
240	6.58	14.82	15.36	15.89	16.40	16.90	17.39
280	8.96	20.17	20.91	21.62	22.32	23.00	23.67
320	11.70	26.34	27.30	28.24	29.15	30.04	30.91
360	14.81	33.34	34.56	35.74	36.90	38.02	39.12
400	18.28	41.16	42.66	44.13	45.55	46.94	48.30
440	22.12	49.80	51.62	53.39	55.12	56.80	58.44
480	26.33	59.27	61.44	63.54	65.59	67.59	69.55
520	30.90	69.56	72.10	74.57	76.98	79.33	81.62

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	33	of	98	

Appendix 14a - 200mm AAAC (Poplar) - (90m Basic)- High altitude

Conductor Code Name	POPLAR
Basic / Recommended Span (m)	90
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	754.7	0.41	0.56	0.73	0.93	1.14	1.38	1.65	1.93	3.70
0	675.0	0.46	0.63	0.82	1.03	1.28	1.55	1.84	2.16	4.14
5	615.6	0.50	0.69	0.90	1.13	1.40	1.70	2.02	2.37	4.54
10	565.5	0.55	0.75	0.98	1.24	1.52	1.85	2.20	2.58	4.94
15	523.4	0.59	0.81	1.05	1.33	1.65	1.99	2.37	2.78	5.34
20	487.7	0.64	0.87	1.13	1.43	1.77	2.14	2.55	2.99	5.73
25	457.4	0.68	0.92	1.21	1.53	1.89	2.28	2.72	3.19	6.11
30	431.2	0.72	0.98	1.28	1.62	2.00	2.42	2.88	3.38	6.48
35	408.6	0.76	1.03	1.35	1.71	2.11	2.55	3.04	3.57	6.84
40	388.8	0.80	1.09	1.42	1.80	2.22	2.68	3.19	3.75	7.19
45	371.3	0.84	1.14	1.49	1.88	2.32	2.81	3.34	3.92	7.52
50	355.9	0.87	1.19	1.55	1.96	2.42	2.93	3.49	4.10	7.85
55	342.0	0.91	1.24	1.61	2.04	2.52	3.05	3.63	4.26	8.17
60	329.5	0.94	1.28	1.67	2.12	2.62	3.17	3.77	4.42	8.48
65	318.3	0.98	1.33	1.73	2.19	2.71	3.28	3.90	4.58	8.78
70	308.0	1.01	1.37	1.79	2.27	2.80	3.39	4.03	4.73	9.07
75	298.6	1.04	1.42	1.85	2.34	2.89	3.49	4.16	4.88	9.36
80	290.0	1.07	1.46	1.90	2.41	2.97	3.60	4.28	5.03	9.64

Conductor Code Name	POPLAR
Basic / Recommended Span (m)	90
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	10.0 at 15°C
Required Percentage Increase in Tension (%)	10 at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	858.4	0.36	0.49	0.64	0.81	1.00	1.22	1.45	1.70	3.25
0	762.4	0.41	0.55	0.72	0.92	1.13	1.37	1.63	1.91	3.67
5	689.5	0.45	0.61	0.80	1.01	1.25	1.51	1.80	2.11	4.05
10	627.7	0.49	0.67	0.88	1.11	1.37	1.66	1.98	2.32	4.45
15	575.7	0.54	0.73	0.96	1.21	1.50	1.81	2.16	2.53	4.85
20	532.0	0.58	0.79	1.04	1.31	1.62	1.96	2.33	2.74	5.25
25	495.1	0.63	0.85	1.11	1.41	1.74	2.11	2.51	2.94	5.64
30	463.6	0.67	0.91	1.19	1.51	1.86	2.25	2.68	3.14	6.03
35	436.6	0.71	0.97	1.26	1.60	1.98	2.39	2.84	3.34	6.40
40	413.3	0.75	1.02	1.34	1.69	2.09	2.52	3.00	3.53	6.76
45	392.9	0.79	1.08	1.40	1.78	2.19	2.66	3.16	3.71	7.11
50	375.0	0.83	1.13	1.47	1.86	2.30	2.78	3.31	3.89	7.45
55	359.1	0.86	1.18	1.54	1.95	2.40	2.91	3.46	4.06	7.78
60	344.9	0.90	1.23	1.60	2.03	2.50	3.03	3.60	4.23	8.10
65	332.2	0.93	1.27	1.66	2.10	2.60	3.14	3.74	4.39	8.41
70	320.6	0.97	1.32	1.72	2.18	2.69	3.25	3.87	4.55	8.71
75	310.2	1.00	1.36	1.78	2.25	2.78	3.36	4.00	4.70	9.01
80	300.6	1.03	1.41	1.84	2.32	2.87	3.47	4.13	4.85	9.30

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	34	of	98

Appendix 15 - 200mm AAAC (Poplar) - (100m Basic)- High altitude

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	POPLAR
Greased Conductor Weight (kg/m)	0.68992
Cross Sectional Area of Conductor (mm²)	239.4
Conductor Overall Diameter (mm)	20.1
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm²)	5700
Rated Breaking Strength of Conductor (kgf)	7200.2
Basic / Recommended Span (m)	100
Wind Pressure on Conductor (N/m²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	2379.3
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1440.04
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	2183.2
Maximum Conductor Weight (MCW) (kg/m)			1.496
Maximum Conductor Pressure (MCP) (kg/m)			1.515
Freezing Point Tension (FPT) (kgf)	at	0°C	1575.5

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	POPLAR	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	10

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.08	0.24	0.26	0.28	0.29	0.31	0.32
80	0.32	0.97	1.04	1.10	1.17	1.23	1.29
120	0.72	2.18	2.33	2.48	2.63	2.77	2.90
160	1.27	3.88	4.15	4.41	4.67	4.92	5.16
200	1.99	6.06	6.48	6.89	7.29	7.68	8.06
240	2.87	8.73	9.33	9.93	10.50	11.07	11.61
280	3.90	11.88	12.71	13.51	14.30	15.06	15.81
320	5.10	15.52	16.59	17.65	18.67	19.67	20.64
360	6.45	19.64	21.00	22.34	23.64	24.90	26.13
400	7.96	24.25	25.93	27.57	29.18	30.74	32.26
440	9.64	29.34	31.37	33.37	35.31	37.20	39.03
480	11.47	34.92	37.34	39.71	42.02	44.27	46.45
520	13.46	40.98	43.82	46.60	49.31	51.95	54.51

Document reference	NSP/004/042/001							
Version:-	2.0	Date of Issue:-	Sept 10	Page	35	of	98	

Appendix 15a - 200mm AAAC (Poplar) - (100m Basic)- High altitude

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	POPLAR
Basic / Recommended Span (m)	100
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	1732.4	0.18	0.24	0.32	0.40	0.50	0.60	0.72	0.84	0.98
0	1575.5	0.20	0.27	0.35	0.44	0.55	0.66	0.79	0.93	1.07
5	1440.0	0.22	0.29	0.38	0.49	0.60	0.72	0.86	1.01	1.17
10	1310.2	0.24	0.32	0.42	0.53	0.66	0.80	0.95	1.11	1.29
15	1187.6	0.26	0.36	0.46	0.59	0.73	0.88	1.05	1.23	1.42
20	1073.6	0.29	0.39	0.51	0.65	0.80	0.97	1.16	1.36	1.57
25	969.7	0.32	0.44	0.57	0.72	0.89	1.08	1.28	1.50	1.74
30	876.9	0.35	0.48	0.63	0.80	0.98	1.19	1.42	1.66	1.93
35	795.6	0.39	0.53	0.69	0.88	1.08	1.31	1.56	1.83	2.12
40	725.4	0.43	0.58	0.76	0.96	1.19	1.44	1.71	2.01	2.33
45	665.4	0.47	0.64	0.83	1.05	1.30	1.57	1.87	2.19	2.54
50	614.3	0.51	0.69	0.90	1.14	1.40	1.70	2.02	2.37	2.75
55	570.8	0.54	0.74	0.97	1.22	1.51	1.83	2.18	2.55	2.96
60	533.7	0.58	0.79	1.03	1.31	1.62	1.96	2.33	2.73	3.17
65	501.7	0.62	0.84	1.10	1.39	1.72	2.08	2.48	2.91	3.37
70	474.0	0.65	0.89	1.16	1.47	1.82	2.20	2.62	3.07	3.57
75	449.9	0.69	0.94	1.23	1.55	1.92	2.32	2.76	3.24	3.76
80	428.6	0.72	0.99	1.29	1.63	2.01	2.43	2.90	3.40	3.94

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	POPLAR
Basic / Recommended Span (m)	100
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	10.0 at 15°C
Required Percentage Increase in Tension (%)	10 at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	140
-5.6	1871.5	0.17	0.23	0.29	0.37	0.46	0.56	0.66	0.78	0.90
0	1711.0	0.18	0.25	0.32	0.41	0.50	0.61	0.73	0.85	0.99
5	1571.2	0.20	0.27	0.35	0.44	0.55	0.66	0.79	0.93	1.08
10	1435.9	0.22	0.29	0.38	0.49	0.60	0.73	0.86	1.01	1.18
15	1306.3	0.24	0.32	0.42	0.53	0.66	0.80	0.95	1.12	1.29
20	1183.9	0.26	0.36	0.47	0.59	0.73	0.88	1.05	1.23	1.43
25	1070.2	0.29	0.39	0.52	0.65	0.81	0.98	1.16	1.36	1.58
30	966.6	0.32	0.44	0.57	0.72	0.89	1.08	1.28	1.51	1.75
35	874.2	0.36	0.48	0.63	0.80	0.99	1.19	1.42	1.67	1.93
40	793.2	0.39	0.53	0.70	0.88	1.09	1.32	1.57	1.84	2.13
45	723.4	0.43	0.58	0.76	0.97	1.19	1.44	1.72	2.01	2.34
50	663.7	0.47	0.64	0.83	1.05	1.30	1.57	1.87	2.20	2.55
55	612.9	0.51	0.69	0.90	1.14	1.41	1.70	2.03	2.38	2.76
60	569.6	0.55	0.74	0.97	1.23	1.51	1.83	2.18	2.56	2.97
65	532.6	0.58	0.79	1.04	1.31	1.62	1.96	2.33	2.74	3.17
70	500.8	0.62	0.84	1.10	1.39	1.72	2.08	2.48	2.91	3.38
75	473.2	0.66	0.89	1.17	1.48	1.82	2.21	2.62	3.08	3.57
80	449.2	0.69	0.94	1.23	1.56	1.92	2.32	2.76	3.24	3.76

Document reference		NSP/004/042/001			
Version:-	2.0	Date of Issue:-	Sept 10	Page	36 of 98

Appendix 16 - Section 2 - Sag charts for the renovation of existing overhead lines

Conductor Type	Basic Span	Altitude or Loading	Appendix No.
.017" / 13mm HDBC	80	All Altitudes	17
	90		18
	100		19
	120		20
.05" / 32mm HDBC	80	Wind = 760(N/m ²), Ice = 0mm	21
	90		22
	100		23
	120		24
.05" / 50mm AAAC	80		25
	100		26
	120		27
.25" / 16mm HDBC	90		28
	100		29
3 swg HDBC	90		30
	100		31
5 swg HDBC	90		32
	100		33
6 swg HDBC	90		34
	100		35
7swg HDBC	90		36
	100		37
.175" / 175mm ACSR (CE/C/36)	76.2 (250ft)	Normal Altitude	38
.1" / 100mm ACSR	90	Wind = 380(N/m ²), Ice = 19mm (Dia.)	39
.1" / 100mm ACSR	100		40
.1" / 70mm HDBC	90		41
.15" / 100mm HDBC	90		42
.2" / 125mm HDBC	100	High Altitude	43
.1" / 100mm ACSR	90		44
.1" / 70mm HDBC	90	Wind = 570(N/m ²), Ice = 25mm (Dia.)	45
.15" / 100mm HDBC	90		46
.2" / 125mm HDBC	100		47

Document reference	NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	37	of	98

Appendix 17 - .017" or 13mm² CACU - (All altitudes) - 80m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	13mm CACU
Greased Conductor Weight (kg/m)	0.11856
Cross Sectional Area of Conductor (mm ²)	13.1475
Conductor Overall Diameter (mm)	5.08
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.3
Rated Breaking Strength of Conductor (kgf)	807.4
Basic / Recommended Span (m)	80
Wind Pressure on Conductor (N/m ²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	323.0
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	269.1
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	323.0
Maximum Conductor Weight (MCW) (kg/m)			0.119
Maximum Conductor Pressure (MCP) (kg/m)			0.394
Freezing Point Tension (FPT) (kgf)	at	0°C	245.6

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor: 13mm CACU	Temperature Shift (Deg. C)	0
	Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.09	0.19	0.21	0.22	0.24	0.25	0.27
80	0.36	0.77	0.83	0.88	0.94	1.00	1.06
120	0.82	1.73	1.86	1.98	2.12	2.25	2.39
160	1.46	3.08	3.30	3.53	3.76	4.00	4.25
200	2.28	4.82	5.16	5.51	5.88	6.26	6.64
240	3.28	6.94	7.43	7.94	8.47	9.01	9.56
280	4.46	9.44	10.11	10.81	11.53	12.26	13.01
320	5.83	12.33	13.20	14.11	15.06	16.02	16.99
360	7.38	15.61	16.71	17.86	19.05	20.27	21.50
400	9.11	19.27	20.63	22.05	23.52	25.03	26.55
440	11.02	23.32	24.97	26.69	28.46	30.28	32.12
480	13.12	27.75	29.71	31.76	33.87	36.04	38.23
520	15.40	32.57	34.87	37.27	39.76	42.29	44.87

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	38	of	98	

Appendix 17a - .017" or 13mm² CACU - (All altitudes) - 80m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	13mm CACU
Basic / Recommended Span (m)	80
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	260.3	0.20	0.28	0.36	0.46	0.57	0.69	0.82	0.96	1.84
0	245.6	0.22	0.30	0.39	0.49	0.60	0.73	0.87	1.02	1.96
5	232.6	0.23	0.31	0.41	0.52	0.64	0.77	0.92	1.08	2.06
10	219.8	0.24	0.33	0.43	0.55	0.67	0.82	0.97	1.14	2.18
15	207.3	0.26	0.35	0.46	0.58	0.71	0.87	1.03	1.21	2.32
20	195.0	0.27	0.37	0.49	0.62	0.76	0.92	1.09	1.28	2.46
25	183.1	0.29	0.40	0.52	0.66	0.81	0.98	1.17	1.37	2.62
30	171.5	0.31	0.42	0.55	0.70	0.86	1.05	1.24	1.46	2.80
35	160.4	0.33	0.45	0.59	0.75	0.92	1.12	1.33	1.56	2.99
40	149.8	0.36	0.48	0.63	0.80	0.99	1.20	1.42	1.67	3.20
45	139.8	0.38	0.52	0.68	0.86	1.06	1.28	1.53	1.79	3.43
50	130.4	0.41	0.56	0.73	0.92	1.14	1.37	1.64	1.92	3.68
55	121.7	0.44	0.60	0.78	0.99	1.22	1.47	1.75	2.06	3.94
60	113.7	0.47	0.64	0.83	1.06	1.30	1.58	1.88	2.20	4.22
65	106.4	0.50	0.68	0.89	1.13	1.39	1.69	2.01	2.35	4.51
70	99.8	0.53	0.73	0.95	1.20	1.48	1.80	2.14	2.51	4.81
75	93.9	0.57	0.77	1.01	1.28	1.58	1.91	2.27	2.67	5.12
80	88.5	0.60	0.82	1.07	1.36	1.67	2.03	2.41	2.83	5.42

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	39	of	98

Appendix 18 - .017" or 13mm² CACU - (All altitudes) - 90m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	13mm CACU
Greased Conductor Weight (kg/m)	0.11856
Cross Sectional Area of Conductor (mm ²)	13.1475
Conductor Overall Diameter (mm)	5.08
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.3
Rated Breaking Strength of Conductor (kgf)	807.4
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m ²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	323.0
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	269.1
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	323.0
Maximum Conductor Weight (MCW) (kg/m)			0.119
Maximum Conductor Pressure (MCP) (kg/m)			0.394
Freezing Point Tension (FPT) (kgf)	at	0°C	230.9

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor: 13mm CACU	Temperature Shift (Deg. C)	0
	Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.10	0.20	0.21	0.22	0.24	0.25	0.26
80	0.39	0.80	0.85	0.90	0.95	1.00	1.06
120	0.87	1.80	1.91	2.02	2.14	2.26	2.38
160	1.55	3.20	3.39	3.60	3.81	4.01	4.22
200	2.42	4.99	5.30	5.62	5.95	6.27	6.60
240	3.48	7.19	7.64	8.10	8.56	9.03	9.50
280	4.74	9.79	10.40	11.02	11.65	12.29	12.94
320	6.19	12.78	13.58	14.39	15.22	16.06	16.90
360	7.84	16.18	17.19	18.22	19.27	20.32	21.38
400	9.67	19.97	21.22	22.49	23.79	25.09	26.40
440	11.71	24.17	25.67	27.21	28.78	30.36	31.94
480	13.93	28.76	30.55	32.39	34.25	36.13	38.01
520	16.35	33.76	35.86	38.01	40.20	42.40	44.61

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	40	of	98	

Appendix 18a - .017" or 13mm² CACU - (All altitudes) - 90m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	13mm CACU
Basic / Recommended Span (m)	90
Temperature Shift for Creep (Deg. C)	0 ` Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	245.1	0.22	0.30	0.39	0.49	0.60	0.73	0.87	1.02	1.96
0	230.9	0.23	0.31	0.41	0.52	0.64	0.78	0.92	1.08	2.08
5	218.5	0.24	0.33	0.43	0.55	0.68	0.82	0.98	1.15	2.20
10	206.4	0.26	0.35	0.46	0.58	0.72	0.87	1.03	1.21	2.33
15	194.5	0.27	0.37	0.49	0.62	0.76	0.92	1.10	1.29	2.47
20	183.1	0.29	0.40	0.52	0.66	0.81	0.98	1.17	1.37	2.62
25	172.1	0.31	0.42	0.55	0.70	0.86	1.04	1.24	1.46	2.79
30	161.5	0.33	0.45	0.59	0.74	0.92	1.11	1.32	1.55	2.97
35	151.5	0.35	0.48	0.63	0.79	0.98	1.18	1.41	1.65	3.17
40	142.1	0.38	0.51	0.67	0.84	1.04	1.26	1.50	1.76	3.38
45	133.3	0.40	0.54	0.71	0.90	1.11	1.35	1.60	1.88	3.60
50	125.1	0.43	0.58	0.76	0.96	1.18	1.43	1.71	2.00	3.84
55	117.6	0.45	0.62	0.81	1.02	1.26	1.52	1.81	2.13	4.08
60	110.7	0.48	0.66	0.86	1.08	1.34	1.62	1.93	2.26	4.34
65	104.5	0.51	0.69	0.91	1.15	1.42	1.72	2.04	2.40	4.59
70	98.9	0.54	0.73	0.96	1.21	1.50	1.81	2.16	2.53	4.86
75	93.8	0.57	0.77	1.01	1.28	1.58	1.91	2.28	2.67	5.12
80	89.1	0.60	0.81	1.06	1.35	1.66	2.01	2.39	2.81	5.39

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	41	of	98

Appendix 19 - .017" or 13mm² CACU - (All altitudes) - 100m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	13mm CACU
Greased Conductor Weight (kg/m)	0.11856
Cross Sectional Area of Conductor (mm ²)	13.1475
Conductor Overall Diameter (mm)	5.08
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.3
Rated Breaking Strength of Conductor (kgf)	807.4
Basic / Recommended Span (m)	100
Wind Pressure on Conductor (N/m ²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	323.0
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	269.1
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	323.0
Maximum Conductor Weight (MCW) (kg/m)			0.119
Maximum Conductor Pressure (MCP) (kg/m)			0.394
Freezing Point Tension (FPT) (kgf)	at	0°C	215.7

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor: 13mm CACU	Temperature Shift (Deg. C)	0
	Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.10	0.21	0.22	0.23	0.24	0.25	0.26
80	0.41	0.83	0.87	0.92	0.96	1.01	1.05
120	0.93	1.86	1.96	2.06	2.16	2.26	2.37
160	1.66	3.30	3.48	3.66	3.84	4.03	4.21
200	2.59	5.16	5.44	5.72	6.01	6.29	6.57
240	3.73	7.43	7.83	8.24	8.65	9.06	9.47
280	5.07	10.11	10.66	11.22	11.77	12.33	12.88
320	6.62	13.21	13.93	14.65	15.38	16.10	16.83
360	8.38	16.72	17.63	18.54	19.46	20.38	21.30
400	10.35	20.64	21.76	22.89	24.03	25.16	26.29
440	12.52	24.97	26.33	27.70	29.07	30.45	31.82
480	14.90	29.72	31.33	32.96	34.60	36.24	37.86
520	17.49	34.88	36.77	38.69	40.61	42.53	44.44

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	42	of	98	

Appendix 19a - .017" or 13mm² CACU - (All altitudes) - 100m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	13mm CACU
Basic / Recommended Span (m)	100
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	229.2	0.23	0.32	0.41	0.52	0.65	0.78	0.93	1.09	2.10
0	215.7	0.25	0.34	0.44	0.56	0.69	0.83	0.99	1.16	2.23
5	204.0	0.26	0.36	0.46	0.59	0.73	0.88	1.05	1.23	2.35
10	192.7	0.28	0.38	0.49	0.62	0.77	0.93	1.11	1.30	2.49
15	181.8	0.29	0.40	0.52	0.66	0.82	0.99	1.17	1.38	2.64
20	171.4	0.31	0.42	0.55	0.70	0.86	1.05	1.25	1.46	2.80
25	161.4	0.33	0.45	0.59	0.74	0.92	1.11	1.32	1.55	2.97
30	152.1	0.35	0.48	0.62	0.79	0.97	1.18	1.40	1.65	3.16
35	143.2	0.37	0.51	0.66	0.84	1.03	1.25	1.49	1.75	3.35
40	135.0	0.40	0.54	0.70	0.89	1.10	1.33	1.58	1.85	3.56
45	127.5	0.42	0.57	0.74	0.94	1.16	1.41	1.67	1.97	3.77
50	120.5	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	3.99
55	114.1	0.47	0.64	0.83	1.05	1.30	1.57	1.87	2.20	4.21
60	108.2	0.49	0.67	0.88	1.11	1.37	1.66	1.97	2.31	4.44
65	102.9	0.52	0.71	0.92	1.17	1.44	1.74	2.07	2.43	4.67
70	98.1	0.54	0.74	0.97	1.22	1.51	1.83	2.18	2.55	4.90
75	93.7	0.57	0.78	1.01	1.28	1.58	1.91	2.28	2.67	5.13
80	89.7	0.59	0.81	1.06	1.34	1.65	2.00	2.38	2.79	5.35

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	43	of	98

Appendix 20 - .017" or 13mm² CACU - (All altitudes) - 120m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	13mm CACU
Greased Conductor Weight (kg/m)	0.11856
Cross Sectional Area of Conductor (mm ²)	13.1475
Conductor Overall Diameter (mm)	5.08
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.3
Rated Breaking Strength of Conductor (kgf)	807.4
Basic / Recommended Span (m)	120
Wind Pressure on Conductor (N/m ²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	323.0
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	269.1
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	323.0
Maximum Conductor Weight (MCW) (kg/m)			0.119
Maximum Conductor Pressure (MCP) (kg/m)			0.394
Freezing Point Tension (FPT) (kgf)	at	0°C	185.9

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor: 13mm CACU	Temperature Shift (Deg. C)	0
	Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.12	0.22	0.23	0.24	0.24	0.25	0.26
80	0.48	0.87	0.91	0.94	0.98	1.01	1.05
120	1.08	1.96	2.04	2.12	2.20	2.28	2.36
160	1.92	3.49	3.63	3.77	3.91	4.05	4.19
200	3.01	5.45	5.68	5.90	6.12	6.33	6.55
240	4.33	7.85	8.17	8.49	8.81	9.12	9.43
280	5.89	10.69	11.12	11.56	11.99	12.42	12.84
320	7.69	13.96	14.53	15.10	15.66	16.22	16.77
360	9.74	17.67	18.39	19.11	19.82	20.52	21.22
400	12.02	21.81	22.70	23.59	24.47	25.34	26.20
440	14.55	26.40	27.47	28.54	29.61	30.66	31.70
480	17.31	31.41	32.69	33.97	35.23	36.49	37.72
520	20.32	36.87	38.37	39.87	41.35	42.82	44.27

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	44	of	98	

Appendix 20a - .017" or 13mm² CACU - (All altitudes) - 120m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	13mm CACU
Basic / Recommended Span (m)	120
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	197.2	0.27	0.37	0.48	0.61	0.75	0.91	1.08	1.27	2.43
0	185.9	0.29	0.39	0.51	0.65	0.80	0.96	1.15	1.35	2.58
5	176.3	0.30	0.41	0.54	0.68	0.84	1.02	1.21	1.42	2.72
10	167.2	0.32	0.43	0.57	0.72	0.89	1.07	1.28	1.50	2.87
15	158.6	0.34	0.46	0.60	0.76	0.93	1.13	1.35	1.58	3.03
20	150.6	0.35	0.48	0.63	0.80	0.98	1.19	1.42	1.66	3.19
25	143.1	0.37	0.51	0.66	0.84	1.04	1.25	1.49	1.75	3.36
30	136.2	0.39	0.53	0.70	0.88	1.09	1.32	1.57	1.84	3.53
35	129.7	0.41	0.56	0.73	0.93	1.14	1.38	1.65	1.93	3.70
40	123.8	0.43	0.59	0.77	0.97	1.20	1.45	1.72	2.02	3.88
45	118.3	0.45	0.61	0.80	1.01	1.25	1.52	1.80	2.12	4.06
50	113.3	0.47	0.64	0.84	1.06	1.31	1.58	1.88	2.21	4.24
55	108.6	0.49	0.67	0.87	1.10	1.36	1.65	1.96	2.31	4.42
60	104.4	0.51	0.70	0.91	1.15	1.42	1.72	2.04	2.40	4.60
65	100.5	0.53	0.72	0.94	1.19	1.48	1.78	2.12	2.49	4.78
70	96.9	0.55	0.75	0.98	1.24	1.53	1.85	2.20	2.59	4.96
75	93.5	0.57	0.78	1.01	1.28	1.58	1.92	2.28	2.68	5.13
80	90.5	0.59	0.80	1.05	1.33	1.64	1.98	2.36	2.77	5.31

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	45	of	98

Appendix 21 - .05" or 32mm² HDBC - (All altitudes) - 80m Basic

Conductor Code Name (if any)	32mm HDBC
Greased Conductor Weight (kg/m)	0.2969
Cross Sectional Area of Conductor (mm ²)	33.134
Conductor Overall Diameter (mm)	8.05
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.3
Rated Breaking Strength of Conductor (kgf)	1296.05
Basic / Recommended Span (m)	80
Wind Pressure on Conductor (N/m ²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	518.4
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	432.0
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	518.4
Maximum Conductor Weight (MCW) (kg/m)			0.297
Maximum Conductor Pressure (MCP) (kg/m)			0.624
Freezing Point Tension (FPT) (kgf)	at	0°C	357.2

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor: 32mm HDBC	Temperature Shift (Deg. C)	0
	Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.15	0.32	0.34	0.35	0.37	0.38	0.39
80	0.62	1.29	1.34	1.40	1.46	1.52	1.57
120	1.38	2.89	3.03	3.16	3.29	3.42	3.54
160	2.46	5.14	5.38	5.61	5.85	6.07	6.30
200	3.85	8.03	8.40	8.77	9.13	9.49	9.84
240	5.54	11.57	12.10	12.63	13.15	13.66	14.17
280	7.54	15.74	16.47	17.19	17.90	18.60	19.28
320	9.85	20.56	21.52	22.46	23.38	24.29	25.18
360	12.46	26.02	27.23	28.42	29.59	30.74	31.87
400	15.38	32.13	33.62	35.09	36.54	37.95	39.35
440	18.62	38.87	40.68	42.46	44.21	45.93	47.61
480	22.15	46.26	48.41	50.53	52.61	54.65	56.66
520	26.00	54.29	56.82	59.30	61.74	64.14	66.50

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	46	of	98	

Appendix 21a - .05" or 32mm² HDBC - (All altitudes) - 80m Basic

Common Design & Erection Tables

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	32mm HDBC
Basic / Recommended Span (m)	80
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	386.0	0.35	0.47	0.62	0.78	0.96	1.16	1.38	1.62	3.12
0	357.2	0.37	0.51	0.67	0.84	1.04	1.26	1.50	1.76	3.37
5	333.1	0.40	0.55	0.71	0.90	1.11	1.35	1.60	1.88	3.61
10	310.7	0.43	0.59	0.76	0.97	1.19	1.45	1.72	2.02	3.87
15	290.1	0.46	0.63	0.82	1.04	1.28	1.55	1.84	2.16	4.15
20	271.3	0.49	0.67	0.88	1.11	1.37	1.66	1.97	2.31	4.43
25	254.2	0.53	0.72	0.93	1.18	1.46	1.77	2.10	2.47	4.73
30	238.8	0.56	0.76	0.99	1.26	1.55	1.88	2.24	2.63	5.03
35	225.0	0.59	0.81	1.06	1.34	1.65	2.00	2.37	2.79	5.34
40	212.7	0.63	0.86	1.12	1.41	1.75	2.11	2.51	2.95	5.65
45	201.6	0.66	0.90	1.18	1.49	1.84	2.23	2.65	3.11	5.96
50	191.7	0.70	0.95	1.24	1.57	1.94	2.34	2.79	3.27	6.27
55	182.8	0.73	0.99	1.30	1.64	2.03	2.46	2.92	3.43	6.58
60	174.8	0.76	1.04	1.36	1.72	2.12	2.57	3.06	3.59	6.88
65	167.6	0.80	1.09	1.42	1.79	2.21	2.68	3.19	3.74	7.18
70	161.0	0.83	1.13	1.48	1.87	2.30	2.79	3.32	3.89	7.47
75	155.1	0.86	1.17	1.53	1.94	2.39	2.90	3.45	4.04	7.75
80	149.7	0.89	1.22	1.59	2.01	2.48	3.00	3.57	4.19	8.03

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	47	of	98

Appendix 22 - .05" or 32mm² HDBC - (All altitudes) - 90m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	32mm HDBC
Greased Conductor Weight (kg/m)	0.2969
Cross Sectional Area of Conductor (mm ²)	33.134
Conductor Overall Diameter (mm)	8.05
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.3
Rated Breaking Strength of Conductor (kgf)	1296.05
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m ²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	518.4
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	432.0
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	518.4
Maximum Conductor Weight (MCW) (kg/m)			0.297
Maximum Conductor Pressure (MCP) (kg/m)			0.624
Freezing Point Tension (FPT) (kgf)	at	0°C	337.0

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor: 32mm HDBC	Temperature Shift (Deg. C)	0
	Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.16	0.31	0.33	0.34	0.35	0.36	0.37
80	0.66	1.26	1.31	1.36	1.40	1.45	1.50
120	1.48	2.83	2.94	3.05	3.16	3.27	3.37
160	2.62	5.03	5.23	5.43	5.62	5.81	6.00
200	4.10	7.86	8.17	8.48	8.78	9.08	9.37
240	5.90	11.31	11.76	12.21	12.64	13.07	13.49
280	8.03	15.40	16.01	16.62	17.21	17.79	18.37
320	10.49	20.11	20.91	21.70	22.48	23.24	23.99
360	13.28	25.46	26.47	27.47	28.45	29.41	30.36
400	16.40	31.43	32.68	33.91	35.12	36.31	37.48
440	19.84	38.03	39.54	41.03	42.50	43.94	45.36
480	23.61	45.26	47.06	48.83	50.58	52.29	53.98
520	27.71	53.11	55.23	57.31	59.36	61.37	63.35

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	48	of	98	

Appendix 22a - .05" or 32mm² HDBC - (All altitudes) - 90m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	32mm HDBC
Basic / Recommended Span (m)	90
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	362.2	0.37	0.50	0.66	0.83	1.02	1.24	1.48	1.73	3.32
0	337.0	0.40	0.54	0.70	0.89	1.10	1.33	1.59	1.86	3.57
5	316.2	0.42	0.58	0.75	0.95	1.17	1.42	1.69	1.98	3.80
10	297.1	0.45	0.61	0.80	1.01	1.25	1.51	1.80	2.11	4.05
15	279.7	0.48	0.65	0.85	1.07	1.33	1.61	1.91	2.24	4.30
20	263.8	0.51	0.69	0.90	1.14	1.41	1.70	2.03	2.38	4.56
25	249.4	0.54	0.73	0.95	1.21	1.49	1.80	2.14	2.51	4.82
30	236.4	0.57	0.77	1.00	1.27	1.57	1.90	2.26	2.65	5.09
35	224.7	0.59	0.81	1.06	1.34	1.65	2.00	2.38	2.79	5.35
40	214.1	0.62	0.85	1.11	1.40	1.73	2.10	2.50	2.93	5.62
45	204.5	0.65	0.89	1.16	1.47	1.81	2.20	2.61	3.07	5.88
50	195.9	0.68	0.93	1.21	1.53	1.89	2.29	2.73	3.20	6.14
55	188.0	0.71	0.97	1.26	1.60	1.97	2.39	2.84	3.34	6.40
60	180.9	0.74	1.01	1.31	1.66	2.05	2.48	2.96	3.47	6.65
65	174.3	0.77	1.04	1.36	1.72	2.13	2.58	3.07	3.60	6.90
70	168.3	0.79	1.08	1.41	1.79	2.20	2.67	3.17	3.73	7.14
75	162.9	0.82	1.12	1.46	1.85	2.28	2.76	3.28	3.85	7.38
80	157.8	0.85	1.15	1.51	1.90	2.35	2.85	3.39	3.97	7.62

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	49	of	98

Appendix 23 - .05" or 32mm² HDBC - (All altitudes) - 100m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	32mm HDBC
Greased Conductor Weight (kg/m)	0.2969
Cross Sectional Area of Conductor (mm ²)	33.134
Conductor Overall Diameter (mm)	8.05
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.3
Rated Breaking Strength of Conductor (kgf)	1296.05
Basic / Recommended Span (m)	100
Wind Pressure on Conductor (N/m ²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	518.4
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	432.0
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	518.4
Maximum Conductor Weight (MCW) (kg/m)			0.297
Maximum Conductor Pressure (MCP) (kg/m)			0.624
Freezing Point Tension (FPT) (kgf)	at	0°C	319.3

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor: 32mm HDBC	Temperature Shift (Deg. C)	0
	Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.17	0.31	0.32	0.33	0.34	0.35	0.36
80	0.70	1.24	1.28	1.32	1.36	1.40	1.44
120	1.57	2.78	2.88	2.97	3.06	3.15	3.24
160	2.79	4.94	5.11	5.28	5.45	5.61	5.77
200	4.36	7.72	7.99	8.25	8.51	8.76	9.01
240	6.27	11.12	11.50	11.88	12.25	12.62	12.98
280	8.54	15.13	15.66	16.17	16.68	17.17	17.66
320	11.15	19.77	20.45	21.12	21.78	22.43	23.07
360	14.12	25.02	25.88	26.73	27.57	28.39	29.19
400	17.43	30.89	31.95	33.00	34.03	35.05	36.04
440	21.09	37.37	38.66	39.93	41.18	42.40	43.61
480	25.09	44.48	46.01	47.52	49.01	50.47	51.90
520	29.45	52.20	54.00	55.77	57.51	59.23	60.91

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	50	of	98	

Appendix 23a - .05" or 32mm² HDBC - (All altitudes) - 100m Basic

Common Design & Erection Tables

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	32mm HDBC
Basic / Recommended Span (m)	100
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	340.7	0.39	0.53	0.70	0.88	1.09	1.32	1.57	1.84	3.53
0	319.3	0.42	0.57	0.74	0.94	1.16	1.41	1.67	1.96	3.77
5	301.7	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	3.99
10	285.6	0.47	0.64	0.83	1.05	1.30	1.57	1.87	2.20	4.21
15	271.0	0.49	0.67	0.88	1.11	1.37	1.66	1.97	2.31	4.44
20	257.6	0.52	0.71	0.92	1.17	1.44	1.74	2.07	2.43	4.67
25	245.5	0.54	0.74	0.97	1.22	1.51	1.83	2.18	2.55	4.90
30	234.5	0.57	0.78	1.01	1.28	1.58	1.92	2.28	2.67	5.13
35	224.4	0.60	0.81	1.06	1.34	1.65	2.00	2.38	2.79	5.36
40	215.3	0.62	0.84	1.10	1.40	1.72	2.09	2.48	2.91	5.59
45	207.0	0.65	0.88	1.15	1.45	1.79	2.17	2.58	3.03	5.81
50	199.3	0.67	0.91	1.19	1.51	1.86	2.25	2.68	3.15	6.03
55	192.4	0.69	0.95	1.23	1.56	1.93	2.33	2.78	3.26	6.25
60	185.9	0.72	0.98	1.28	1.62	2.00	2.42	2.87	3.37	6.47
65	180.0	0.74	1.01	1.32	1.67	2.06	2.49	2.97	3.48	6.68
70	174.6	0.77	1.04	1.36	1.72	2.13	2.57	3.06	3.59	6.89
75	169.5	0.79	1.07	1.40	1.77	2.19	2.65	3.15	3.70	7.09
80	164.8	0.81	1.10	1.44	1.82	2.25	2.72	3.24	3.81	7.30

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	51	of	98

Appendix 24 - .05" or 32mm² HDBC - (All altitudes) - 120m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	32mm HDBC
Greased Conductor Weight (kg/m)	0.2969
Cross Sectional Area of Conductor (mm ²)	33.134
Conductor Overall Diameter (mm)	8.05
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.3
Rated Breaking Strength of Conductor (kgf)	1296.05
Basic / Recommended Span (m)	120
Wind Pressure on Conductor (N/m ²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	518.4
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	432.0
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	518.4
Maximum Conductor Weight (MCW) (kg/m)			0.297
Maximum Conductor Pressure (MCP) (kg/m)			0.624
Freezing Point Tension (FPT) (kgf)	at	0°C	291.9

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor: 32mm HDBC	Temperature Shift (Deg. C)	0
	Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.19	0.30	0.31	0.32	0.32	0.33	0.34
80	0.77	1.20	1.24	1.27	1.30	1.33	1.36
120	1.74	2.71	2.78	2.85	2.92	2.99	3.06
160	3.10	4.82	4.95	5.07	5.20	5.32	5.44
200	4.84	7.53	7.73	7.93	8.12	8.31	8.50
240	6.96	10.84	11.13	11.41	11.69	11.97	12.24
280	9.48	14.76	15.15	15.54	15.91	16.29	16.65
320	12.38	19.28	19.79	20.29	20.79	21.27	21.75
360	15.67	24.40	25.05	25.68	26.31	26.92	27.53
400	19.35	30.12	30.92	31.71	32.48	33.24	33.99
440	23.41	36.45	37.41	38.36	39.30	40.22	41.13
480	27.86	43.38	44.53	45.66	46.77	47.86	48.94
520	32.69	50.91	52.26	53.58	54.89	56.17	57.44

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	52	of	98	

Appendix 24a - .05" or 32mm² HDBC - (All altitudes) - 120m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	32mm HDBC
Basic / Recommended Span (m)	120
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	306.9	0.44	0.59	0.77	0.98	1.21	1.46	1.74	2.04	3.92
0	291.9	0.46	0.62	0.81	1.03	1.27	1.54	1.83	2.15	4.12
5	279.6	0.48	0.65	0.85	1.08	1.33	1.61	1.91	2.24	4.30
10	268.3	0.50	0.68	0.89	1.12	1.38	1.67	1.99	2.34	4.48
15	258.0	0.52	0.70	0.92	1.17	1.44	1.74	2.07	2.43	4.66
20	248.4	0.54	0.73	0.96	1.21	1.49	1.81	2.15	2.52	4.84
25	239.6	0.56	0.76	0.99	1.25	1.55	1.87	2.23	2.62	5.02
30	231.5	0.58	0.79	1.03	1.30	1.60	1.94	2.31	2.71	5.19
35	224.0	0.60	0.81	1.06	1.34	1.66	2.00	2.39	2.80	5.37
40	217.1	0.62	0.84	1.09	1.38	1.71	2.07	2.46	2.89	5.54
45	210.7	0.63	0.86	1.13	1.43	1.76	2.13	2.54	2.98	5.71
50	204.7	0.65	0.89	1.16	1.47	1.81	2.19	2.61	3.06	5.88
55	199.1	0.67	0.91	1.19	1.51	1.86	2.26	2.68	3.15	6.04
60	193.9	0.69	0.94	1.23	1.55	1.91	2.32	2.76	3.23	6.20
65	189.0	0.71	0.96	1.26	1.59	1.96	2.38	2.83	3.32	6.36
70	184.5	0.72	0.99	1.29	1.63	2.01	2.43	2.90	3.40	6.52
75	180.2	0.74	1.01	1.32	1.67	2.06	2.49	2.97	3.48	6.67
80	176.1	0.76	1.03	1.35	1.71	2.11	2.55	3.03	3.56	6.83

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	53	of	98

Appendix 25 - .05" or 50mm² AAAC - (All altitudes) - 80m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	HAZEL
Greased Conductor Weight (kg/m)	0.1648
Cross Sectional Area of Conductor (mm ²)	59.9
Conductor Overall Diameter (mm)	9.9
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm ²)	6016.33
Rated Breaking Strength of Conductor (kgf)	1800.8
Basic / Recommended Span (m)	80
Wind Pressure on Conductor (N/m ²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	720.3
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	360.2
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	595.0
Maximum Conductor Weight (MCW) (kg/m)			0.165
Maximum Conductor Pressure (MCP) (kg/m)			0.767
Freezing Point Tension (FPT) (kgf)	at	0°C	398.0

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	HAZEL	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.07	0.27	0.29	0.31	0.34	0.36	0.38
80	0.30	1.07	1.17	1.26	1.35	1.43	1.52
120	0.67	2.42	2.63	2.83	3.03	3.23	3.41
160	1.20	4.30	4.67	5.04	5.39	5.73	6.06
200	1.87	6.72	7.30	7.87	8.43	8.96	9.47
240	2.69	9.67	10.52	11.34	12.13	12.90	13.64
280	3.66	13.17	14.31	15.43	16.51	17.56	18.57
320	4.78	17.20	18.70	20.16	21.57	22.93	24.25
360	6.05	21.76	23.66	25.51	27.30	29.03	30.69
400	7.47	26.87	29.21	31.49	33.70	35.83	37.89
440	9.04	32.51	35.35	38.11	40.78	43.36	45.85
480	10.76	38.69	42.07	45.35	48.53	51.60	54.57
520	12.62	45.41	49.37	53.23	56.96	60.56	64.04

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	54	of	98	

Appendix 25a - .05" or 50mm² AAAC - (All altitudes) - 80m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	HAZEL
Basic / Recommended Span (m)	80
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	441.3	0.17	0.23	0.30	0.38	0.47	0.56	0.67	0.79	1.51
0	398.0	0.19	0.25	0.33	0.42	0.52	0.63	0.75	0.87	1.68
5	360.2	0.21	0.28	0.37	0.46	0.57	0.69	0.82	0.97	1.85
10	323.5	0.23	0.31	0.41	0.52	0.64	0.77	0.92	1.08	2.06
15	288.5	0.26	0.35	0.46	0.58	0.71	0.86	1.03	1.21	2.31
20	255.6	0.29	0.39	0.52	0.65	0.81	0.98	1.16	1.36	2.61
25	225.6	0.33	0.45	0.58	0.74	0.91	1.11	1.32	1.54	2.96
30	198.8	0.37	0.51	0.66	0.84	1.04	1.25	1.49	1.75	3.36
35	175.8	0.42	0.57	0.75	0.95	1.17	1.42	1.69	1.98	3.80
40	156.5	0.47	0.64	0.84	1.07	1.32	1.59	1.90	2.22	4.26
45	140.6	0.53	0.72	0.94	1.19	1.47	1.77	2.11	2.48	4.75
50	127.5	0.58	0.79	1.03	1.31	1.62	1.95	2.33	2.73	5.23
55	116.8	0.63	0.86	1.13	1.43	1.76	2.13	2.54	2.98	5.71
60	108.0	0.69	0.93	1.22	1.55	1.91	2.31	2.75	3.22	6.18
65	100.6	0.74	1.00	1.31	1.66	2.05	2.48	2.95	3.46	6.63
70	94.4	0.79	1.07	1.40	1.77	2.18	2.64	3.14	3.69	7.07
75	89.0	0.83	1.13	1.48	1.87	2.31	2.80	3.33	3.91	7.50
80	84.4	0.88	1.20	1.56	1.98	2.44	2.95	3.51	4.12	7.90

Document reference	NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	55	of	98

Appendix 26 - .05" or 50mm² AAAC - (All altitudes) - 100m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	HAZEL
Greased Conductor Weight (kg/m)	0.1648
Cross Sectional Area of Conductor (mm ²)	59.9
Conductor Overall Diameter (mm)	9.9
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm ²)	6016.33
Rated Breaking Strength of Conductor (kgf)	1800.8
Basic / Recommended Span (m)	100
Wind Pressure on Conductor (N/m ²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	720.3
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	360.2
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	641.4
Maximum Conductor Weight (MCW) (kg/m)			0.165
Maximum Conductor Pressure (MCP) (kg/m)			0.767
Freezing Point Tension (FPT) (kgf)	at	0°C	396.2

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	HAZEL	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.08	0.23	0.25	0.27	0.28	0.30	0.32
80	0.30	0.93	1.00	1.07	1.14	1.20	1.27
120	0.68	2.10	2.26	2.41	2.56	2.71	2.85
160	1.20	3.74	4.02	4.29	4.55	4.81	5.06
200	1.88	5.84	6.28	6.70	7.12	7.52	7.91
240	2.71	8.41	9.04	9.65	10.25	10.83	11.39
280	3.69	11.45	12.31	13.14	13.95	14.73	15.50
320	4.82	14.96	16.07	17.16	18.22	19.24	20.24
360	6.10	18.93	20.34	21.72	23.06	24.36	25.62
400	7.53	23.37	25.11	26.81	28.47	30.07	31.63
440	9.11	28.28	30.39	32.44	34.44	36.38	38.27
480	10.84	33.66	36.16	38.61	40.99	43.30	45.54
520	12.72	39.50	42.44	45.31	48.11	50.82	53.45

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	56	of	98	

Appendix 26a - .05" or 50mm² AAAC - (All altitudes) - 100m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	HAZEL	
Basic / Recommended Span (m)	100	
Temperature Shift for Creep (Deg. C)	0	Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0	at 15°C
Required Percentage Increase in Tension (%)	0	at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	437.9	0.17	0.23	0.30	0.38	0.47	0.57	0.68	0.80	1.52
0	396.2	0.19	0.25	0.33	0.42	0.52	0.63	0.75	0.88	1.68
5	360.2	0.21	0.28	0.37	0.46	0.57	0.69	0.82	0.97	1.85
10	325.7	0.23	0.31	0.40	0.51	0.63	0.77	0.91	1.07	2.05
15	293.3	0.25	0.34	0.45	0.57	0.70	0.85	1.01	1.19	2.28
20	263.2	0.28	0.38	0.50	0.63	0.78	0.95	1.13	1.32	2.54
25	236.1	0.31	0.43	0.56	0.71	0.87	1.06	1.26	1.47	2.83
30	212.1	0.35	0.48	0.62	0.79	0.97	1.18	1.40	1.64	3.15
35	191.4	0.39	0.53	0.69	0.87	1.08	1.30	1.55	1.82	3.49
40	173.7	0.43	0.58	0.76	0.96	1.19	1.43	1.71	2.00	3.84
45	158.8	0.47	0.64	0.83	1.05	1.30	1.57	1.87	2.19	4.20
50	146.3	0.51	0.69	0.90	1.14	1.41	1.70	2.03	2.38	4.56
55	135.7	0.55	0.74	0.97	1.23	1.52	1.84	2.19	2.57	4.92
60	126.7	0.59	0.80	1.04	1.32	1.63	1.97	2.34	2.75	5.27
65	119.1	0.62	0.85	1.11	1.40	1.73	2.09	2.49	2.92	5.61
70	112.5	0.66	0.90	1.17	1.48	1.83	2.22	2.64	3.10	5.94
75	106.7	0.70	0.95	1.24	1.56	1.93	2.34	2.78	3.26	6.26
80	101.7	0.73	0.99	1.30	1.64	2.03	2.45	2.92	3.42	6.57

Document reference	NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	57	of	98

Appendix 27 - .05" or 50mm² AAAC - (All altitudes) - 120m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	HAZEL
Greased Conductor Weight (kg/m)	0.1648
Cross Sectional Area of Conductor (mm ²)	59.9
Conductor Overall Diameter (mm)	9.9
Coefficient of Linear Expansion (/Degree C)	2.3E-05
Modulus of Elasticity (kg/mm ²)	6016.33
Rated Breaking Strength of Conductor (kgf)	1800.8
Basic / Recommended Span (m)	120
Wind Pressure on Conductor (N/m ²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	720.3
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	360.2
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	685.8
Maximum Conductor Weight (MCW) (kg/m)			0.165
Maximum Conductor Pressure (MCP) (kg/m)			0.767
Freezing Point Tension (FPT) (kgf)	at	0°C	394.1

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	HAZEL	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.08	0.21	0.22	0.24	0.25	0.26	0.27
80	0.30	0.84	0.89	0.95	1.00	1.05	1.10
120	0.68	1.89	2.01	2.13	2.25	2.36	2.47
160	1.22	3.35	3.57	3.78	3.99	4.20	4.39
200	1.90	5.24	5.58	5.91	6.24	6.55	6.86
240	2.73	7.54	8.03	8.51	8.98	9.44	9.88
280	3.72	10.27	10.94	11.59	12.23	12.85	13.45
320	4.86	13.41	14.28	15.14	15.97	16.78	17.57
360	6.15	16.98	18.08	19.16	20.21	21.24	22.24
400	7.60	20.96	22.32	23.65	24.95	26.22	27.45
440	9.19	25.36	27.01	28.62	30.19	31.73	33.22
480	10.94	30.18	32.14	34.06	35.93	37.76	39.53
520	12.84	35.42	37.72	39.97	42.17	44.31	46.40

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	58	of	98	

Appendix 27a - .05" or 50mm² AAAC - (All altitudes) - 120m Basic

Conductor Code Name	HAZEL
Basic / Recommended Span (m)	120
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	433.9	0.17	0.23	0.30	0.38	0.47	0.57	0.68	0.80	1.54
0	394.1	0.19	0.26	0.33	0.42	0.52	0.63	0.75	0.88	1.69
5	360.2	0.21	0.28	0.37	0.46	0.57	0.69	0.82	0.97	1.85
10	328.0	0.23	0.31	0.40	0.51	0.63	0.76	0.90	1.06	2.03
15	298.1	0.25	0.34	0.44	0.56	0.69	0.84	1.00	1.17	2.24
20	270.7	0.27	0.37	0.49	0.62	0.76	0.92	1.10	1.29	2.47
25	246.1	0.30	0.41	0.54	0.68	0.84	1.01	1.21	1.41	2.71
30	224.3	0.33	0.45	0.59	0.74	0.92	1.11	1.32	1.55	2.98
35	205.4	0.36	0.49	0.64	0.81	1.00	1.21	1.44	1.69	3.25
40	189.1	0.39	0.53	0.70	0.88	1.09	1.32	1.57	1.84	3.53
45	175.0	0.42	0.58	0.75	0.95	1.18	1.42	1.69	1.99	3.81
50	163.0	0.46	0.62	0.81	1.02	1.26	1.53	1.82	2.14	4.10
55	152.6	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	4.37
60	143.6	0.52	0.70	0.92	1.16	1.43	1.74	2.07	2.42	4.65
65	135.8	0.55	0.74	0.97	1.23	1.52	1.83	2.18	2.56	4.91
70	129.0	0.57	0.78	1.02	1.29	1.60	1.93	2.30	2.70	5.17
75	123.0	0.60	0.82	1.07	1.36	1.68	2.03	2.41	2.83	5.43
80	117.6	0.63	0.86	1.12	1.42	1.75	2.12	2.52	2.96	5.67

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	59	of	98

Appendix 28 - .025" or 16mm² HDBC - (All altitudes) - 90m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	.025" HDBC
Greased Conductor Weight (kg/m)	0.1473
Cross Sectional Area of Conductor (mm ²)	16.4416
Conductor Overall Diameter (mm)	5.689
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.3
Rated Breaking Strength of Conductor (kgf)	689.4
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m ²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	275.8
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	229.8
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	275.8
Maximum Conductor Weight (MCW) (kg/m)			0.147
Maximum Conductor Pressure (MCP) (kg/m)			0.441
Freezing Point Tension (FPT) (kgf)	at	0°C	137.3

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor: .025" HDBC	Temperature Shift (Deg. C)	0
	Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.20	0.36	0.37	0.38	0.39	0.40	0.41
80	0.80	1.42	1.47	1.51	1.56	1.60	1.65
120	1.81	3.20	3.30	3.41	3.51	3.61	3.71
160	3.21	5.68	5.87	6.06	6.24	6.42	6.60
200	5.01	8.88	9.17	9.46	9.75	10.03	10.31
240	7.22	12.78	13.21	13.63	14.04	14.44	14.84
280	9.83	17.40	17.98	18.55	19.11	19.66	20.20
320	12.84	22.73	23.49	24.23	24.96	25.68	26.38
360	16.25	28.77	29.72	30.67	31.59	32.50	33.39
400	20.06	35.51	36.70	37.86	39.00	40.12	41.22
440	24.27	42.97	44.40	45.81	47.19	48.55	49.88
480	28.89	51.14	52.84	54.52	56.16	57.78	59.36
520	33.90	60.02	62.02	63.98	65.91	67.81	69.67

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	60	of	98	

Appendix 28a - .025" or 16mm HDBC - (All altitudes) - 90m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name .025" HDBC
 Basic / Recommended Span (m) 90
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C
 Required Percentage Increase in Tension (%) 0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	146.9	0.45	0.61	0.80	1.02	1.25	1.52	1.81	2.12	4.06
0	137.3	0.48	0.66	0.86	1.09	1.34	1.62	1.93	2.27	4.35
5	129.5	0.51	0.70	0.91	1.15	1.42	1.72	2.05	2.40	4.61
10	122.5	0.54	0.74	0.96	1.22	1.50	1.82	2.16	2.54	4.87
15	116.2	0.57	0.78	1.01	1.28	1.58	1.92	2.28	2.68	5.13
20	110.5	0.60	0.82	1.07	1.35	1.67	2.02	2.40	2.82	5.40
25	105.3	0.63	0.86	1.12	1.42	1.75	2.12	2.52	2.95	5.66
30	100.7	0.66	0.90	1.17	1.48	1.83	2.21	2.63	3.09	5.93
35	96.4	0.69	0.94	1.22	1.55	1.91	2.31	2.75	3.23	6.19
40	92.6	0.72	0.97	1.27	1.61	1.99	2.41	2.86	3.36	6.44
45	89.1	0.74	1.01	1.32	1.67	2.07	2.50	2.98	3.49	6.69
50	85.9	0.77	1.05	1.37	1.74	2.14	2.59	3.09	3.62	6.94
55	83.0	0.80	1.09	1.42	1.80	2.22	2.68	3.19	3.75	7.19
60	80.3	0.83	1.12	1.47	1.86	2.29	2.77	3.30	3.87	7.43
65	77.9	0.85	1.16	1.51	1.92	2.36	2.86	3.41	4.00	7.66
70	75.6	0.88	1.19	1.56	1.97	2.44	2.95	3.51	4.12	7.89
75	73.5	0.90	1.23	1.60	2.03	2.51	3.03	3.61	4.24	8.12
80	71.5	0.93	1.26	1.65	2.09	2.58	3.12	3.71	4.35	8.34

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	61	of	98

Appendix 29 - .025" or 16mm² HDBC - (All altitudes) - 100m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	.025" HDBC
Greased Conductor Weight (kg/m)	0.1473
Cross Sectional Area of Conductor (mm ²)	16.4416
Conductor Overall Diameter (mm)	5.689
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.3
Rated Breaking Strength of Conductor (kgf)	689.4
Basic / Recommended Span (m)	100
Wind Pressure on Conductor (N/m ²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	275.8
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	229.8
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	275.8
Maximum Conductor Weight (MCW) (kg/m)			0.147
Maximum Conductor Pressure (MCP) (kg/m)			0.441
Freezing Point Tension (FPT) (kgf)	at	0°C	126.7

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor: .025" HDBC	Temperature Shift (Deg. C)	0
	Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.22	0.35	0.36	0.37	0.38	0.39	0.40
80	0.88	1.42	1.46	1.50	1.54	1.57	1.61
120	1.98	3.19	3.28	3.37	3.45	3.54	3.62
160	3.52	5.67	5.83	5.99	6.14	6.29	6.44
200	5.49	8.86	9.11	9.36	9.59	9.83	10.06
240	7.91	12.76	13.12	13.47	13.82	14.16	14.49
280	10.77	17.37	17.86	18.34	18.81	19.27	19.72
320	14.06	22.69	23.33	23.95	24.56	25.16	25.76
360	17.80	28.72	29.52	30.31	31.09	31.85	32.60
400	21.97	35.46	36.45	37.42	38.38	39.32	40.25
440	26.59	42.90	44.10	45.28	46.44	47.58	48.70
480	31.64	51.06	52.49	53.89	55.27	56.62	57.95
520	37.14	59.92	61.60	63.24	64.86	66.45	68.01

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	62	of	98	

Appendix 29a - .025" or 16mm² HDBC - (All altitudes) - 100m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name .025" HDBC
 Basic / Recommended Span (m) 100
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C
 Required Percentage Increase in Tension (%) 0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	134.1	0.49	0.67	0.88	1.11	1.37	1.66	1.98	2.32	4.45
0	126.7	0.52	0.71	0.93	1.18	1.45	1.76	2.09	2.46	4.71
5	120.8	0.55	0.75	0.98	1.23	1.52	1.84	2.19	2.58	4.94
10	115.4	0.57	0.78	1.02	1.29	1.60	1.93	2.30	2.70	5.17
15	110.5	0.60	0.82	1.07	1.35	1.67	2.02	2.40	2.82	5.40
20	106.1	0.62	0.85	1.11	1.41	1.74	2.10	2.50	2.93	5.62
25	102.0	0.65	0.88	1.16	1.46	1.81	2.18	2.60	3.05	5.85
30	98.3	0.67	0.92	1.20	1.52	1.87	2.27	2.70	3.17	6.07
35	94.9	0.70	0.95	1.24	1.57	1.94	2.35	2.80	3.28	6.29
40	91.7	0.72	0.98	1.28	1.63	2.01	2.43	2.89	3.39	6.50
45	88.8	0.75	1.02	1.33	1.68	2.07	2.51	2.98	3.50	6.72
50	86.2	0.77	1.05	1.37	1.73	2.14	2.59	3.08	3.61	6.92
55	83.7	0.79	1.08	1.41	1.78	2.20	2.66	3.17	3.72	7.13
60	81.4	0.81	1.11	1.45	1.83	2.26	2.74	3.26	3.82	7.33
65	79.2	0.84	1.14	1.49	1.88	2.32	2.81	3.35	3.93	7.53
70	77.2	0.86	1.17	1.53	1.93	2.38	2.88	3.43	4.03	7.72
75	75.4	0.88	1.20	1.56	1.98	2.44	2.96	3.52	4.13	7.91
80	73.6	0.90	1.23	1.60	2.03	2.50	3.03	3.60	4.23	8.10

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	63	of	98

Appendix 30 - 3 SWG HDBC - (All altitudes) - 90m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	3SWG
Greased Conductor Weight (kg/m)	0.2861
Cross Sectional Area of Conductor (mm²)	32.18
Conductor Overall Diameter (mm)	6.4
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm²)	12655.3
Rated Breaking Strength of Conductor (kgf)	1301.9
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	520.8
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	434.0
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	520.8
Maximum Conductor Weight (MCW) (kg/m)			0.286
Maximum Conductor Pressure (MCP) (kg/m)			0.496
Freezing Point Tension (FPT) (kgf)	at	0°C	389.8

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	3SWG	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.14	0.28	0.29	0.30	0.32	0.33	0.34
80	0.55	1.11	1.16	1.21	1.26	1.31	1.36
120	1.23	2.50	2.61	2.73	2.84	2.96	3.07
160	2.19	4.44	4.65	4.85	5.06	5.26	5.45
200	3.42	6.94	7.26	7.58	7.90	8.21	8.52
240	4.92	9.99	10.46	10.92	11.38	11.83	12.27
280	6.70	13.59	14.23	14.86	15.48	16.10	16.70
320	8.75	17.75	18.59	19.41	20.22	21.02	21.81
360	11.07	22.47	23.52	24.57	25.60	26.61	27.61
400	13.66	27.74	29.04	30.33	31.60	32.85	34.08
440	16.53	33.57	35.14	36.70	38.24	39.75	41.24
480	19.68	39.95	41.82	43.67	45.50	47.31	49.08
520	23.09	46.88	49.08	51.26	53.40	55.52	57.60

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	64	of	98	

Appendix 30a - 3 SWG HDBC - (All altitudes) - 90m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name 3SWG
 Basic / Recommended Span (m) 90
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C
 Required Percentage Increase in Tension (%) 0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	418.8	0.31	0.42	0.55	0.69	0.85	1.03	1.23	1.44	2.77
0	389.8	0.33	0.45	0.59	0.74	0.92	1.11	1.32	1.55	2.97
5	365.4	0.35	0.48	0.63	0.79	0.98	1.18	1.41	1.65	3.17
10	342.5	0.38	0.51	0.67	0.85	1.04	1.26	1.50	1.76	3.38
15	321.1	0.40	0.55	0.71	0.90	1.11	1.35	1.60	1.88	3.61
20	301.3	0.43	0.58	0.76	0.96	1.19	1.44	1.71	2.01	3.85
25	283.1	0.45	0.62	0.81	1.02	1.26	1.53	1.82	2.13	4.09
30	266.5	0.48	0.66	0.86	1.09	1.34	1.62	1.93	2.27	4.35
35	251.4	0.51	0.70	0.91	1.15	1.42	1.72	2.05	2.40	4.61
40	237.8	0.54	0.74	0.96	1.22	1.50	1.82	2.17	2.54	4.87
45	225.5	0.57	0.78	1.01	1.28	1.59	1.92	2.28	2.68	5.14
50	214.4	0.60	0.82	1.07	1.35	1.67	2.02	2.40	2.82	5.40
55	204.4	0.63	0.86	1.12	1.42	1.75	2.12	2.52	2.96	5.67
60	195.3	0.66	0.90	1.17	1.48	1.83	2.22	2.64	3.09	5.93
65	187.1	0.69	0.94	1.22	1.55	1.91	2.31	2.75	3.23	6.19
70	179.7	0.72	0.98	1.27	1.61	1.99	2.41	2.87	3.36	6.45
75	172.9	0.74	1.01	1.32	1.68	2.07	2.50	2.98	3.50	6.70
80	166.7	0.77	1.05	1.37	1.74	2.15	2.60	3.09	3.63	6.95

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	65	of	98

Appendix 31 - 3 SWG HDBC - (All altitude) - 100m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	3SWG
Greased Conductor Weight (kg/m)	0.2861
Cross Sectional Area of Conductor (mm²)	32.18
Conductor Overall Diameter (mm)	6.4
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm²)	12655.3
Rated Breaking Strength of Conductor (kgf)	1301.9
Basic / Recommended Span (m)	100
Wind Pressure on Conductor (N/m²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	520.8
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	434.0
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	520.8
Maximum Conductor Weight (MCW) (kg/m)			0.286
Maximum Conductor Pressure (MCP) (kg/m)			0.496
Freezing Point Tension (FPT) (kgf)	at	0°C	375.4

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	3SWG	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.14	0.27	0.28	0.29	0.30	0.31	0.33
80	0.57	1.08	1.13	1.17	1.22	1.26	1.30
120	1.28	2.44	2.54	2.64	2.74	2.83	2.93
160	2.28	4.34	4.51	4.69	4.87	5.04	5.21
200	3.56	6.77	7.05	7.33	7.60	7.87	8.13
240	5.13	9.75	10.16	10.55	10.95	11.33	11.71
280	6.98	13.28	13.82	14.37	14.90	15.43	15.94
320	9.12	17.34	18.06	18.76	19.46	20.15	20.83
360	11.54	21.95	22.85	23.75	24.63	25.50	26.36
400	14.24	27.09	28.21	29.32	30.41	31.48	32.54
440	17.24	32.78	34.14	35.48	36.79	38.09	39.37
480	20.51	39.02	40.63	42.22	43.79	45.33	46.86
520	24.07	45.79	47.68	49.55	51.39	53.20	54.99

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	66	of	98	

Appendix 31 - a3 SWG HDBC - (All altitudes) - 100m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name 3SWG
 Basic / Recommended Span (m) 100
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C
 Required Percentage Increase in Tension (%) 0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	401.7	0.32	0.44	0.57	0.72	0.89	1.08	1.28	1.50	2.88
0	375.4	0.34	0.47	0.61	0.77	0.95	1.15	1.37	1.61	3.09
5	353.4	0.36	0.50	0.65	0.82	1.01	1.22	1.46	1.71	3.28
10	332.9	0.39	0.53	0.69	0.87	1.07	1.30	1.55	1.82	3.48
15	313.9	0.41	0.56	0.73	0.92	1.14	1.38	1.64	1.93	3.69
20	296.4	0.43	0.59	0.77	0.98	1.21	1.46	1.74	2.04	3.91
25	280.4	0.46	0.62	0.82	1.03	1.28	1.54	1.84	2.16	4.13
30	265.8	0.48	0.66	0.86	1.09	1.35	1.63	1.94	2.27	4.36
35	252.5	0.51	0.69	0.91	1.15	1.42	1.71	2.04	2.39	4.59
40	240.4	0.54	0.73	0.95	1.21	1.49	1.80	2.14	2.51	4.82
45	229.4	0.56	0.76	1.00	1.26	1.56	1.89	2.25	2.64	5.05
50	219.4	0.59	0.80	1.04	1.32	1.63	1.97	2.35	2.76	5.28
55	210.3	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.87	5.51
60	202.0	0.64	0.87	1.13	1.43	1.77	2.14	2.55	2.99	5.74
65	194.4	0.66	0.90	1.18	1.49	1.84	2.23	2.65	3.11	5.96
70	187.5	0.69	0.93	1.22	1.55	1.91	2.31	2.75	3.22	6.18
75	181.1	0.71	0.97	1.26	1.60	1.97	2.39	2.84	3.34	6.40
80	175.2	0.73	1.00	1.31	1.65	2.04	2.47	2.94	3.45	6.61

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	67	of	98

Appendix 32 - 5SWG HDBC - (All altitudes) - 90m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	5SWG
Greased Conductor Weight (kg/m)	0.2024
Cross Sectional Area of Conductor (mm²)	22.77
Conductor Overall Diameter (mm)	5.38
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm²)	12655.3
Rated Breaking Strength of Conductor (kgf)	945
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	378.0
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	315.0
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	378.0
Maximum Conductor Weight (MCW) (kg/m)			0.202
Maximum Conductor Pressure (MCP) (kg/m)			0.417
Freezing Point Tension (FPT) (kgf)	at	0°C	262.3

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	5SWG	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.14	0.29	0.30	0.31	0.33	0.34	0.35
80	0.57	1.15	1.20	1.25	1.30	1.35	1.40
120	1.29	2.59	2.71	2.82	2.93	3.04	3.15
160	2.30	4.60	4.81	5.01	5.21	5.41	5.61
200	3.59	7.19	7.52	7.83	8.15	8.46	8.76
240	5.17	10.36	10.82	11.28	11.73	12.18	12.62
280	7.04	14.10	14.73	15.36	15.97	16.58	17.17
320	9.19	18.41	19.24	20.06	20.86	21.65	22.43
360	11.63	23.31	24.35	25.38	26.40	27.40	28.39
400	14.36	28.77	30.06	31.34	32.59	33.83	35.04
440	17.37	34.82	36.38	37.92	39.44	40.93	42.40
480	20.68	41.43	43.29	45.13	46.93	48.71	50.46
520	24.27	48.63	50.81	52.96	55.08	57.17	59.22

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	68	of	98	

Appendix 32a - 5SWG HDBC - (All altitudes) - 90m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name 5SWG
 Basic / Recommended Span (m) 90
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C
 Required Percentage Increase in Tension (%) 0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	281.9	0.32	0.44	0.57	0.73	0.90	1.09	1.29	1.52	2.91
0	262.3	0.35	0.47	0.62	0.78	0.96	1.17	1.39	1.63	3.13
5	245.8	0.37	0.50	0.66	0.83	1.03	1.25	1.48	1.74	3.33
10	230.4	0.40	0.54	0.70	0.89	1.10	1.33	1.58	1.86	3.56
15	216.2	0.42	0.57	0.75	0.95	1.17	1.42	1.69	1.98	3.79
20	203.0	0.45	0.61	0.80	1.01	1.25	1.51	1.79	2.11	4.04
25	191.0	0.48	0.65	0.85	1.07	1.32	1.60	1.91	2.24	4.29
30	180.2	0.51	0.69	0.90	1.14	1.40	1.70	2.02	2.37	4.55
35	170.3	0.53	0.73	0.95	1.20	1.49	1.80	2.14	2.51	4.81
40	161.4	0.56	0.77	1.00	1.27	1.57	1.90	2.26	2.65	5.08
45	153.3	0.59	0.81	1.06	1.34	1.65	2.00	2.38	2.79	5.35
50	146.1	0.62	0.85	1.11	1.40	1.73	2.10	2.49	2.93	5.61
55	139.5	0.65	0.89	1.16	1.47	1.81	2.19	2.61	3.06	5.87
60	133.6	0.68	0.93	1.21	1.53	1.89	2.29	2.73	3.20	6.14
65	128.2	0.71	0.97	1.26	1.60	1.97	2.39	2.84	3.33	6.39
70	123.3	0.74	1.01	1.31	1.66	2.05	2.48	2.95	3.47	6.65
75	118.9	0.77	1.04	1.36	1.72	2.13	2.58	3.06	3.60	6.90
80	114.8	0.79	1.08	1.41	1.79	2.20	2.67	3.17	3.72	7.14

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	69	of	98

Appendix 33 - 5SWG HDBC - (All altitude) - 100m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	5SWG
Greased Conductor Weight (kg/m)	0.2024
Cross Sectional Area of Conductor (mm²)	22.77
Conductor Overall Diameter (mm)	5.38
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm²)	12655.3
Rated Breaking Strength of Conductor (kgf)	945
Basic / Recommended Span (m)	100
Wind Pressure on Conductor (N/m²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	378.0
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	315.0
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	378.0
Maximum Conductor Weight (MCW) (kg/m)			0.202
Maximum Conductor Pressure (MCP) (kg/m)			0.417
Freezing Point Tension (FPT) (kgf)	at	0°C	249.2

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	5SWG	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.15	0.28	0.29	0.30	0.32	0.33	0.34
80	0.61	1.13	1.18	1.22	1.26	1.31	1.35
120	1.37	2.55	2.65	2.74	2.84	2.94	3.03
160	2.43	4.53	4.71	4.88	5.05	5.22	5.39
200	3.80	7.08	7.35	7.62	7.89	8.16	8.42
240	5.46	10.19	10.59	10.98	11.37	11.75	12.12
280	7.44	13.87	14.41	14.94	15.47	15.99	16.50
320	9.72	18.12	18.82	19.52	20.21	20.88	21.55
360	12.30	22.93	23.82	24.70	25.57	26.43	27.27
400	15.18	28.31	29.41	30.50	31.57	32.63	33.67
440	18.37	34.25	35.59	36.90	38.20	39.48	40.73
480	21.86	40.76	42.35	43.92	45.46	46.98	48.48
520	25.65	47.84	49.70	51.54	53.35	55.14	56.89

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	70	of	98	

Appendix 33a - 5SWG HDBC - (All altitudes) - 100m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name 5SWG
 Basic / Recommended Span (m) 100
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C
 Required Percentage Increase in Tension (%) 0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	266.7	0.34	0.46	0.61	0.77	0.95	1.15	1.37	1.60	3.07
0	249.2	0.37	0.50	0.65	0.82	1.02	1.23	1.46	1.72	3.29
5	234.8	0.39	0.53	0.69	0.87	1.08	1.30	1.55	1.82	3.49
10	221.4	0.41	0.56	0.73	0.93	1.14	1.38	1.65	1.93	3.70
15	209.1	0.44	0.59	0.77	0.98	1.21	1.46	1.74	2.04	3.92
20	197.8	0.46	0.63	0.82	1.04	1.28	1.55	1.84	2.16	4.14
25	187.5	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	4.37
30	178.1	0.51	0.70	0.91	1.15	1.42	1.72	2.05	2.40	4.60
35	169.6	0.54	0.73	0.95	1.21	1.49	1.80	2.15	2.52	4.83
40	161.9	0.56	0.77	1.00	1.27	1.56	1.89	2.25	2.64	5.06
45	154.8	0.59	0.80	1.05	1.32	1.63	1.98	2.35	2.76	5.29
50	148.4	0.61	0.84	1.09	1.38	1.70	2.06	2.45	2.88	5.52
55	142.6	0.64	0.87	1.14	1.44	1.77	2.15	2.56	3.00	5.75
60	137.2	0.66	0.90	1.18	1.49	1.84	2.23	2.65	3.12	5.97
65	132.4	0.69	0.94	1.22	1.55	1.91	2.31	2.75	3.23	6.19
70	127.9	0.71	0.97	1.27	1.60	1.98	2.39	2.85	3.34	6.41
75	123.8	0.74	1.00	1.31	1.66	2.04	2.47	2.94	3.45	6.62
80	120.0	0.76	1.03	1.35	1.71	2.11	2.55	3.04	3.56	6.83

Document reference	NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	71	of	98

Appendix 34 - 6SWG HDBC - (All altitudes) - 90m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	6SWG
Greased Conductor Weight (kg/m)	0.16606
Cross Sectional Area of Conductor (mm²)	18.68
Conductor Overall Diameter (mm)	4.88
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm²)	12655.29
Rated Breaking Strength of Conductor (kgf)	787
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	314.8
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	262.3
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	314.8
Maximum Conductor Weight (MCW) (kg/m)			0.166
Maximum Conductor Pressure (MCP) (kg/m)			0.378
Freezing Point Tension (FPT) (kgf)	at	0°C	206.5

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	6SWG	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.15	0.30	0.31	0.32	0.33	0.35	0.36
80	0.60	1.18	1.24	1.29	1.34	1.38	1.43
120	1.35	2.66	2.78	2.89	3.01	3.12	3.22
160	2.39	4.74	4.94	5.14	5.34	5.54	5.73
200	3.74	7.40	7.72	8.04	8.35	8.65	8.95
240	5.38	10.66	11.12	11.57	12.02	12.46	12.89
280	7.33	14.51	15.14	15.75	16.36	16.96	17.55
320	9.57	18.95	19.77	20.58	21.37	22.15	22.92
360	12.11	23.98	25.02	26.04	27.05	28.04	29.01
400	14.96	29.61	30.89	32.15	33.39	34.62	35.82
440	18.10	35.83	37.38	38.90	40.41	41.89	43.34
480	21.54	42.64	44.48	46.30	48.09	49.85	51.58
520	25.28	50.04	52.20	54.34	56.44	58.50	60.53

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	72	of	98	

Appendix 34a - 6SWG HDBC - (All altitudes) - 90m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	6SWG
Basic / Recommended Span (m)	90
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	222.1	0.34	0.46	0.60	0.76	0.93	1.13	1.35	1.58	3.03
0	206.5	0.36	0.49	0.64	0.81	1.01	1.22	1.45	1.70	3.26
5	193.6	0.39	0.53	0.69	0.87	1.07	1.30	1.54	1.81	3.47
10	181.5	0.41	0.56	0.73	0.93	1.14	1.38	1.65	1.93	3.71
15	170.4	0.44	0.60	0.78	0.99	1.22	1.47	1.75	2.06	3.95
20	160.2	0.47	0.63	0.83	1.05	1.30	1.57	1.87	2.19	4.20
25	151.0	0.49	0.67	0.88	1.11	1.37	1.66	1.98	2.32	4.45
30	142.6	0.52	0.71	0.93	1.18	1.46	1.76	2.10	2.46	4.72
35	135.0	0.55	0.75	0.98	1.25	1.54	1.86	2.21	2.60	4.98
40	128.2	0.58	0.79	1.04	1.31	1.62	1.96	2.33	2.74	5.25
45	122.0	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.88	5.51
50	116.4	0.64	0.87	1.14	1.44	1.78	2.16	2.57	3.01	5.78
55	111.4	0.67	0.91	1.19	1.51	1.86	2.26	2.68	3.15	6.04
60	106.8	0.70	0.95	1.24	1.57	1.94	2.35	2.80	3.29	6.30
65	102.6	0.73	0.99	1.29	1.64	2.02	2.45	2.91	3.42	6.55
70	98.8	0.76	1.03	1.34	1.70	2.10	2.54	3.02	3.55	6.80
75	95.4	0.78	1.07	1.39	1.76	2.18	2.63	3.13	3.68	7.05
80	92.2	0.81	1.10	1.44	1.82	2.25	2.72	3.24	3.80	7.29

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	73	of	98

Appendix 35 - 6SWG HDBC - (All altitudes) - 100m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	6SWG
Greased Conductor Weight (kg/m)	0.16606
Cross Sectional Area of Conductor (mm²)	18.68
Conductor Overall Diameter (mm)	4.88
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm²)	12655.29
Rated Breaking Strength of Conductor (kgf)	787
Basic / Recommended Span (m)	100
Wind Pressure on Conductor (N/m²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	314.8
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	262.3
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	314.8
Maximum Conductor Weight (MCW) (kg/m)			0.166
Maximum Conductor Pressure (MCP) (kg/m)			0.378
Freezing Point Tension (FPT) (kgf)	at	0°C	194.5

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	6SWG	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.16	0.29	0.30	0.31	0.32	0.34	0.35
80	0.64	1.17	1.21	1.26	1.30	1.34	1.38
120	1.44	2.63	2.73	2.83	2.92	3.02	3.11
160	2.55	4.68	4.86	5.03	5.20	5.36	5.53
200	3.99	7.32	7.59	7.86	8.12	8.38	8.64
240	5.75	10.53	10.93	11.31	11.69	12.07	12.44
280	7.82	14.34	14.87	15.40	15.92	16.43	16.93
320	10.22	18.73	19.43	20.11	20.79	21.46	22.11
360	12.93	23.70	24.59	25.46	26.31	27.16	27.99
400	15.97	29.26	30.35	31.43	32.49	33.53	34.55
440	19.32	35.41	36.73	38.03	39.31	40.57	41.81
480	22.99	42.14	43.71	45.26	46.78	48.28	49.75
520	26.98	49.45	51.30	53.11	54.90	56.66	58.39

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	74	of	98	

Appendix 35a - 6SWG HDBC - (All altitudes) - 100m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name 6SWG
 Basic / Recommended Span (m) 100
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C
 Required Percentage Increase in Tension (%) 0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	208.0	0.36	0.49	0.64	0.81	1.00	1.21	1.44	1.69	3.23
0	194.5	0.38	0.52	0.68	0.86	1.07	1.29	1.54	1.80	3.46
5	183.4	0.41	0.55	0.72	0.92	1.13	1.37	1.63	1.91	3.67
10	173.1	0.43	0.59	0.77	0.97	1.20	1.45	1.73	2.03	3.88
15	163.7	0.46	0.62	0.81	1.03	1.27	1.53	1.83	2.14	4.11
20	155.2	0.48	0.66	0.86	1.08	1.34	1.62	1.93	2.26	4.33
25	147.4	0.51	0.69	0.90	1.14	1.41	1.70	2.03	2.38	4.56
30	140.2	0.53	0.73	0.95	1.20	1.48	1.79	2.13	2.50	4.80
35	133.8	0.56	0.76	0.99	1.26	1.55	1.88	2.23	2.62	5.03
40	127.9	0.58	0.80	1.04	1.31	1.62	1.96	2.34	2.74	5.26
45	122.6	0.61	0.83	1.08	1.37	1.69	2.05	2.44	2.86	5.49
50	117.7	0.63	0.86	1.13	1.43	1.76	2.13	2.54	2.98	5.71
55	113.3	0.66	0.90	1.17	1.48	1.83	2.22	2.64	3.10	5.94
60	109.2	0.68	0.93	1.22	1.54	1.90	2.30	2.74	3.21	6.16
65	105.5	0.71	0.96	1.26	1.59	1.97	2.38	2.83	3.32	6.37
70	102.1	0.73	1.00	1.30	1.65	2.03	2.46	2.93	3.44	6.59
75	98.9	0.76	1.03	1.34	1.70	2.10	2.54	3.02	3.55	6.80
80	96.0	0.78	1.06	1.38	1.75	2.16	2.62	3.11	3.65	7.01

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	75	of	98

Appendix 36 - 7SWG HDBC - (All altitudes) - 90m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	7SWG
Greased Conductor Weight (kg/m)	0.13953
Cross Sectional Area of Conductor (mm²)	15.7
Conductor Overall Diameter (mm)	4.47
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm²)	12655.3
Rated Breaking Strength of Conductor (kgf)	671
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	268.4
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	223.7
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	268.4
Maximum Conductor Weight (MCW) (kg/m)			0.140
Maximum Conductor Pressure (MCP) (kg/m)			0.346
Freezing Point Tension (FPT) (kgf)	at	0°C	166.7

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	7SWG	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.16	0.30	0.32	0.33	0.34	0.35	0.37
80	0.62	1.22	1.27	1.32	1.37	1.42	1.46
120	1.40	2.74	2.85	2.97	3.08	3.19	3.29
160	2.49	4.87	5.07	5.27	5.47	5.66	5.85
200	3.89	7.61	7.93	8.24	8.55	8.85	9.14
240	5.61	10.96	11.41	11.86	12.30	12.74	13.17
280	7.63	14.91	15.53	16.15	16.75	17.34	17.92
320	9.97	19.48	20.29	21.09	21.88	22.65	23.41
360	12.61	24.65	25.68	26.69	27.69	28.67	29.63
400	15.57	30.43	31.70	32.95	34.18	35.39	36.58
440	18.84	36.82	38.36	39.87	41.36	42.82	44.26
480	22.42	43.82	45.65	47.45	49.22	50.96	52.67
520	26.32	51.43	53.57	55.69	57.76	59.81	61.82

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	76	of	98	

Appendix 36a - 7SWG HDBC - (All altitudes) - 90m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	7SWG
Basic / Recommended Span (m)	90
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	179.2	0.35	0.48	0.62	0.79	0.97	1.18	1.40	1.64	3.15
0	166.7	0.38	0.51	0.67	0.85	1.05	1.27	1.51	1.77	3.39
5	156.2	0.40	0.55	0.71	0.90	1.12	1.35	1.61	1.89	3.62
10	146.6	0.43	0.58	0.76	0.96	1.19	1.44	1.71	2.01	3.85
15	137.8	0.46	0.62	0.81	1.03	1.27	1.53	1.82	2.14	4.10
20	129.7	0.48	0.66	0.86	1.09	1.34	1.63	1.94	2.27	4.36
25	122.4	0.51	0.70	0.91	1.15	1.43	1.72	2.05	2.41	4.62
30	115.8	0.54	0.74	0.96	1.22	1.51	1.82	2.17	2.55	4.88
35	109.8	0.57	0.78	1.02	1.29	1.59	1.92	2.29	2.69	5.15
40	104.4	0.60	0.82	1.07	1.35	1.67	2.02	2.41	2.82	5.41
45	99.5	0.63	0.86	1.12	1.42	1.75	2.12	2.52	2.96	5.68
50	95.1	0.66	0.90	1.17	1.49	1.83	2.22	2.64	3.10	5.94
55	91.1	0.69	0.94	1.23	1.55	1.91	2.32	2.76	3.23	6.20
60	87.5	0.72	0.98	1.28	1.61	1.99	2.41	2.87	3.37	6.46
65	84.2	0.75	1.01	1.33	1.68	2.07	2.51	2.98	3.50	6.71
70	81.2	0.77	1.05	1.37	1.74	2.15	2.60	3.09	3.63	6.96
75	78.5	0.80	1.09	1.42	1.80	2.22	2.69	3.20	3.76	7.20
80	75.9	0.83	1.13	1.47	1.86	2.30	2.78	3.31	3.88	7.44

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	77	of	98

Appendix 37 - 7SWG HDBC - (All altitude) - 100m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	7SWG
Greased Conductor Weight (kg/m)	0.13953
Cross Sectional Area of Conductor (mm²)	15.7
Conductor Overall Diameter (mm)	4.47
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm²)	12655.3
Rated Breaking Strength of Conductor (kgf)	671
Basic / Recommended Span (m)	100
Wind Pressure on Conductor (N/m²)	760
Radial Ice Thickness (mm)	0
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	268.4
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	223.7
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	268.4
Maximum Conductor Weight (MCW) (kg/m)			0.140
Maximum Conductor Pressure (MCP) (kg/m)			0.346
Freezing Point Tension (FPT) (kgf)	at	0°C	155.7

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	7SWG	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.17	0.30	0.31	0.32	0.33	0.34	0.35
80	0.67	1.21	1.25	1.29	1.34	1.38	1.42
120	1.51	2.72	2.81	2.91	3.00	3.10	3.19
160	2.68	4.83	5.00	5.17	5.34	5.50	5.67
200	4.19	7.55	7.82	8.08	8.34	8.60	8.85
240	6.04	10.87	11.26	11.64	12.02	12.39	12.75
280	8.22	14.80	15.32	15.84	16.36	16.86	17.35
320	10.74	19.33	20.02	20.69	21.36	22.02	22.66
360	13.59	24.46	25.33	26.19	27.04	27.87	28.68
400	16.77	30.20	31.27	32.34	33.38	34.40	35.41
440	20.30	36.54	37.84	39.13	40.39	41.63	42.85
480	24.16	43.48	45.04	46.56	48.07	49.54	51.00
520	28.35	51.03	52.85	54.65	56.41	58.14	59.85

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	78	of	98	

Appendix 37a - 7SWG HDBC - (All altitude) - 100m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name 7SWG
 Basic / Recommended Span (m) 100
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C
 Required Percentage Increase in Tension (%) 0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	166.4	0.38	0.51	0.67	0.85	1.05	1.27	1.51	1.77	3.40
0	155.7	0.40	0.55	0.72	0.91	1.12	1.36	1.61	1.89	3.63
5	147.0	0.43	0.58	0.76	0.96	1.19	1.44	1.71	2.01	3.85
10	138.9	0.45	0.62	0.80	1.02	1.26	1.52	1.81	2.12	4.07
15	131.6	0.48	0.65	0.85	1.07	1.33	1.60	1.91	2.24	4.29
20	124.9	0.50	0.68	0.89	1.13	1.40	1.69	2.01	2.36	4.52
25	118.9	0.53	0.72	0.94	1.19	1.47	1.78	2.11	2.48	4.75
30	113.3	0.55	0.75	0.98	1.25	1.54	1.86	2.22	2.60	4.99
35	108.3	0.58	0.79	1.03	1.30	1.61	1.95	2.32	2.72	5.22
40	103.8	0.61	0.82	1.08	1.36	1.68	2.03	2.42	2.84	5.45
45	99.6	0.63	0.86	1.12	1.42	1.75	2.12	2.52	2.96	5.67
50	95.8	0.66	0.89	1.16	1.47	1.82	2.20	2.62	3.08	5.90
55	92.4	0.68	0.93	1.21	1.53	1.89	2.28	2.72	3.19	6.12
60	89.2	0.70	0.96	1.25	1.58	1.96	2.37	2.82	3.31	6.34
65	86.3	0.73	0.99	1.29	1.64	2.02	2.45	2.91	3.42	6.55
70	83.6	0.75	1.02	1.34	1.69	2.09	2.53	3.01	3.53	6.76
75	81.1	0.77	1.05	1.38	1.74	2.15	2.60	3.10	3.64	6.97
80	78.8	0.80	1.09	1.42	1.79	2.21	2.68	3.19	3.74	7.17

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	79	of	98

Appendix 38 - .175" ACSR / 175mm ACSR (Lynx) - (All altitudes) - 76.2mm Basic
Matched to Table 5 & 6 of CE/C/36 Specification.

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	LYNX
Greased Conductor Weight (kg/m)	0.8626
Cross Sectional Area of Conductor (mm²)	226.2
Conductor Overall Diameter (mm)	19.5
Coefficient of Linear Expansion (/Degree C)	1.8E-05
Modulus of Elasticity (kg/mm²)	8157.73
Rated Breaking Strength of Conductor (kgf)	8137.3
Basic / Recommended Span (m)	76.2
Wind Pressure on Conductor (N/m²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	2312.8
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1627.5
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	2277.1
Maximum Conductor Weight (MCW) (kg/m)			1.653
Maximum Conductor Pressure (MCP) (kg/m)			1.492
Freezing Point Tension (FPT) (kgf)	at	0°C	1772.1

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	LYNX	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	10

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.09	0.27	0.29	0.31	0.32	0.34	0.36
80	0.36	1.07	1.15	1.22	1.30	1.37	1.44
120	0.80	2.41	2.58	2.76	2.92	3.09	3.25
160	1.42	4.28	4.59	4.90	5.20	5.49	5.77
200	2.22	6.69	7.17	7.65	8.12	8.58	9.02
240	3.20	9.63	10.33	11.02	11.69	12.35	12.99
280	4.36	13.11	14.06	15.00	15.92	16.81	17.68
320	5.70	17.12	18.37	19.59	20.79	21.96	23.09
360	7.21	21.67	23.25	24.80	26.31	27.79	29.23
400	8.90	26.75	28.70	30.61	32.48	34.31	36.08
440	10.77	32.37	34.73	37.04	39.31	41.51	43.66
480	12.81	38.52	41.33	44.08	46.78	49.40	51.96
520	15.04	45.21	48.50	51.74	54.90	57.98	60.98

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	80	of	98	

**Appendix 38a - .175" ACSR / 175mm ACSR (Lynx) - (All altitudes) - 76.2mm Basic
Matched to Table 5 & 6 of CE/C/36 Specification.**

Table 6 - Existing conductors re-sagged

Conductor Code Name	LYNX
Basic / Recommended Span (m)	76.2
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	75	80	90	100	110	120	130	180
-5.6	1938.6	0.20	0.31	0.36	0.45	0.56	0.67	0.80	0.94	1.80
0	1772.1	0.22	0.34	0.39	0.49	0.61	0.74	0.88	1.03	1.97
5	1627.5	0.24	0.37	0.42	0.54	0.66	0.80	0.95	1.12	2.15
10	1487.9	0.26	0.41	0.46	0.59	0.72	0.88	1.04	1.22	2.35
15	1354.6	0.29	0.45	0.51	0.64	0.80	0.96	1.15	1.35	2.58
20	1229.2	0.32	0.49	0.56	0.71	0.88	1.06	1.26	1.48	2.84
25	1113.2	0.35	0.54	0.62	0.78	0.97	1.17	1.39	1.64	3.14
30	1007.9	0.39	0.60	0.68	0.87	1.07	1.29	1.54	1.81	3.47
35	914.2	0.42	0.66	0.75	0.96	1.18	1.43	1.70	1.99	3.82
40	832.1	0.47	0.73	0.83	1.05	1.30	1.57	1.87	2.19	4.20
45	761.3	0.51	0.80	0.91	1.15	1.42	1.71	2.04	2.39	4.59
50	700.7	0.55	0.87	0.98	1.25	1.54	1.86	2.22	2.60	4.99
55	648.8	0.60	0.93	1.06	1.35	1.66	2.01	2.39	2.81	5.38
60	604.5	0.64	1.00	1.14	1.44	1.78	2.16	2.57	3.01	5.78
65	566.5	0.69	1.07	1.22	1.54	1.90	2.30	2.74	3.22	6.17
70	533.6	0.73	1.14	1.29	1.64	2.02	2.44	2.91	3.41	6.55
75	505.1	0.77	1.20	1.37	1.73	2.13	2.58	3.07	3.61	6.92
80	480.1	0.81	1.26	1.44	1.82	2.25	2.72	3.23	3.80	7.28

Table 5 - New conductors with 8% overtension

Conductor Code Name	LYNX
Basic / Recommended Span (m)	76.2
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	8.0 at 15°C
Required Percentage Increase in Tension (%)	8 at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	75	80	90	100	110	120	130	180
-5.6	2062.5	0.19	0.29	0.33	0.42	0.52	0.63	0.75	0.88	1.69
0	1893.2	0.21	0.32	0.36	0.46	0.57	0.69	0.82	0.96	1.85
5	1745.3	0.22	0.35	0.40	0.50	0.62	0.75	0.89	1.04	2.00
10	1601.5	0.24	0.38	0.43	0.55	0.67	0.81	0.97	1.14	2.18
15	1463.0	0.27	0.41	0.47	0.60	0.74	0.89	1.06	1.25	2.39
20	1331.1	0.29	0.46	0.52	0.66	0.81	0.98	1.17	1.37	2.62
25	1207.2	0.32	0.50	0.57	0.72	0.89	1.08	1.29	1.51	2.89
30	1093.1	0.36	0.55	0.63	0.80	0.99	1.19	1.42	1.67	3.20
35	989.9	0.39	0.61	0.70	0.88	1.09	1.32	1.57	1.84	3.53
40	898.3	0.43	0.68	0.77	0.97	1.20	1.45	1.73	2.03	3.89
45	818.4	0.47	0.74	0.84	1.07	1.32	1.59	1.90	2.23	4.27
50	749.5	0.52	0.81	0.92	1.17	1.44	1.74	2.07	2.43	4.66
55	690.6	0.56	0.88	1.00	1.26	1.56	1.89	2.25	2.64	5.06
60	640.2	0.61	0.95	1.08	1.36	1.68	2.04	2.43	2.85	5.46
65	597.1	0.65	1.02	1.16	1.46	1.81	2.18	2.60	3.05	5.85
70	560.1	0.69	1.08	1.23	1.56	1.93	2.33	2.77	3.25	6.24
75	528.1	0.74	1.15	1.31	1.65	2.04	2.47	2.94	3.45	6.62
80	500.2	0.78	1.21	1.38	1.75	2.16	2.61	3.10	3.64	6.98

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	81	of	98

Appendix 39 - .1" / 100mm² ACSR (Normal altitude) - 90m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	DOG
Greased Conductor Weight (kg/m)	0.3962
Cross Sectional Area of Conductor (mm ²)	118.5
Conductor Overall Diameter (mm)	14.2
Coefficient of Linear Expansion (/Degree C)	2E-05
Modulus of Elasticity (kg/mm ²)	7699
Rated Breaking Strength of Conductor (kgf)	3329.4
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m ²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1664.7
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	665.9
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1270.9
Maximum Conductor Weight (MCW) (kg/m)			1.042
Maximum Conductor Pressure (MCP) (kg/m)			1.286
Freezing Point Tension (FPT) (kgf)	at	0°C	736.3

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	DOG	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	10

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.10	0.29	0.30	0.32	0.33	0.35	0.36
80	0.39	1.14	1.21	1.27	1.33	1.40	1.45
120	0.87	2.57	2.72	2.86	3.00	3.14	3.27
160	1.55	4.56	4.83	5.09	5.34	5.58	5.82
200	2.42	7.13	7.55	7.95	8.34	8.72	9.09
240	3.48	10.27	10.87	11.45	12.01	12.56	13.09
280	4.73	13.98	14.79	15.58	16.35	17.10	17.82
320	6.18	18.26	19.32	20.35	21.36	22.33	23.28
360	7.83	23.11	24.45	25.76	27.03	28.26	29.46
400	9.66	28.53	30.19	31.80	33.37	34.89	36.37
440	11.69	34.52	36.53	38.48	40.38	42.22	44.01
480	13.91	41.08	43.47	45.80	48.05	50.25	52.37
520	16.33	48.22	51.02	53.75	56.40	58.97	61.47

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	82	of	98	

Appendix 39a - .1" / 100mm2 ACSR (Normal altitude) - 90m Basic

Conductor Code Name	DOG
Basic / Recommended Span (m)	90
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	820.2	0.22	0.30	0.39	0.49	0.60	0.73	0.87	1.02	1.96
0	736.3	0.24	0.33	0.43	0.54	0.67	0.81	0.97	1.14	2.18
5	665.9	0.27	0.36	0.48	0.60	0.74	0.90	1.07	1.26	2.41
10	600.6	0.30	0.40	0.53	0.67	0.82	1.00	1.19	1.39	2.67
15	541.2	0.33	0.45	0.59	0.74	0.92	1.11	1.32	1.55	2.96
20	488.5	0.36	0.50	0.65	0.82	1.01	1.23	1.46	1.71	3.28
25	442.5	0.40	0.55	0.72	0.91	1.12	1.35	1.61	1.89	3.63
30	402.9	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	3.98
35	369.3	0.48	0.66	0.86	1.09	1.34	1.62	1.93	2.27	4.34
40	340.8	0.52	0.71	0.93	1.18	1.45	1.76	2.09	2.46	4.71
45	316.6	0.56	0.77	1.00	1.27	1.56	1.89	2.25	2.64	5.07
50	295.9	0.60	0.82	1.07	1.36	1.67	2.03	2.41	2.83	5.42
55	278.2	0.64	0.87	1.14	1.44	1.78	2.15	2.56	3.01	5.77
60	262.9	0.68	0.92	1.21	1.53	1.88	2.28	2.71	3.18	6.10
65	249.5	0.71	0.97	1.27	1.61	1.99	2.40	2.86	3.35	6.43
70	237.8	0.75	1.02	1.33	1.69	2.08	2.52	3.00	3.52	6.75
75	227.4	0.78	1.07	1.39	1.76	2.18	2.64	3.14	3.68	7.06
80	218.1	0.82	1.11	1.45	1.84	2.27	2.75	3.27	3.84	7.36

Conductor Code Name	DOG
Basic / Recommended Span (m)	90
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	10.0 at 15°C
Required Percentage Increase in Tension (%)	10 at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	891.9	0.20	0.27	0.36	0.45	0.56	0.67	0.80	0.94	1.80
0	804.6	0.22	0.30	0.39	0.50	0.62	0.74	0.89	1.04	1.99
5	730.3	0.24	0.33	0.43	0.55	0.68	0.82	0.98	1.15	2.20
10	660.2	0.27	0.37	0.48	0.61	0.75	0.91	1.08	1.27	2.43
15	595.3	0.30	0.41	0.53	0.67	0.83	1.01	1.20	1.41	2.70
20	536.6	0.33	0.45	0.59	0.75	0.92	1.12	1.33	1.56	2.99
25	484.4	0.37	0.50	0.65	0.83	1.02	1.24	1.47	1.73	3.31
30	438.9	0.41	0.55	0.72	0.91	1.13	1.37	1.62	1.91	3.66
35	399.9	0.45	0.61	0.79	1.00	1.24	1.50	1.78	2.09	4.01
40	366.7	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	4.38
45	338.6	0.53	0.72	0.94	1.18	1.46	1.77	2.11	2.47	4.74
50	314.7	0.57	0.77	1.01	1.27	1.57	1.90	2.27	2.66	5.10
55	294.3	0.61	0.82	1.08	1.36	1.68	2.04	2.42	2.84	5.45
60	276.8	0.64	0.88	1.14	1.45	1.79	2.16	2.58	3.02	5.80
65	261.7	0.68	0.93	1.21	1.53	1.89	2.29	2.73	3.20	6.13
70	248.5	0.72	0.98	1.28	1.61	1.99	2.41	2.87	3.37	6.46
75	236.8	0.75	1.02	1.34	1.69	2.09	2.53	3.01	3.53	6.78
80	226.5	0.79	1.07	1.40	1.77	2.19	2.65	3.15	3.69	7.08

Document reference	NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	83	of	98

Appendix 40 - .1" / 100mm² ACSR (Normal altitude) - 100m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	DOG
Greased Conductor Weight (kg/m)	0.3962
Cross Sectional Area of Conductor (mm ²)	118.5
Conductor Overall Diameter (mm)	14.2
Coefficient of Linear Expansion (/Degree C)	2E-05
Modulus of Elasticity (kg/mm ²)	7699
Rated Breaking Strength of Conductor (kgf)	3329.4
Basic / Recommended Span (m)	100
Wind Pressure on Conductor (N/m ²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1664.7
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	665.9
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1320.4
Maximum Conductor Weight (MCW) (kg/m)			1.042
Maximum Conductor Pressure (MCP) (kg/m)			1.286
Freezing Point Tension (FPT) (kgf)	at	0°C	732.8

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	DOG	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	10

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.10	0.27	0.28	0.30	0.31	0.32	0.34
80	0.39	1.07	1.13	1.18	1.24	1.29	1.34
120	0.88	2.41	2.54	2.66	2.78	2.90	3.02
160	1.56	4.28	4.51	4.73	4.95	5.16	5.37
200	2.44	6.68	7.04	7.39	7.74	8.07	8.39
240	3.51	9.63	10.14	10.65	11.14	11.62	12.08
280	4.78	13.10	13.81	14.49	15.16	15.81	16.45
320	6.24	17.11	18.03	18.93	19.80	20.65	21.48
360	7.89	21.66	22.82	23.96	25.06	26.14	27.19
400	9.75	26.74	28.18	29.58	30.94	32.27	33.56
440	11.79	32.35	34.09	35.79	37.44	39.05	40.61
480	14.03	38.50	40.57	42.59	44.56	46.47	48.33
520	16.47	45.18	47.62	49.99	52.30	54.54	56.72

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	84	of	98	

Appendix 40a - .1" / 100mm² ACSR (Normal altitude) - 100m Basic

Conductor Code Name	DOG
Basic / Recommended Span (m)	100
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0 at 15°C
Required Percentage Increase in Tension (%)	0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	813.0	0.22	0.30	0.39	0.49	0.61	0.74	0.88	1.03	1.97
0	732.8	0.24	0.33	0.43	0.55	0.68	0.82	0.97	1.14	2.19
5	665.9	0.27	0.36	0.48	0.60	0.74	0.90	1.07	1.26	2.41
10	604.4	0.30	0.40	0.52	0.66	0.82	0.99	1.18	1.38	2.66
15	548.8	0.32	0.44	0.58	0.73	0.90	1.09	1.30	1.53	2.92
20	499.5	0.36	0.49	0.63	0.80	0.99	1.20	1.43	1.68	3.21
25	456.4	0.39	0.53	0.69	0.88	1.09	1.31	1.56	1.83	3.52
30	419.2	0.43	0.58	0.76	0.96	1.18	1.43	1.70	2.00	3.83
35	387.3	0.46	0.63	0.82	1.04	1.28	1.55	1.84	2.16	4.14
40	359.9	0.50	0.67	0.88	1.11	1.38	1.67	1.98	2.33	4.46
45	336.3	0.53	0.72	0.94	1.19	1.47	1.78	2.12	2.49	4.77
50	316.0	0.56	0.77	1.00	1.27	1.57	1.90	2.26	2.65	5.08
55	298.4	0.60	0.81	1.06	1.34	1.66	2.01	2.39	2.81	5.38
60	283.0	0.63	0.86	1.12	1.42	1.75	2.12	2.52	2.96	5.67
65	269.4	0.66	0.90	1.18	1.49	1.84	2.22	2.65	3.11	5.96
70	257.4	0.69	0.94	1.23	1.56	1.92	2.33	2.77	3.25	6.23
75	246.8	0.72	0.98	1.28	1.63	2.01	2.43	2.89	3.39	6.50
80	237.2	0.75	1.02	1.34	1.69	2.09	2.53	3.01	3.53	6.77

Conductor Code Name	DOG
Basic / Recommended Span (m)	100
Temperature Shift for Creep (Deg. C)	0 Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	10.0 at 15°C
Required Percentage Increase in Tension (%)	10 at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	887.8	0.20	0.27	0.36	0.45	0.56	0.68	0.80	0.94	1.81
0	803.3	0.22	0.30	0.39	0.50	0.62	0.75	0.89	1.04	2.00
5	731.9	0.24	0.33	0.43	0.55	0.68	0.82	0.97	1.14	2.19
10	665.1	0.27	0.36	0.48	0.60	0.74	0.90	1.07	1.26	2.41
15	603.7	0.30	0.40	0.53	0.66	0.82	0.99	1.18	1.39	2.66
20	548.2	0.33	0.44	0.58	0.73	0.90	1.09	1.30	1.53	2.93
25	499.0	0.36	0.49	0.64	0.80	0.99	1.20	1.43	1.68	3.22
30	456.0	0.39	0.53	0.70	0.88	1.09	1.31	1.56	1.84	3.52
35	418.8	0.43	0.58	0.76	0.96	1.18	1.43	1.70	2.00	3.83
40	386.9	0.46	0.63	0.82	1.04	1.28	1.55	1.84	2.16	4.15
45	359.6	0.50	0.67	0.88	1.12	1.38	1.67	1.98	2.33	4.46
50	336.1	0.53	0.72	0.94	1.19	1.47	1.78	2.12	2.49	4.77
55	315.8	0.56	0.77	1.00	1.27	1.57	1.90	2.26	2.65	5.08
60	298.2	0.60	0.81	1.06	1.35	1.66	2.01	2.39	2.81	5.38
65	282.8	0.63	0.86	1.12	1.42	1.75	2.12	2.52	2.96	5.67
70	269.3	0.66	0.90	1.18	1.49	1.84	2.23	2.65	3.11	5.96
75	257.3	0.69	0.94	1.23	1.56	1.92	2.33	2.77	3.25	6.24
80	246.6	0.72	0.98	1.29	1.63	2.01	2.43	2.89	3.39	6.51

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	85	of	98

Appendix 41 - .1" / 70mm² HDBC (Normal altitude) - 90m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	COPPER 70
Greased Conductor Weight (kg/m)	0.621
Cross Sectional Area of Conductor (mm ²)	69.28
Conductor Overall Diameter (mm)	10.65
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.3
Rated Breaking Strength of Conductor (kgf)	2740.9
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m ²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1370.5
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	913.6
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1363.1
Maximum Conductor Weight (MCW) (kg/m)			1.170
Maximum Conductor Pressure (MCP) (kg/m)			1.149
Freezing Point Tension (FPT) (kgf)	at	0°C	972.2

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor: COPPER 70	Temperature Shift (Deg. C)	0
	Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.12	0.25	0.26	0.27	0.29	0.30	0.31
80	0.48	0.99	1.04	1.10	1.15	1.20	1.25
120	1.07	2.23	2.35	2.47	2.58	2.70	2.81
160	1.91	3.96	4.17	4.38	4.59	4.80	5.00
200	2.98	6.20	6.52	6.85	7.18	7.50	7.82
240	4.30	8.92	9.39	9.86	10.33	10.80	11.25
280	5.85	12.14	12.78	13.43	14.06	14.69	15.32
320	7.64	15.86	16.70	17.54	18.37	19.19	20.01
360	9.67	20.07	21.13	22.19	23.25	24.29	25.32
400	11.94	24.78	26.09	27.40	28.70	29.99	31.26
440	14.45	29.98	31.57	33.15	34.73	36.29	37.83
480	17.19	35.68	37.57	39.46	41.33	43.18	45.02
520	20.18	41.88	44.09	46.31	48.50	50.68	52.83

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	86	of	98	

Appendix 41a - .1" / 70mm² HDBC (Normal altitude) - 90m Basic

Common Design & Erection Chart

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	COPPER 70	
Basic / Recommended Span (m)	90	
Temperature Shift for Creep (Deg. C)	0	Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0	at 15°C
Required Percentage Increase in Tension (%)	0	at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	1040.4	0.27	0.37	0.48	0.60	0.75	0.90	1.07	1.26	2.42
0	972.2	0.29	0.39	0.51	0.65	0.80	0.97	1.15	1.35	2.59
5	913.6	0.31	0.42	0.54	0.69	0.85	1.03	1.22	1.44	2.75
10	857.6	0.33	0.44	0.58	0.73	0.91	1.10	1.30	1.53	2.93
15	804.3	0.35	0.47	0.62	0.78	0.97	1.17	1.39	1.63	3.13
20	754.0	0.37	0.50	0.66	0.83	1.03	1.25	1.48	1.74	3.34
25	707.1	0.40	0.54	0.70	0.89	1.10	1.33	1.58	1.86	3.56
30	663.5	0.42	0.57	0.75	0.95	1.17	1.42	1.68	1.98	3.79
35	623.4	0.45	0.61	0.80	1.01	1.25	1.51	1.79	2.10	4.03
40	586.7	0.48	0.65	0.85	1.07	1.32	1.60	1.91	2.24	4.29
45	553.4	0.50	0.69	0.90	1.14	1.40	1.70	2.02	2.37	4.54
50	523.2	0.53	0.73	0.95	1.20	1.48	1.80	2.14	2.51	4.81
55	495.8	0.56	0.77	1.00	1.27	1.57	1.89	2.25	2.65	5.07
60	471.2	0.59	0.81	1.05	1.33	1.65	1.99	2.37	2.78	5.34
65	448.9	0.62	0.85	1.11	1.40	1.73	2.09	2.49	2.92	5.60
70	428.8	0.65	0.89	1.16	1.47	1.81	2.19	2.61	3.06	5.87
75	410.6	0.68	0.93	1.21	1.53	1.89	2.29	2.72	3.20	6.13
80	394.0	0.71	0.97	1.26	1.60	1.97	2.38	2.84	3.33	6.38

Document reference	NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	87	of	98

Appendix 42 - .15" / 100mm² HDBC (Normal altitude) - 90m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	HDBC 100
Greased Conductor Weight (kg/m)	0.911
Cross Sectional Area of Conductor (mm ²)	101.654
Conductor Overall Diameter (mm)	12.9
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.3
Rated Breaking Strength of Conductor (kgf)	3838.2
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m ²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1814.4
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1279.4
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1803.8
Maximum Conductor Weight (MCW) (kg/m)			1.522
Maximum Conductor Pressure (MCP) (kg/m)			1.236
Freezing Point Tension (FPT) (kgf)	at	0°C	1362.7

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	HDBC 100	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.12	0.26	0.27	0.28	0.30	0.31	0.32
80	0.50	1.03	1.08	1.13	1.19	1.24	1.29
120	1.12	2.32	2.44	2.55	2.67	2.78	2.90
160	2.00	4.12	4.33	4.54	4.74	4.95	5.15
200	3.12	6.44	6.76	7.09	7.41	7.73	8.05
240	4.49	9.27	9.74	10.21	10.68	11.14	11.59
280	6.12	12.62	13.26	13.90	14.53	15.16	15.77
320	7.99	16.48	17.32	18.15	18.98	19.80	20.60
360	10.11	20.86	21.92	22.97	24.02	25.05	26.08
400	12.48	25.75	27.06	28.36	29.66	30.93	32.19
440	15.10	31.16	32.74	34.32	35.88	37.43	38.95
480	17.97	37.08	38.97	40.84	42.70	44.54	46.36
520	21.09	43.52	45.73	47.93	50.12	52.28	54.40

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-		Sept 10	Page		88 of 98

Appendix 42a - .15" / 100mm² HDBC (Normal altitude) - 90m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name	HDBC 100	
Basic / Recommended Span (m)	90	
Temperature Shift for Creep (Deg. C)	0	Insert minus sign as necessary
Equivalent Percentage Increase in Tension (%)	0.0	at 15°C
Required Percentage Increase in Tension (%)	0	at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	1460.1	0.28	0.38	0.50	0.63	0.78	0.94	1.12	1.32	2.53
0	1362.7	0.30	0.41	0.53	0.68	0.84	1.01	1.20	1.41	2.71
5	1279.4	0.32	0.44	0.57	0.72	0.89	1.08	1.28	1.50	2.88
10	1200.1	0.34	0.47	0.61	0.77	0.95	1.15	1.37	1.60	3.08
15	1125.2	0.36	0.50	0.65	0.82	1.01	1.22	1.46	1.71	3.28
20	1055.0	0.39	0.53	0.69	0.87	1.08	1.31	1.55	1.82	3.50
25	989.8	0.41	0.56	0.74	0.93	1.15	1.39	1.66	1.94	3.73
30	929.6	0.44	0.60	0.78	0.99	1.23	1.48	1.76	2.07	3.97
35	874.5	0.47	0.64	0.83	1.06	1.30	1.58	1.88	2.20	4.22
40	824.3	0.50	0.68	0.88	1.12	1.38	1.67	1.99	2.34	4.48
45	778.8	0.53	0.72	0.94	1.18	1.46	1.77	2.11	2.47	4.74
50	737.6	0.56	0.76	0.99	1.25	1.54	1.87	2.22	2.61	5.00
55	700.4	0.59	0.80	1.04	1.32	1.63	1.97	2.34	2.75	5.27
60	666.9	0.61	0.84	1.09	1.38	1.71	2.07	2.46	2.89	5.53
65	636.6	0.64	0.88	1.15	1.45	1.79	2.16	2.58	3.02	5.80
70	609.1	0.67	0.92	1.20	1.51	1.87	2.26	2.69	3.16	6.06
75	584.3	0.70	0.96	1.25	1.58	1.95	2.36	2.81	3.29	6.32
80	561.7	0.73	0.99	1.30	1.64	2.03	2.45	2.92	3.43	6.57

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	89	of	98

Appendix 43 - .2" / 125mm² HDBC (Normal altitude) - 100m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	HDBC 125
Greased Conductor Weight (kg/m)	1.130
Cross Sectional Area of Conductor (mm ²)	125.499
Conductor Overall Diameter (mm)	14.5
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.29
Rated Breaking Strength of Conductor (kgf)	4929.3
Basic / Recommended Span (m)	100
Wind Pressure on Conductor (N/m ²)	380
Radial Ice Thickness (mm)	9.5
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1814.4
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1643.1
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1814.4
Maximum Conductor Weight (MCW) (kg/m)			1.784
Maximum Conductor Pressure (MCP) (kg/m)			1.298
Freezing Point Tension (FPT) (kgf)	at	0°C	1237.0

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	HDBC 125	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.17	0.31	0.32	0.33	0.34	0.35	0.36
80	0.68	1.22	1.26	1.31	1.35	1.39	1.43
120	1.54	2.75	2.84	2.94	3.03	3.12	3.21
160	2.74	4.88	5.05	5.22	5.39	5.55	5.71
200	4.28	7.63	7.90	8.16	8.42	8.68	8.93
240	6.16	10.99	11.37	11.75	12.13	12.49	12.85
280	8.39	14.96	15.48	16.00	16.50	17.00	17.49
320	10.95	19.53	20.22	20.89	21.56	22.21	22.85
360	13.86	24.72	25.59	26.44	27.28	28.11	28.92
400	17.11	30.52	31.59	32.64	33.68	34.70	35.70
440	20.71	36.93	38.23	39.50	40.75	41.99	43.20
480	24.65	43.95	45.49	47.01	48.50	49.97	51.41
520	28.92	51.58	53.39	55.17	56.92	58.64	60.34

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	90	of	98	

Appendix 43a - .2" / 125mm² HDBC (Normal altitude) - 100m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name HDBC 125
 Basic / Recommended Span (m) 100
 Temperature Shift for Creep (Deg. C) 0 Insert minus sign as necessary
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C
 Required Percentage Increase in Tension (%) 0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	1320.5	0.39	0.52	0.68	0.87	1.07	1.29	1.54	1.81	3.47
0	1237.0	0.41	0.56	0.73	0.92	1.14	1.38	1.64	1.93	3.70
5	1168.5	0.44	0.59	0.77	0.98	1.21	1.46	1.74	2.04	3.92
10	1105.8	0.46	0.63	0.82	1.03	1.28	1.55	1.84	2.16	4.14
15	1048.4	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	4.37
20	996.2	0.51	0.69	0.91	1.15	1.42	1.72	2.04	2.40	4.59
25	948.6	0.54	0.73	0.95	1.21	1.49	1.80	2.14	2.52	4.82
30	905.4	0.56	0.76	1.00	1.26	1.56	1.89	2.25	2.64	5.05
35	866.1	0.59	0.80	1.04	1.32	1.63	1.97	2.35	2.76	5.28
40	830.3	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.87	5.51
45	797.7	0.64	0.87	1.13	1.43	1.77	2.14	2.55	2.99	5.74
50	767.9	0.66	0.90	1.18	1.49	1.84	2.23	2.65	3.11	5.96
55	740.5	0.69	0.93	1.22	1.54	1.91	2.31	2.75	3.22	6.18
60	715.5	0.71	0.97	1.26	1.60	1.97	2.39	2.84	3.34	6.40
65	692.4	0.73	1.00	1.31	1.65	2.04	2.47	2.94	3.45	6.61
70	671.1	0.76	1.03	1.35	1.70	2.10	2.55	3.03	3.56	6.82
75	651.4	0.78	1.06	1.39	1.76	2.17	2.62	3.12	3.66	7.03
80	633.1	0.80	1.09	1.43	1.81	2.23	2.70	3.21	3.77	7.23

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	91	of	98

Appendix 44 - .1" / 100mm² ACSR (High altitude) - 90m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	DOG
Greased Conductor Weight (kg/m)	0.3962
Cross Sectional Area of Conductor (mm ²)	118.5
Conductor Overall Diameter (mm)	14.2
Coefficient of Linear Expansion (/Degree C)	2E-05
Modulus of Elasticity (kg/mm ²)	7699
Rated Breaking Strength of Conductor (kgf)	3329.4
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m ²)	570
Radial Ice Thickness (mm)	12.5
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1664.7
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	665.9
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1596.6
Maximum Conductor Weight (MCW) (kg/m)			1.353
Maximum Conductor Pressure (MCP) (kg/m)			2.278
Freezing Point Tension (FPT) (kgf)	at	0°C	736.3

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	DOG	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	10

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.10	0.29	0.30	0.32	0.33	0.35	0.36
80	0.39	1.14	1.21	1.27	1.33	1.40	1.45
120	0.87	2.57	2.72	2.86	3.00	3.14	3.27
160	1.55	4.56	4.83	5.09	5.34	5.58	5.82
200	2.42	7.13	7.55	7.95	8.34	8.72	9.09
240	3.48	10.27	10.87	11.45	12.01	12.56	13.09
280	4.73	13.98	14.79	15.58	16.35	17.10	17.82
320	6.18	18.26	19.32	20.35	21.36	22.33	23.28
360	7.83	23.11	24.45	25.76	27.03	28.26	29.46
400	9.66	28.53	30.19	31.80	33.37	34.89	36.37
440	11.69	34.52	36.53	38.48	40.38	42.22	44.01
480	13.91	41.08	43.47	45.80	48.05	50.25	52.37
520	16.33	48.22	51.02	53.75	56.40	58.97	61.47

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	92	of	98	

Appendix 44a - .1" / 100mm2 ACSR (High altitude) - 90m Basic

Conductor Code Name DOG
 Basic / Recommended Span (m) 90
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C
 Required Percentage Increase in Tension (%) 0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	820.2	0.22	0.30	0.39	0.49	0.60	0.73	0.87	1.02	1.96
0	736.3	0.24	0.33	0.43	0.54	0.67	0.81	0.97	1.14	2.18
5	665.9	0.27	0.36	0.48	0.60	0.74	0.90	1.07	1.26	2.41
10	600.6	0.30	0.40	0.53	0.67	0.82	1.00	1.19	1.39	2.67
15	541.2	0.33	0.45	0.59	0.74	0.92	1.11	1.32	1.55	2.96
20	488.5	0.36	0.50	0.65	0.82	1.01	1.23	1.46	1.71	3.28
25	442.5	0.40	0.55	0.72	0.91	1.12	1.35	1.61	1.89	3.63
30	402.9	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	3.98
35	369.3	0.48	0.66	0.86	1.09	1.34	1.62	1.93	2.27	4.34
40	340.8	0.52	0.71	0.93	1.18	1.45	1.76	2.09	2.46	4.71
45	316.6	0.56	0.77	1.00	1.27	1.56	1.89	2.25	2.64	5.07
50	295.9	0.60	0.82	1.07	1.36	1.67	2.03	2.41	2.83	5.42
55	278.2	0.64	0.87	1.14	1.44	1.78	2.15	2.56	3.01	5.77
60	262.9	0.68	0.92	1.21	1.53	1.88	2.28	2.71	3.18	6.10
65	249.5	0.71	0.97	1.27	1.61	1.99	2.40	2.86	3.35	6.43
70	237.8	0.75	1.02	1.33	1.69	2.08	2.52	3.00	3.52	6.75
75	227.4	0.78	1.07	1.39	1.76	2.18	2.64	3.14	3.68	7.06
80	218.1	0.82	1.11	1.45	1.84	2.27	2.75	3.27	3.84	7.36

Conductor Code Name DOG
 Basic / Recommended Span (m) 90
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**
 Equivalent Percentage Increase in Tension (%) 10.0 at 15°C
 Required Percentage Increase in Tension (%) 10 at 15°C

Temp. (Deg. C)	Tension (kgf)	ERECTION TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	891.9	0.20	0.27	0.36	0.45	0.56	0.67	0.80	0.94	1.80
0	804.6	0.22	0.30	0.39	0.50	0.62	0.74	0.89	1.04	1.99
5	730.3	0.24	0.33	0.43	0.55	0.68	0.82	0.98	1.15	2.20
10	660.2	0.27	0.37	0.48	0.61	0.75	0.91	1.08	1.27	2.43
15	595.3	0.30	0.41	0.53	0.67	0.83	1.01	1.20	1.41	2.70
20	536.6	0.33	0.45	0.59	0.75	0.92	1.12	1.33	1.56	2.99
25	484.4	0.37	0.50	0.65	0.83	1.02	1.24	1.47	1.73	3.31
30	438.9	0.41	0.55	0.72	0.91	1.13	1.37	1.62	1.91	3.66
35	399.9	0.45	0.61	0.79	1.00	1.24	1.50	1.78	2.09	4.01
40	366.7	0.49	0.66	0.86	1.09	1.35	1.63	1.94	2.28	4.38
45	338.6	0.53	0.72	0.94	1.18	1.46	1.77	2.11	2.47	4.74
50	314.7	0.57	0.77	1.01	1.27	1.57	1.90	2.27	2.66	5.10
55	294.3	0.61	0.82	1.08	1.36	1.68	2.04	2.42	2.84	5.45
60	276.8	0.64	0.88	1.14	1.45	1.79	2.16	2.58	3.02	5.80
65	261.7	0.68	0.93	1.21	1.53	1.89	2.29	2.73	3.20	6.13
70	248.5	0.72	0.98	1.28	1.61	1.99	2.41	2.87	3.37	6.46
75	236.8	0.75	1.02	1.34	1.69	2.09	2.53	3.01	3.53	6.78
80	226.5	0.79	1.07	1.40	1.77	2.19	2.65	3.15	3.69	7.08

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	93	of	98

Appendix 45 - .1" / 70mm² HDBC (High altitude) - 90m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	COPPER 70
Greased Conductor Weight (kg/m)	0.621
Cross Sectional Area of Conductor (mm ²)	69.28
Conductor Overall Diameter (mm)	10.65
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.3
Rated Breaking Strength of Conductor (kgf)	2740.9
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m ²)	570
Radial Ice Thickness (mm)	12.5
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1370.5
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	913.6
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1370.5
Maximum Conductor Weight (MCW) (kg/m)			1.451
Maximum Conductor Pressure (MCP) (kg/m)			2.072
Freezing Point Tension (FPT) (kgf)	at	0°C	597.9

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor: COPPER 70	Temperature Shift (Deg. C)	0
	Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.19	0.35	0.36	0.37	0.38	0.39	0.41
80	0.78	1.39	1.44	1.49	1.53	1.58	1.62
120	1.75	3.13	3.24	3.35	3.45	3.55	3.65
160	3.10	5.57	5.76	5.95	6.14	6.32	6.49
200	4.85	8.70	9.00	9.30	9.59	9.87	10.15
240	6.98	12.53	12.97	13.39	13.80	14.21	14.61
280	9.50	17.06	17.65	18.22	18.79	19.34	19.89
320	12.41	22.28	23.05	23.80	24.54	25.27	25.98
360	15.71	28.20	29.17	30.12	31.06	31.98	32.88
400	19.39	34.82	36.02	37.19	38.34	39.48	40.59
440	23.46	42.13	43.58	45.00	46.40	47.77	49.11
480	27.93	50.14	51.86	53.55	55.22	56.85	58.45
520	32.77	58.84	60.87	62.85	64.80	66.72	68.60

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	94	of	98	

Appendix 45a - .1" / 70mm² HDBC (High altitude) - 90m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name **COPPER 70**
 Basic / Recommended Span (m) **90**
 Temperature Shift for Creep (Deg. C) **0** **Insert minus sign as necessary**
 Equivalent Percentage Increase in Tension (%) **0.0** **at 15°C**
 Required Percentage Increase in Tension (%) **0** **at 15°C**

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	640.5	0.44	0.59	0.78	0.98	1.21	1.47	1.75	2.05	3.93
0	597.9	0.47	0.64	0.83	1.05	1.30	1.57	1.87	2.19	4.21
5	563.6	0.50	0.67	0.88	1.12	1.38	1.67	1.98	2.33	4.46
10	532.4	0.52	0.71	0.93	1.18	1.46	1.76	2.10	2.46	4.72
15	504.2	0.55	0.75	0.99	1.25	1.54	1.86	2.22	2.60	4.99
20	478.7	0.58	0.79	1.04	1.31	1.62	1.96	2.34	2.74	5.25
25	455.7	0.61	0.83	1.09	1.38	1.70	2.06	2.45	2.88	5.52
30	434.9	0.64	0.87	1.14	1.45	1.78	2.16	2.57	3.02	5.78
35	416.1	0.67	0.91	1.19	1.51	1.87	2.26	2.69	3.15	6.04
40	399.1	0.70	0.95	1.24	1.58	1.95	2.35	2.80	3.29	6.30
45	383.6	0.73	0.99	1.30	1.64	2.02	2.45	2.91	3.42	6.56
50	369.5	0.76	1.03	1.34	1.70	2.10	2.54	3.03	3.55	6.81
55	356.6	0.78	1.07	1.39	1.76	2.18	2.63	3.13	3.68	7.05
60	344.7	0.81	1.10	1.44	1.82	2.25	2.72	3.24	3.81	7.30
65	333.8	0.84	1.14	1.49	1.88	2.33	2.81	3.35	3.93	7.53
70	323.8	0.86	1.17	1.53	1.94	2.40	2.90	3.45	4.05	7.77
75	314.5	0.89	1.21	1.58	2.00	2.47	2.99	3.55	4.17	8.00
80	305.9	0.91	1.24	1.62	2.06	2.54	3.07	3.65	4.29	8.22

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	95	of	98

Appendix 46 - .15" / 100mm² HDBC (High altitude) - 90m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	HDBC 100
Greased Conductor Weight (kg/m)	0.911
Cross Sectional Area of Conductor (mm ²)	101.654
Conductor Overall Diameter (mm)	12.9
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.3
Rated Breaking Strength of Conductor (kgf)	3838.2
Basic / Recommended Span (m)	90
Wind Pressure on Conductor (N/m ²)	570
Radial Ice Thickness (mm)	12.5
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1814.4
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1279.4
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1814.4
Maximum Conductor Weight (MCW) (kg/m)			1.822
Maximum Conductor Pressure (MCP) (kg/m)			2.203
Freezing Point Tension (FPT) (kgf)	at	0°C	985.4

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	HDBC 100	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.17	0.32	0.34	0.35	0.36	0.37	0.38
80	0.69	1.30	1.35	1.39	1.44	1.49	1.54
120	1.55	2.92	3.03	3.14	3.25	3.35	3.46
160	2.75	5.19	5.38	5.58	5.77	5.96	6.14
200	4.30	8.10	8.41	8.72	9.02	9.31	9.60
240	6.20	11.67	12.11	12.55	12.98	13.41	13.82
280	8.44	15.88	16.49	17.08	17.67	18.25	18.81
320	11.02	20.74	21.54	22.31	23.08	23.83	24.57
360	13.95	26.26	27.26	28.24	29.21	30.16	31.10
400	17.22	32.41	33.65	34.87	36.06	37.24	38.39
440	20.83	39.22	40.72	42.19	43.64	45.06	46.45
480	24.79	46.68	48.46	50.21	51.93	53.62	55.28
520	29.10	54.78	56.87	58.93	60.95	62.93	64.88

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	96	of	98	

Appendix 46a - .15" / 100mm² HDBC (High altitude) - 90m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name HDBC 100
 Basic / Recommended Span (m) 90
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C
 Required Percentage Increase in Tension (%) 0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	1058.4	0.39	0.53	0.69	0.87	1.08	1.30	1.55	1.82	3.49
0	985.4	0.42	0.57	0.74	0.94	1.16	1.40	1.66	1.95	3.75
5	925.6	0.44	0.60	0.79	1.00	1.23	1.49	1.77	2.08	3.99
10	870.8	0.47	0.64	0.84	1.06	1.31	1.58	1.88	2.21	4.24
15	820.9	0.50	0.68	0.89	1.12	1.39	1.68	2.00	2.34	4.50
20	775.7	0.53	0.72	0.94	1.19	1.47	1.78	2.11	2.48	4.76
25	734.9	0.56	0.76	0.99	1.26	1.55	1.88	2.23	2.62	5.02
30	698.0	0.59	0.80	1.04	1.32	1.63	1.97	2.35	2.76	5.29
35	664.7	0.62	0.84	1.10	1.39	1.71	2.07	2.47	2.90	5.55
40	634.6	0.65	0.88	1.15	1.45	1.79	2.17	2.58	3.03	5.82
45	607.3	0.68	0.92	1.20	1.52	1.88	2.27	2.70	3.17	6.08
50	582.6	0.70	0.96	1.25	1.58	1.95	2.37	2.82	3.30	6.33
55	560.2	0.73	1.00	1.30	1.65	2.03	2.46	2.93	3.44	6.59
60	539.7	0.76	1.03	1.35	1.71	2.11	2.55	3.04	3.57	6.84
65	520.9	0.79	1.07	1.40	1.77	2.19	2.65	3.15	3.70	7.08
70	503.8	0.81	1.11	1.45	1.83	2.26	2.74	3.26	3.82	7.33
75	487.9	0.84	1.14	1.49	1.89	2.33	2.82	3.36	3.94	7.56
80	473.4	0.87	1.18	1.54	1.95	2.41	2.91	3.46	4.07	7.80

Document reference		NSP/004/042/001					
Version:-	2.0	Date of Issue:-	Sept 10	Page	97	of	98

Appendix 47 - .2" / 125mm² HDBC (High altitude) - 100m Basic

CONDUCTOR SAG/TENSION PARAMETERS for DESIGN

Conductor Code Name (if any)	HDBC 125
Greased Conductor Weight (kg/m)	1.130
Cross Sectional Area of Conductor (mm ²)	125.499
Conductor Overall Diameter (mm)	14.5
Coefficient of Linear Expansion (/Degree C)	1.70E-05
Modulus of Elasticity (kg/mm ²)	12655.29
Rated Breaking Strength of Conductor (kgf)	4929.3
Basic / Recommended Span (m)	100
Wind Pressure on Conductor (N/m ²)	570
Radial Ice Thickness (mm)	12.5
Ice Density (kg/m ³)	913
Absolute Maximum Working Tension (MWT) Limit (kgf)	1814.4
Temperature at MWT Limit (Degrees C)	-5.6
Maximum "Everyday" Tension (EDT) Limit (kgf)	1643.1
Temperature at EDT Limit (Degrees C)	5

Maximum Conductor Tension (MCT) (kgf)	at	-5.6°C	1814.4
Maximum Conductor Weight (MCW) (kg/m)			2.098
Maximum Conductor Pressure (MCP) (kg/m)			2.296
Freezing Point Tension (FPT) (kgf)	at	0°C	860.3

ORDINATES FOR PLOTTING UPLIFT AND GROUND CLEARANCE CURVES

Conductor:	HDBC 125	Temperature Shift (Deg. C)	0
		Tension Reduction at 15°C (%)	5

2X Span (m)	Sag (m) for Cold Curve -5.6	Sag (m) for Range of Hot Curve Temperatures					
		50	55	60	65	70	75
40	0.25	0.38	0.39	0.40	0.41	0.42	0.43
80	1.00	1.53	1.57	1.60	1.64	1.68	1.71
120	2.25	3.44	3.53	3.61	3.69	3.77	3.85
160	4.00	6.12	6.27	6.42	6.56	6.71	6.85
200	6.25	9.56	9.79	10.02	10.25	10.48	10.70
240	9.00	13.76	14.10	14.43	14.76	15.09	15.40
280	12.25	18.73	19.19	19.65	20.09	20.53	20.97
320	16.00	24.46	25.07	25.66	26.25	26.82	27.39
360	20.25	30.96	31.73	32.48	33.22	33.94	34.66
400	25.00	38.22	39.17	40.10	41.01	41.91	42.79
440	30.25	46.25	47.39	48.52	49.62	50.71	51.78
480	36.00	55.04	56.40	57.74	59.05	60.35	61.62
520	42.25	64.60	66.19	67.76	69.30	70.82	72.32

Document reference		NSP/004/042/001						
Version:-	2.0	Date of Issue:-	Sept 10	Page	98	of	98	

Appendix 47a - .2" / 125mm² HDBC (High altitude) - 100m Basic

DESIGN and ERECTION SAG/ TENSION TABLE

Conductor Code Name HDBC 125
 Basic / Recommended Span (m) 100
 Temperature Shift for Creep (Deg. C) 0 **Insert minus sign as necessary**
 Equivalent Percentage Increase in Tension (%) 0.0 at 15°C
 Required Percentage Increase in Tension (%) 0 at 15°C

Temp. (Deg. C)	Tension (kgf)	DESIGN TABLE								
		Sag (m) for Span Length (m)								
		60	70	80	90	100	110	120	130	180
-5.6	904.0	0.56	0.77	1.00	1.27	1.56	1.89	2.25	2.64	5.06
0	860.3	0.59	0.80	1.05	1.33	1.64	1.99	2.36	2.77	5.32
5	825.1	0.62	0.84	1.10	1.39	1.71	2.07	2.47	2.89	5.55
10	792.9	0.64	0.87	1.14	1.44	1.78	2.16	2.57	3.01	5.77
15	763.5	0.67	0.91	1.18	1.50	1.85	2.24	2.66	3.13	5.99
20	736.5	0.69	0.94	1.23	1.55	1.92	2.32	2.76	3.24	6.21
25	711.7	0.71	0.97	1.27	1.61	1.98	2.40	2.86	3.35	6.43
30	689.0	0.74	1.00	1.31	1.66	2.05	2.48	2.95	3.46	6.64
35	667.9	0.76	1.04	1.35	1.71	2.11	2.56	3.05	3.57	6.85
40	648.4	0.78	1.07	1.39	1.76	2.18	2.64	3.14	3.68	7.06
45	630.4	0.81	1.10	1.43	1.82	2.24	2.71	3.23	3.79	7.26
50	613.5	0.83	1.13	1.47	1.86	2.30	2.79	3.32	3.89	7.46
55	597.9	0.85	1.16	1.51	1.91	2.36	2.86	3.40	3.99	7.65
60	583.2	0.87	1.19	1.55	1.96	2.42	2.93	3.49	4.09	7.85
65	569.5	0.89	1.22	1.59	2.01	2.48	3.00	3.57	4.19	8.04
70	556.6	0.91	1.24	1.62	2.06	2.54	3.07	3.65	4.29	8.22
75	544.4	0.93	1.27	1.66	2.10	2.59	3.14	3.74	4.38	8.41
80	533.0	0.95	1.30	1.70	2.15	2.65	3.21	3.82	4.48	8.59